



Department of EARTH SCIENCES

Seismic Image Labarortory

Pre Trained Model Performance on Seismic Image Data

**We are Using here a  PyTorch
Pre-trained model
(Image Classification)**

Timeline

AlexNet (2012)

GoogLeNet (Inception v1) (2014)

VGG (2015)

ResNet (2015)

Inception v3 (2016)

ResNeXt (2016)

DenseNet (2016)

ShuffleNet V2 (2016)

SqueezeNet (2016)

MobileNet V1 (2017)

MobileNet V2 (2017)

MNASNet (2018)

EfficientNet (2018)

EfficientNetV2 (2019)

Wide ResNet (2019)

RegNet (2020)

SwinTransformer (2020)

VisionTransformer (ViT) (2020)

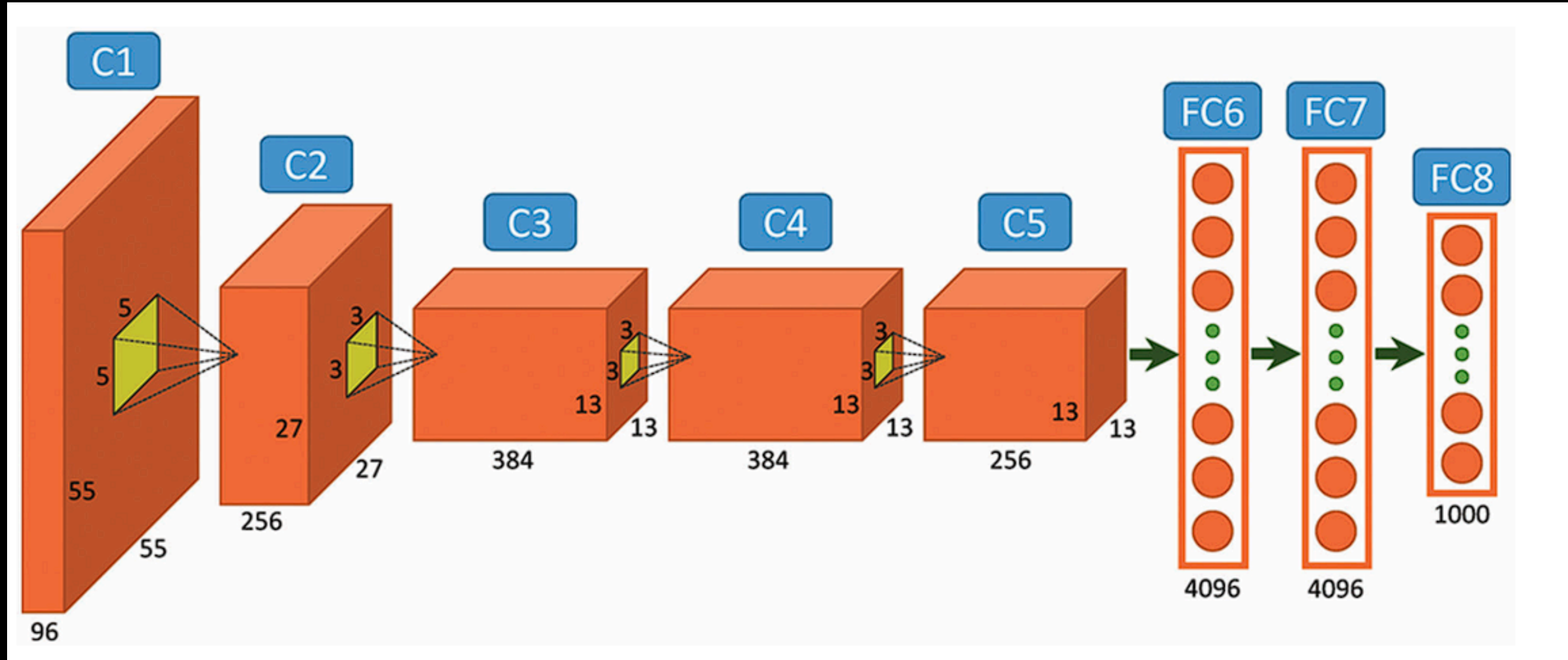
ConvNeXt (2021)

MaxViT (2022)

MobileNet V3 (2022)

AlexNet

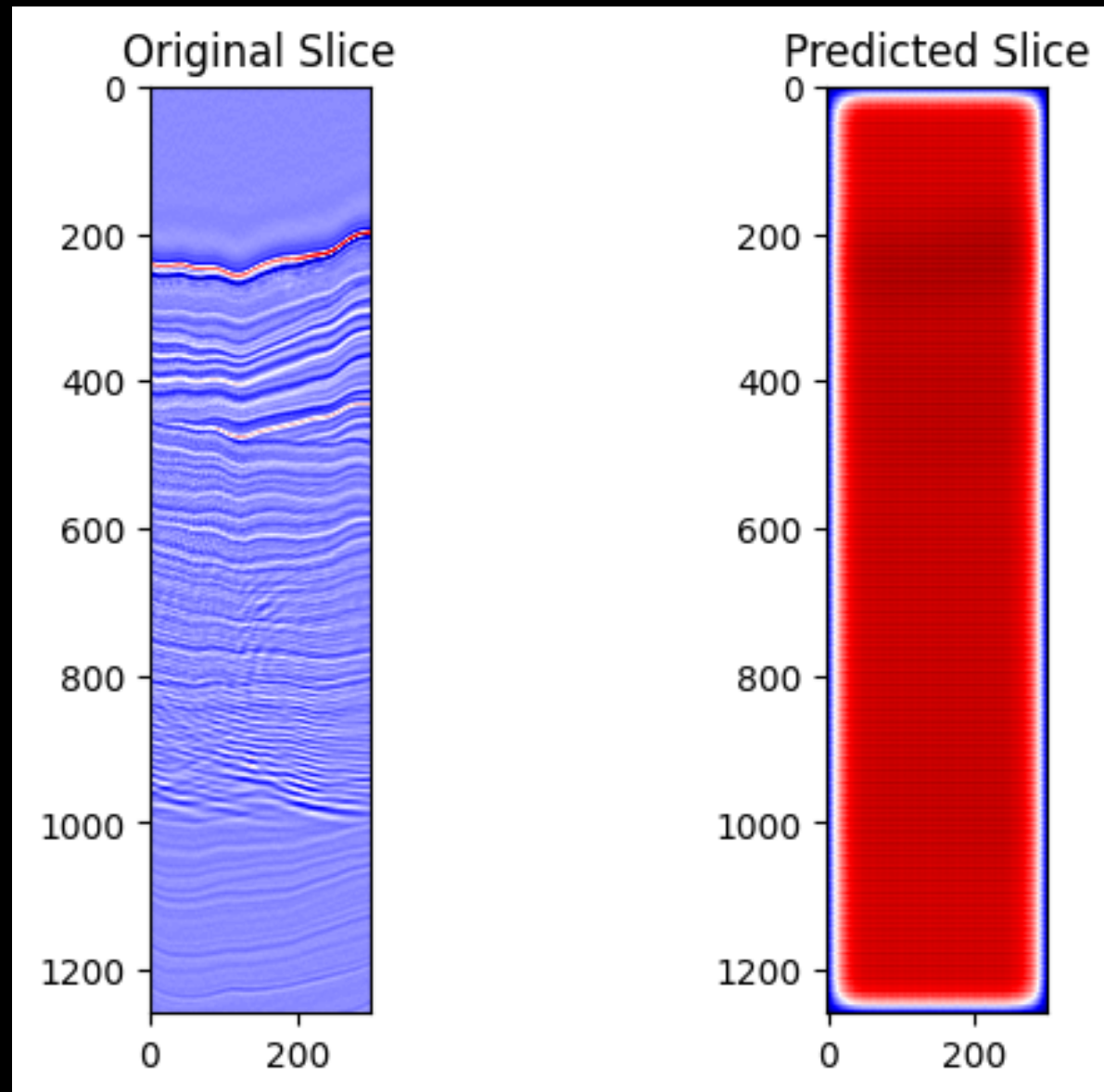
Model Design



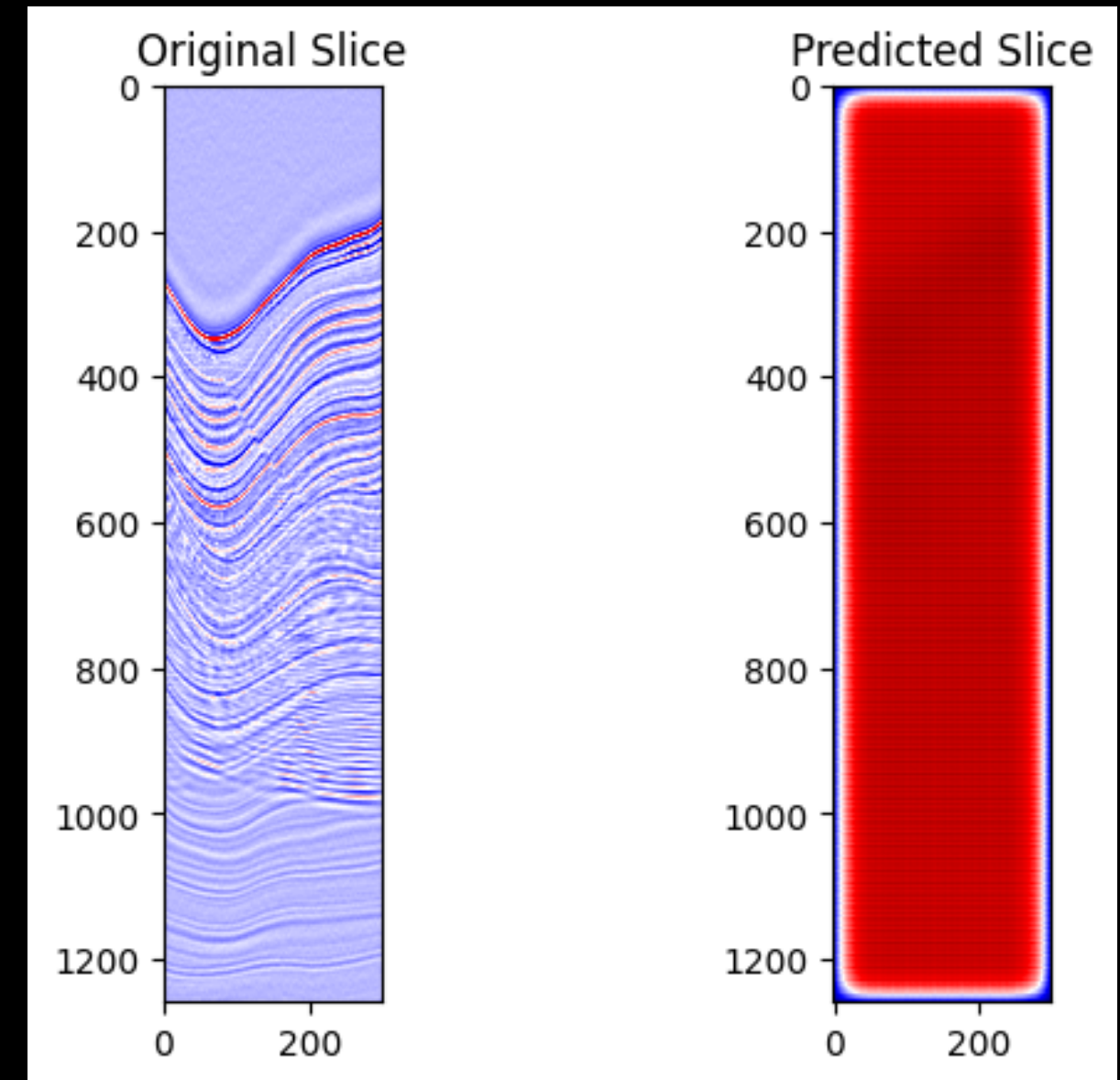
Model Design

Layer		Feature Map	Size	Kernel Size	Stride	Activation
Input	Image	1	227x227x3	-	-	-
1	Convolution	96	55 x 55 x 96	11x11	4	relu
	Max Pooling	96	27 x 27 x 96	3x3	2	relu
2	Convolution	256	27 x 27 x 256	5x5	1	relu
	Max Pooling	256	13 x 13 x 256	3x3	2	relu
3	Convolution	384	13 x 13 x 384	3x3	1	relu
4	Convolution	384	13 x 13 x 384	3x3	1	relu
5	Convolution	256	13 x 13 x 256	3x3	1	relu
	Max Pooling	256	6 x 6 x 256	3x3	2	relu
6	FC	-	9216	-	-	relu
7	FC	-	4096	-	-	relu
8	FC	-	4096	-	-	relu
Output	FC	-	1000	-	-	Softmax

Model Performance



Epoch [2/200],
Train Loss: 9001730764062641968430186496.0000, Train SSIM: nan,
Val Loss: 9116585784941655656159510528.0000, Val SSIM: nan

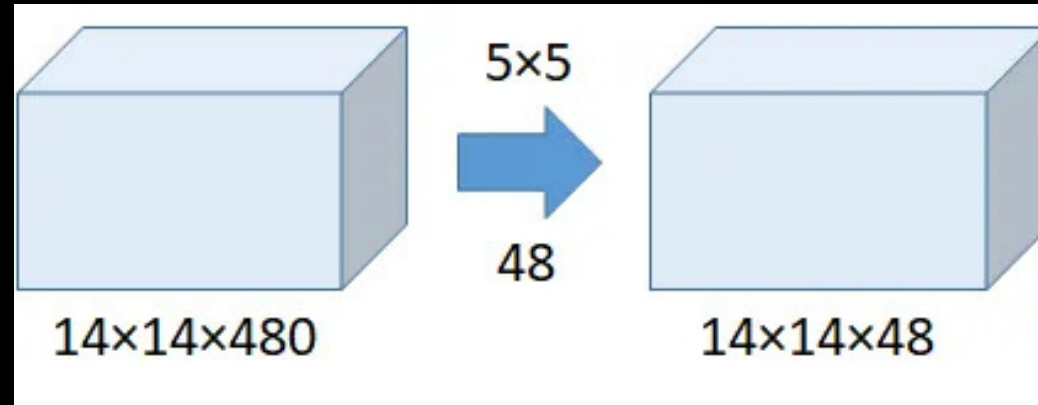


Epoch [3/200],
Train Loss: 9001252065228640735306186752.0000, Train SSIM: nan,
Val Loss: 9116173913807028063458820096.0000, Val SSIM: nan

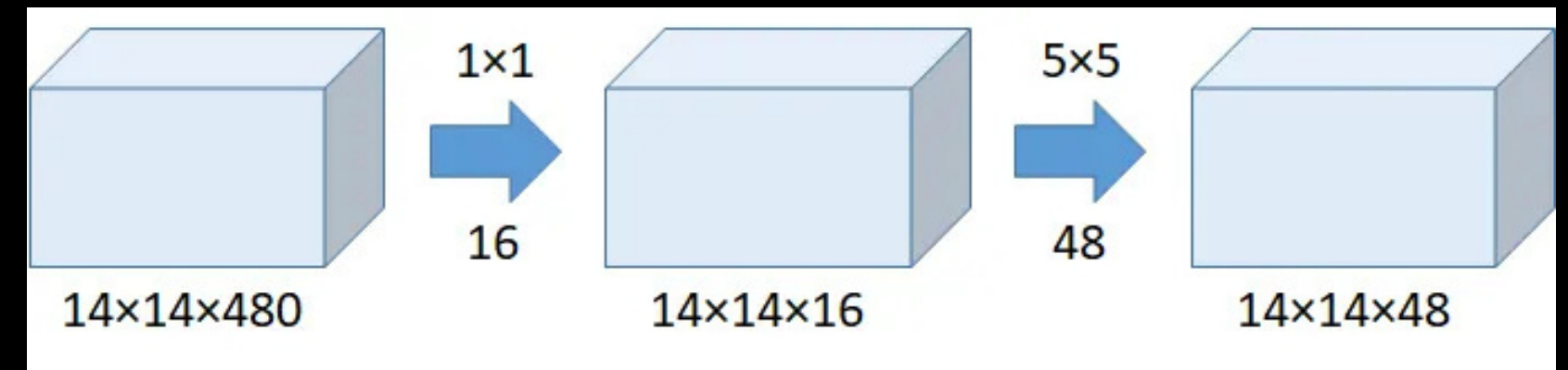
GoLeNet

(Inspection v1)

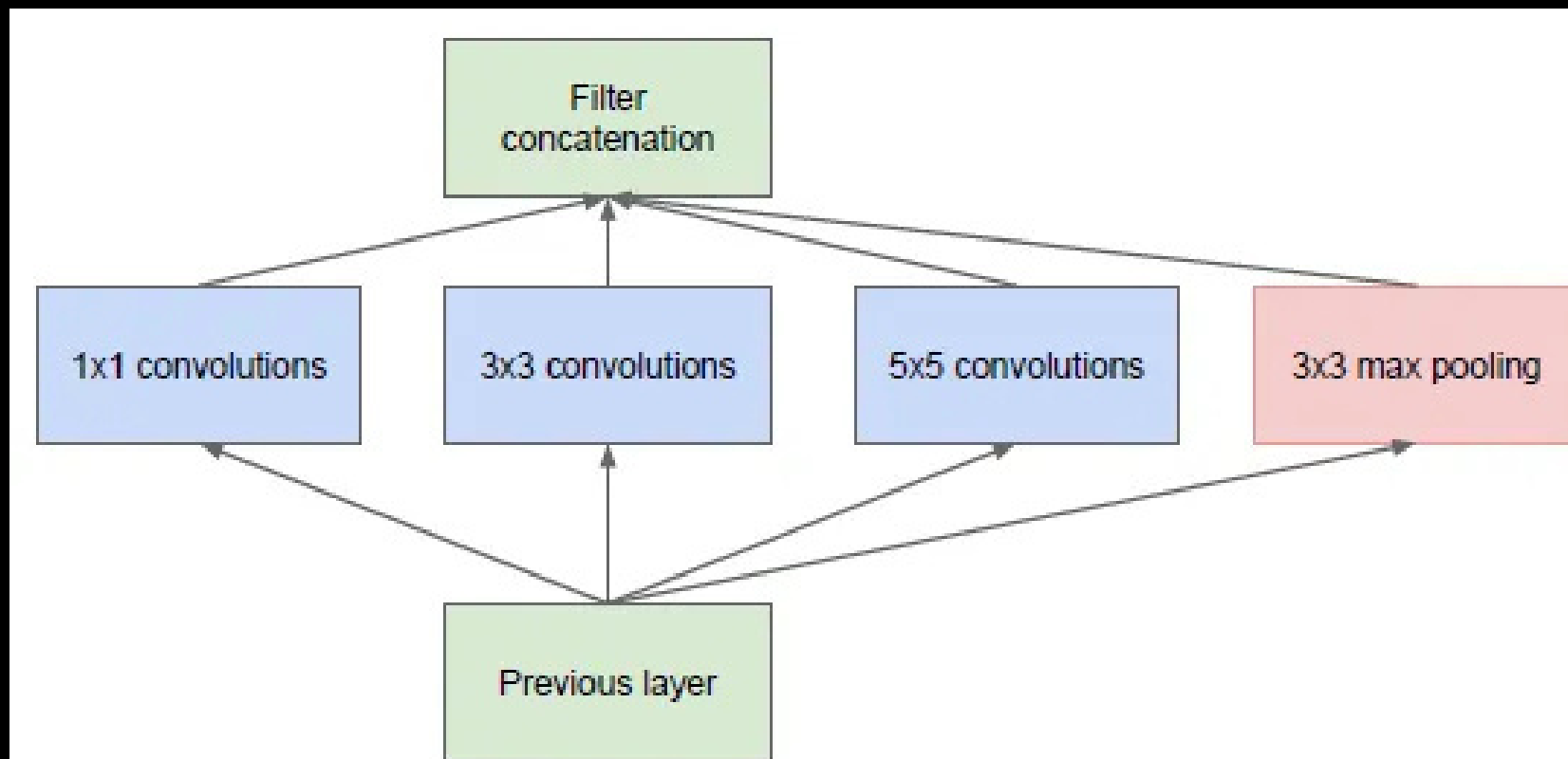
Model Design



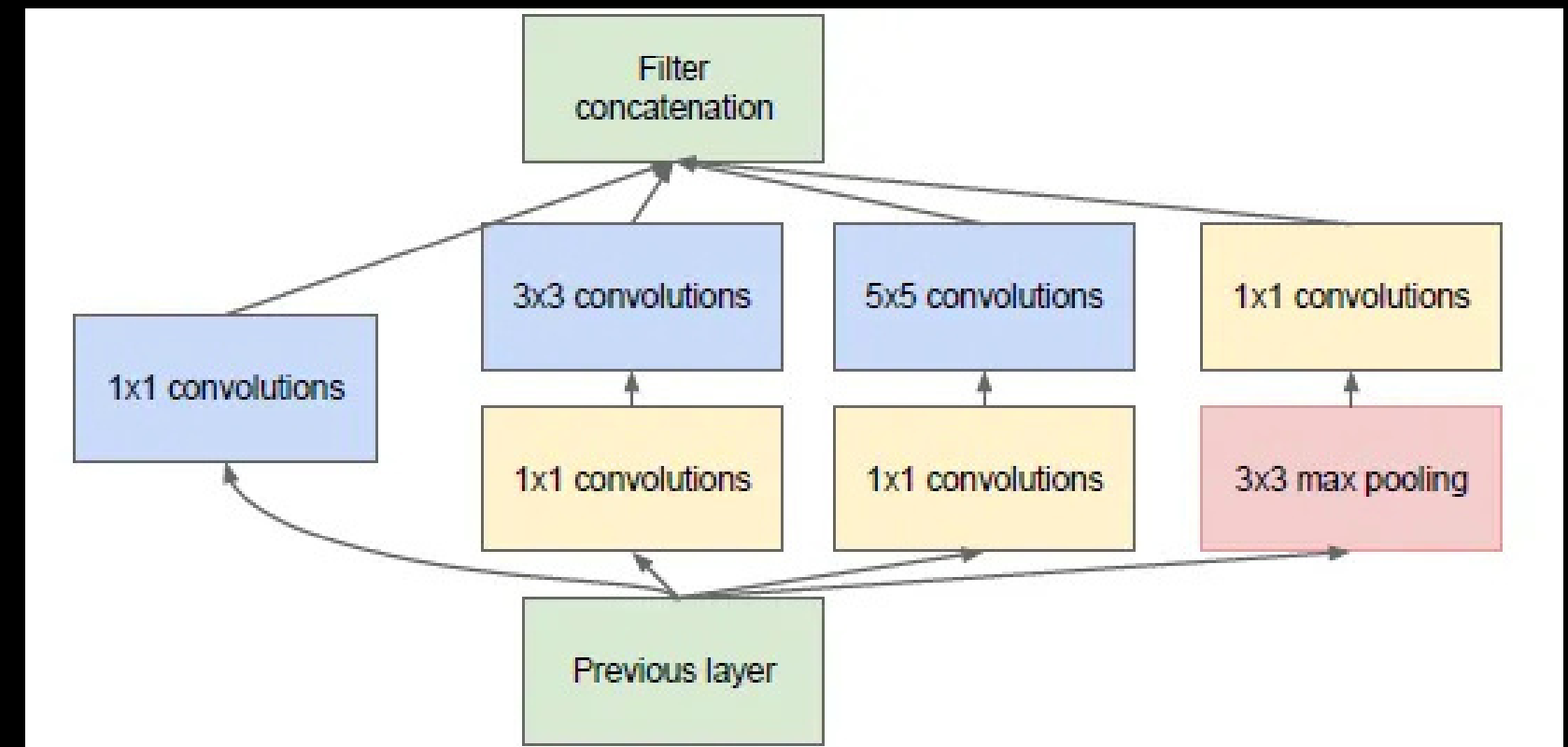
Number of operations = $(14 \times 14 \times 48) \times (5 \times 5 \times 480) = 112.9\text{M}$



Number of operations for 1x1 = $(14 \times 14 \times 16) \times (1 \times 1 \times 480) = 1.5\text{M}$
 Number of operations for 5x5 = $(14 \times 14 \times 48) \times (5 \times 5 \times 16) = 3.8\text{M}$
 Total number of operations = $1.5\text{M} + 3.8\text{M} = 5.3\text{M}$
 which is much much smaller than 112.9M !!!!!!!!!!!!!!!

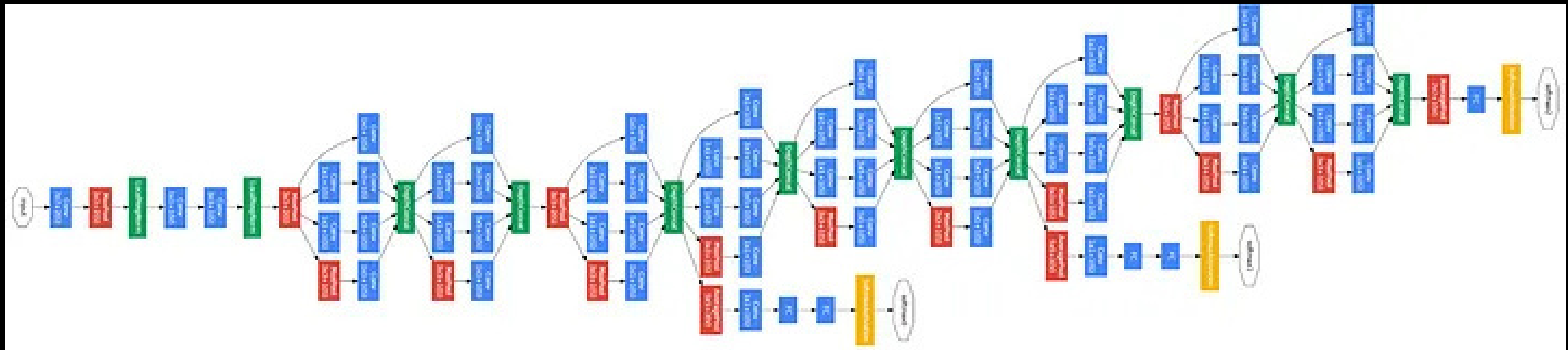
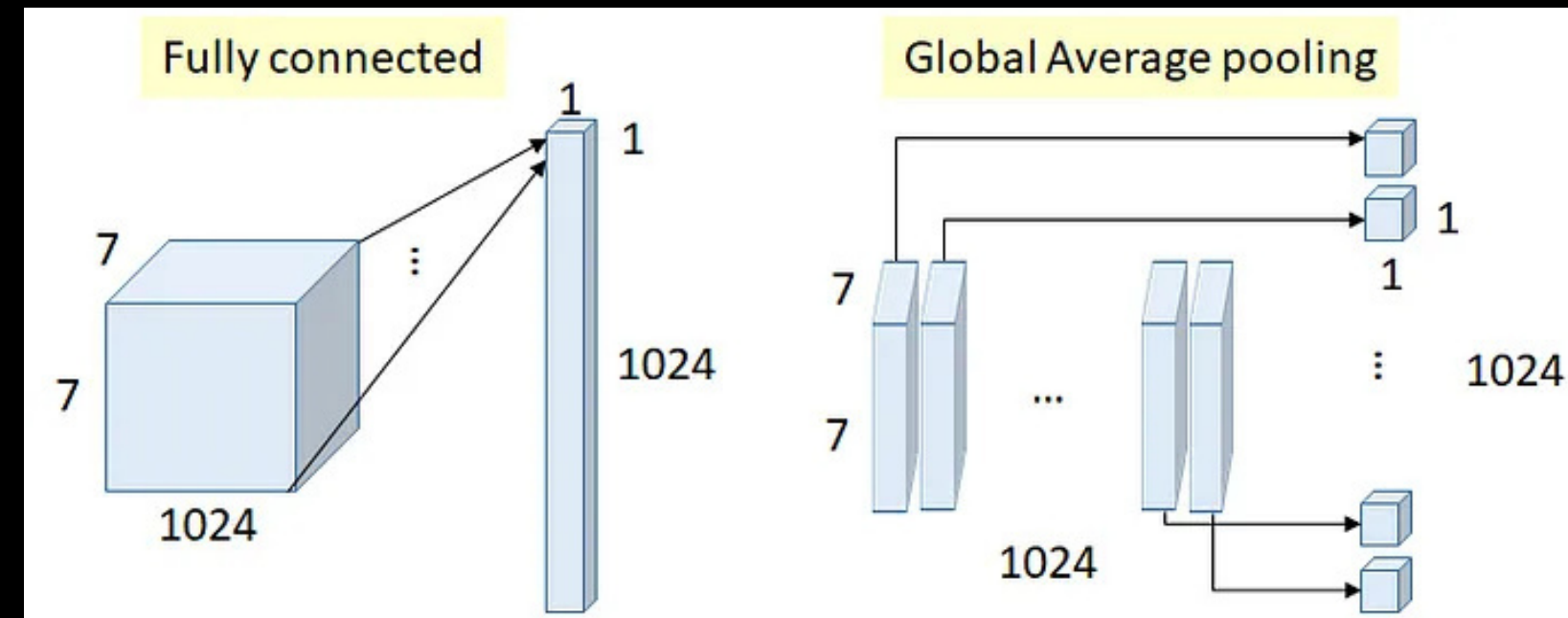


(a) Inception module, naïve version



(b) Inception module with dimensionality reduction

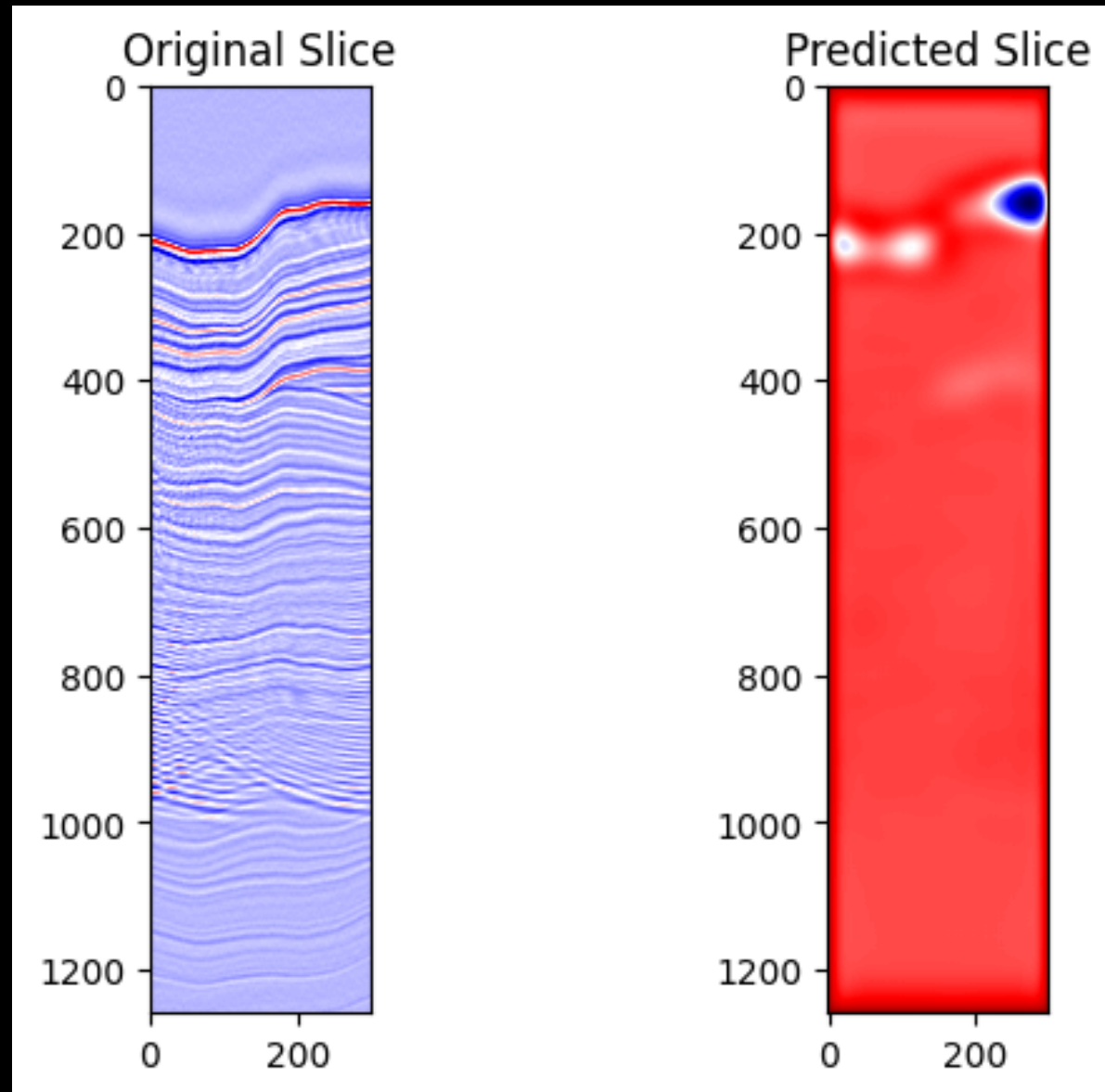
Model Design



Model Design

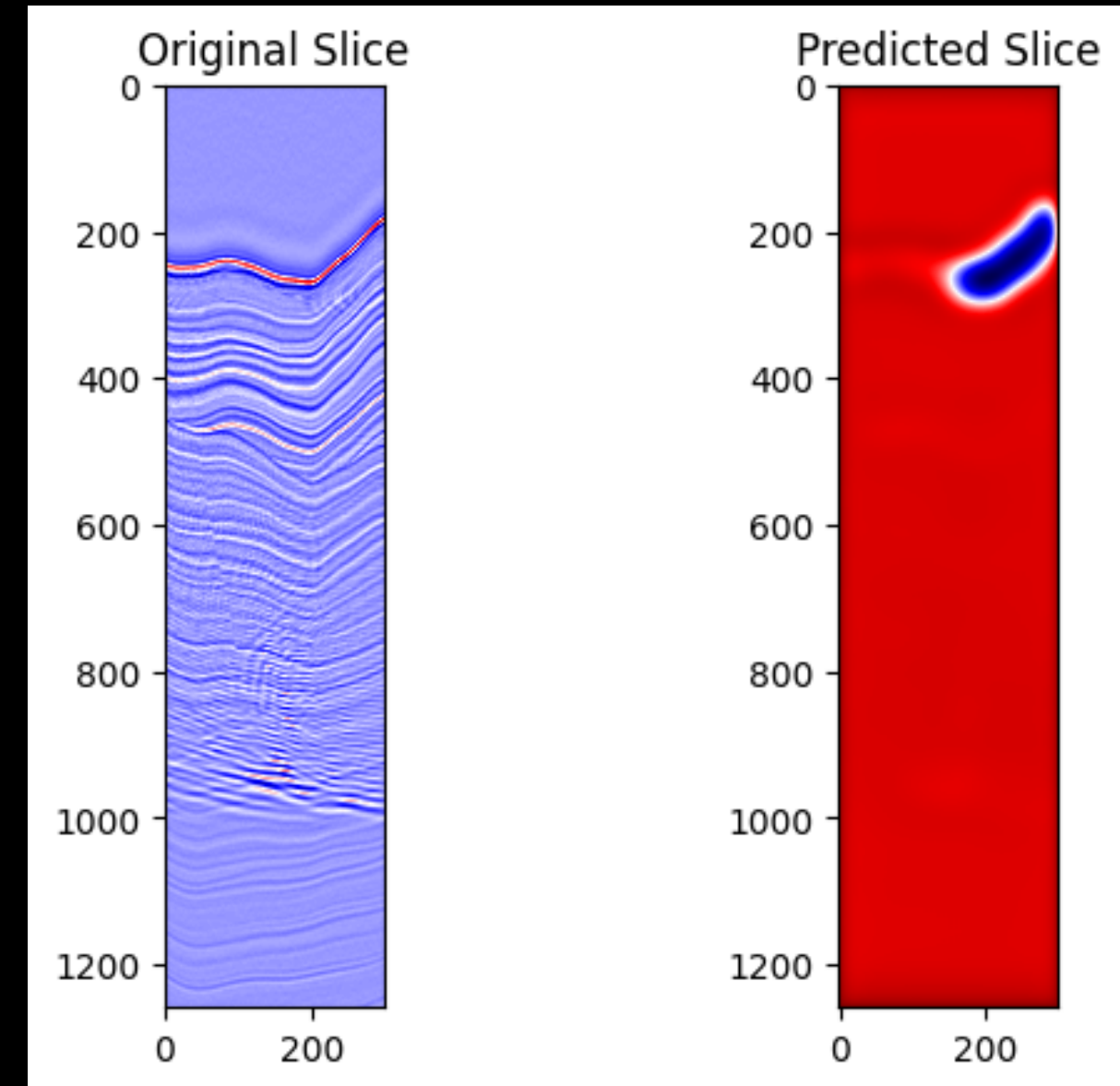
type	patch size/ stride	output size	depth	#1×1	#3×3 reduce	#3×3	#5×5 reduce	#5×5	pool proj	params	ops
convolution	7×7/2	112×112×64	1							2.7K	34M
max pool	3×3/2	56×56×64	0								
convolution	3×3/1	56×56×192	2		64	192				112K	360M
max pool	3×3/2	28×28×192	0								
inception (3a)		28×28×256	2	64	96	128	16	32	32	159K	128M
inception (3b)		28×28×480	2	128	128	192	32	96	64	380K	304M
max pool	3×3/2	14×14×480	0								
inception (4a)		14×14×512	2	192	96	208	16	48	64	364K	73M
inception (4b)		14×14×512	2	160	112	224	24	64	64	437K	88M
inception (4c)		14×14×512	2	128	128	256	24	64	64	463K	100M
inception (4d)		14×14×528	2	112	144	288	32	64	64	580K	119M
inception (4e)		14×14×832	2	256	160	320	32	128	128	840K	170M
max pool	3×3/2	7×7×832	0								
inception (5a)		7×7×832	2	256	160	320	32	128	128	1072K	54M
inception (5b)		7×7×1024	2	384	192	384	48	128	128	1388K	71M
avg pool	7×7/1	1×1×1024	0								
dropout (40%)		1×1×1024	0								
linear		1×1×1000	1							1000K	1M
softmax		1×1×1000	0								

Model Performance



Epoch [1/200],

Train Loss: 541237030937295322827030790144.0000, Train SSIM: nan,
Val Loss: 1387358725187735787791674507264.0000, Val SSIM: nan

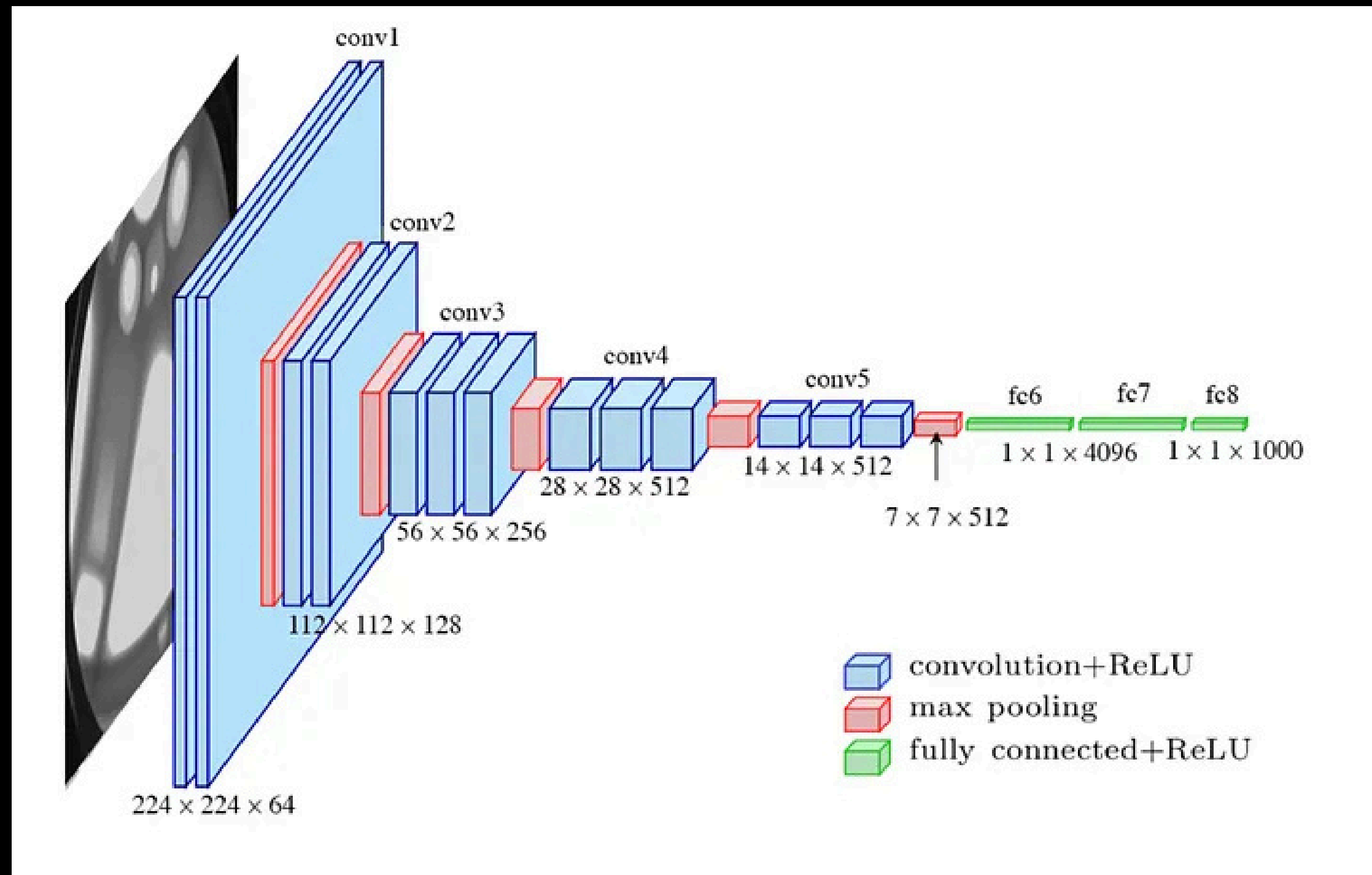


Epoch [2/200],

Train Loss: 9001730764062641968430186496.0000, Train SSIM: nan,
Val Loss: 9116585784941655656159510528.0000, Val SSIM: nan

VGGNet

Model Design



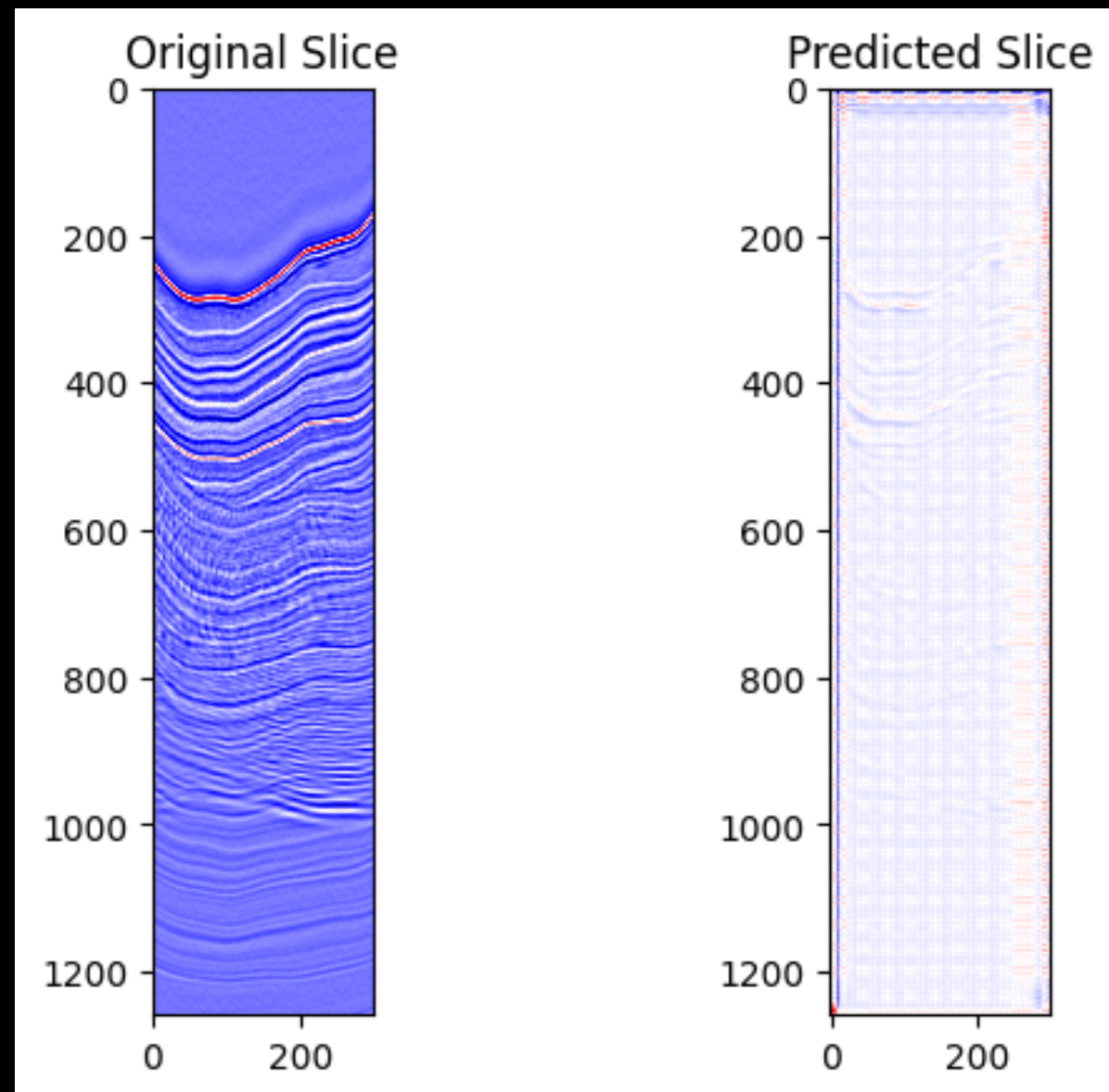
Model Design

ConvNet Configuration					
A	A-LRN	B	C	D	E
11 weight layers	11 weight layers	13 weight layers	16 weight layers	16 weight layers	19 weight layers
input (224×224 RGB image)					
conv3-64	conv3-64 LRN	conv3-64 conv3-64	conv3-64 conv3-64	conv3-64 conv3-64	conv3-64 conv3-64
maxpool					
conv3-128	conv3-128	conv3-128 conv3-128	conv3-128 conv3-128	conv3-128 conv3-128	conv3-128 conv3-128
maxpool					
conv3-256 conv3-256	conv3-256 conv3-256	conv3-256 conv3-256	conv3-256 conv3-256 conv1-256	conv3-256 conv3-256 conv3-256	conv3-256 conv3-256 conv3-256 conv3-256
maxpool					
conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512 conv1-512	conv3-512 conv3-512 conv3-512	conv3-512 conv3-512 conv3-512 conv3-512
maxpool					
conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512 conv1-512	conv3-512 conv3-512 conv3-512	conv3-512 conv3-512 conv3-512 conv3-512
maxpool					
FC-4096					
FC-4096					
FC-1000					
soft-max					

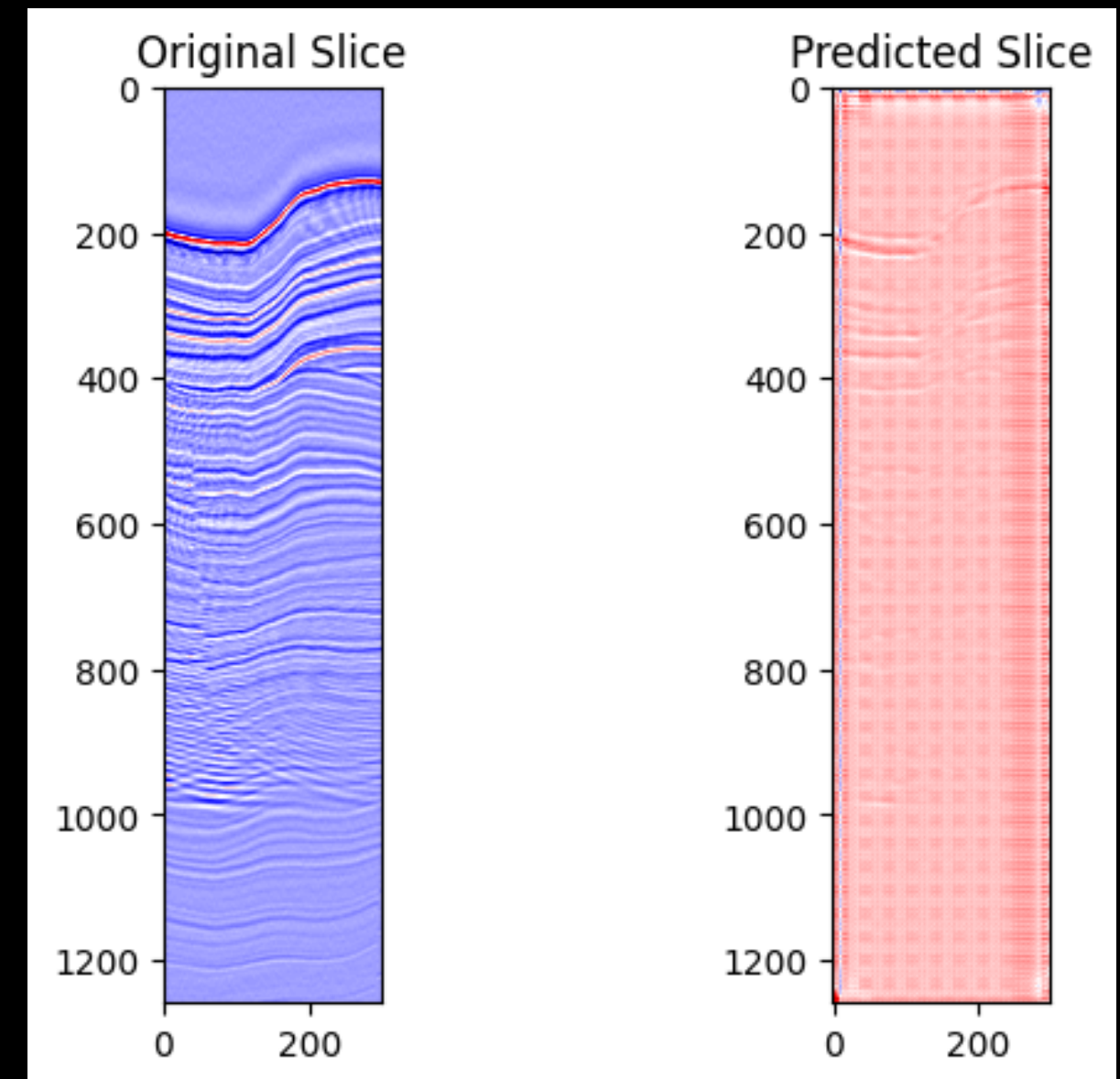
Model Design

	Layer	Feature Map	Size	Kernel Size	Stride	Activation
Input	Image	1	224 x 224 x 3	-	-	-
1	2 X Convolution	64	224 x 224 x 64	3x3	1	relu
	Max Pooling	64	112 x 112 x 64	3x3	2	relu
3	2 X Convolution	128	112 x 112 x 128	3x3	1	relu
	Max Pooling	128	56 x 56 x 128	3x3	2	relu
5	2 X Convolution	256	56 x 56 x 256	3x3	1	relu
	Max Pooling	256	28 x 28 x 256	3x3	2	relu
7	3 X Convolution	512	28 x 28 x 512	3x3	1	relu
	Max Pooling	512	14 x 14 x 512	3x3	2	relu
10	3 X Convolution	512	14 x 14 x 512	3x3	1	relu
	Max Pooling	512	7 x 7 x 512	3x3	2	relu
13	FC	-	25088	-	-	relu
14	FC	-	4096	-	-	relu
15	FC	-	4096	-	-	relu
Output	FC	-	1000	-	-	Softmax

Model Performance



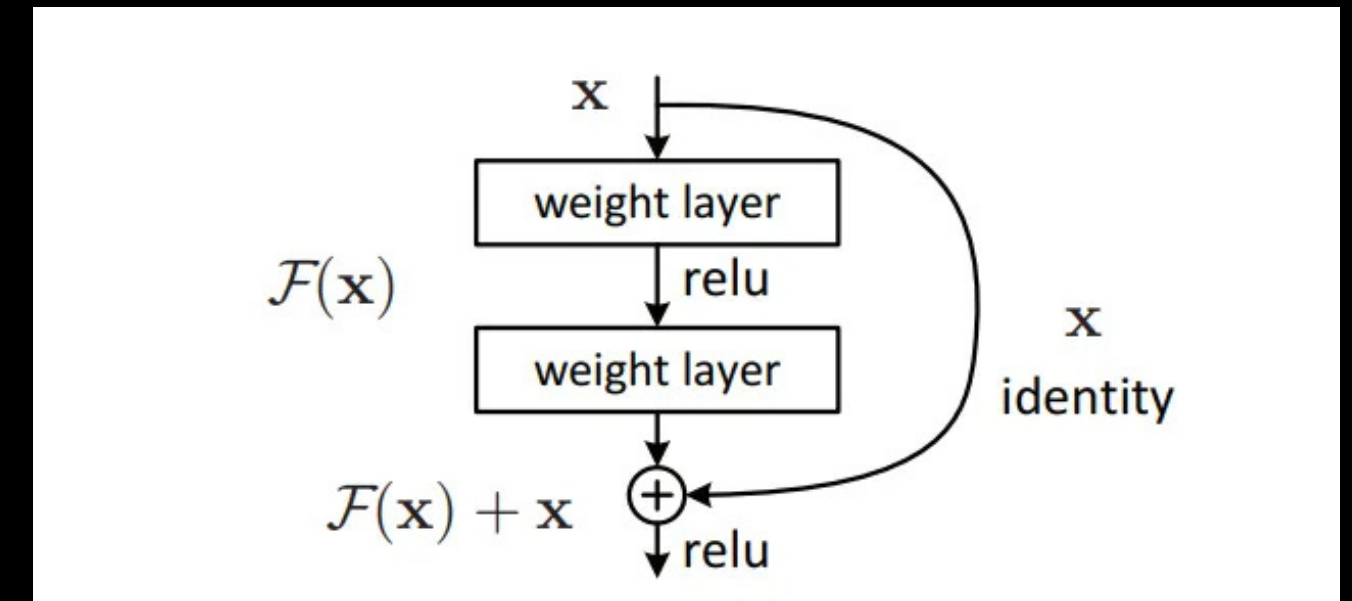
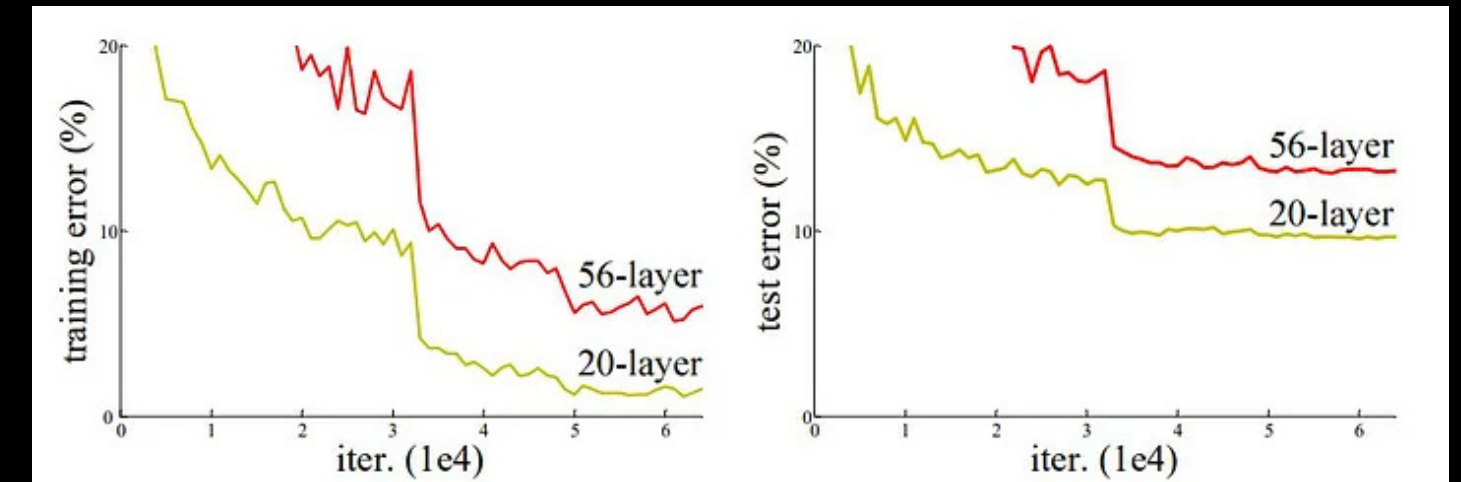
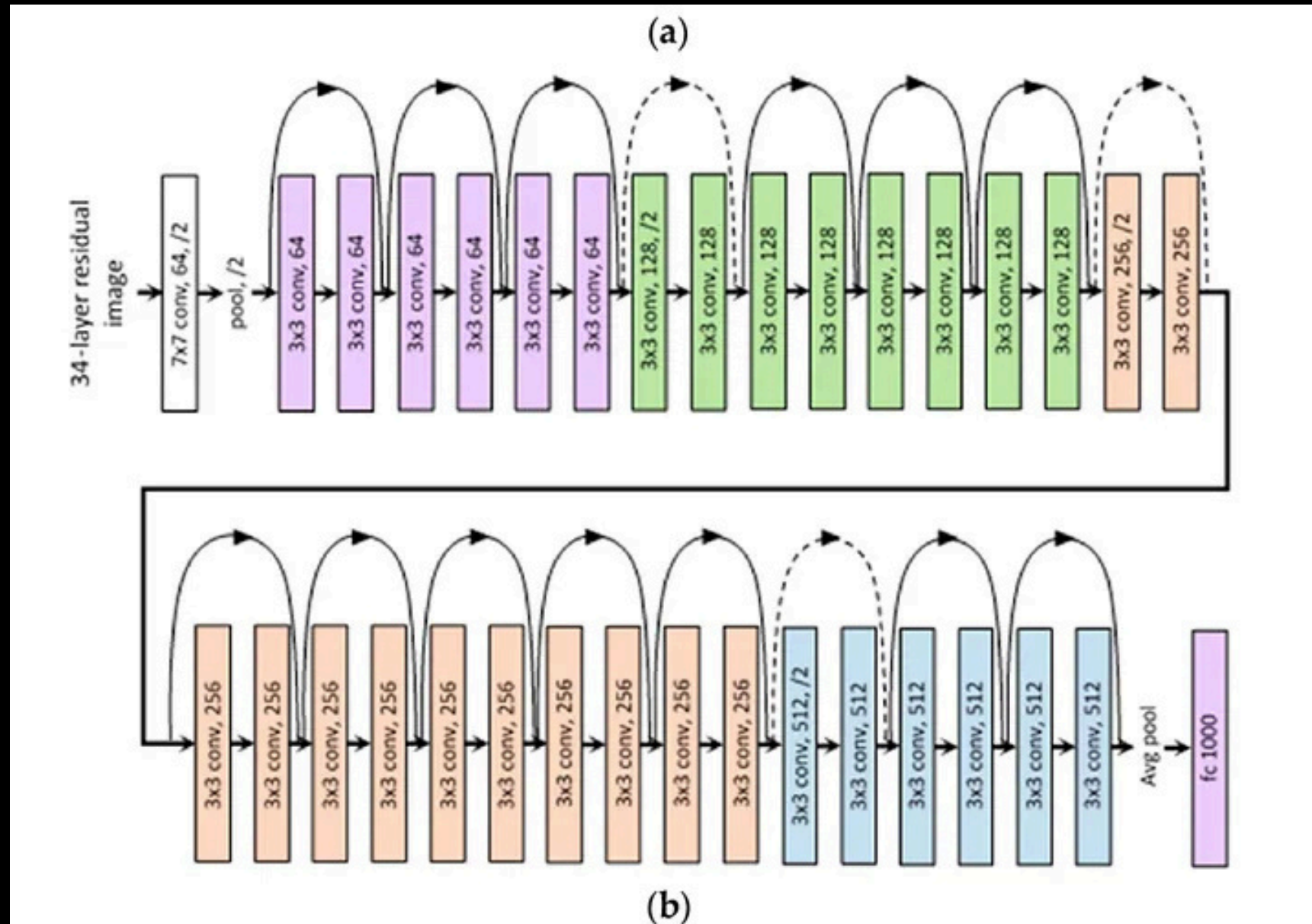
Epoch [5/200],
Train Loss: 51969.2820, Train SSIM: 0.2876,
Val Loss: 52615.7606, Val SSIM: 0.2307



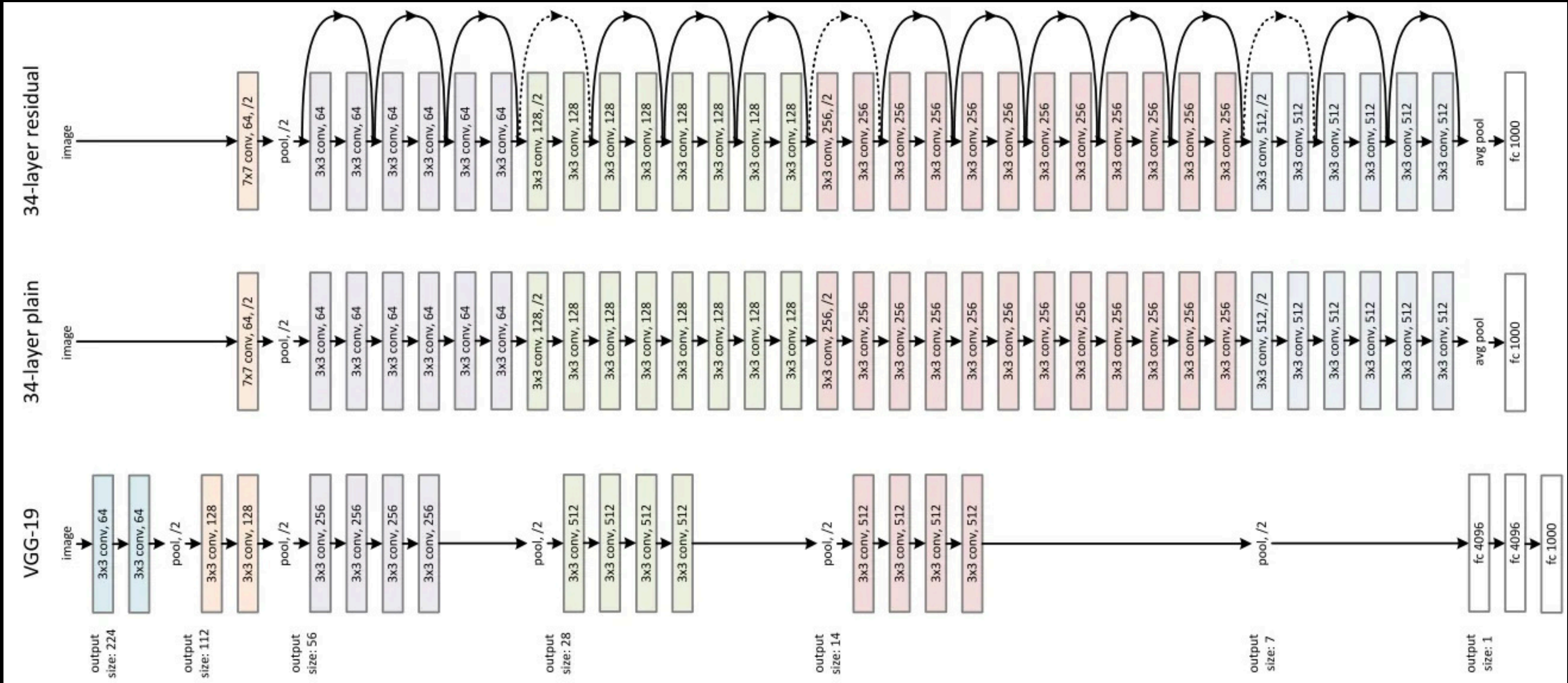
Epoch [6/200],
Train Loss: 36762.0051, Train SSIM: 0.2245,
Val Loss: 41903.7605, Val SSIM: 0.1761

ResNet

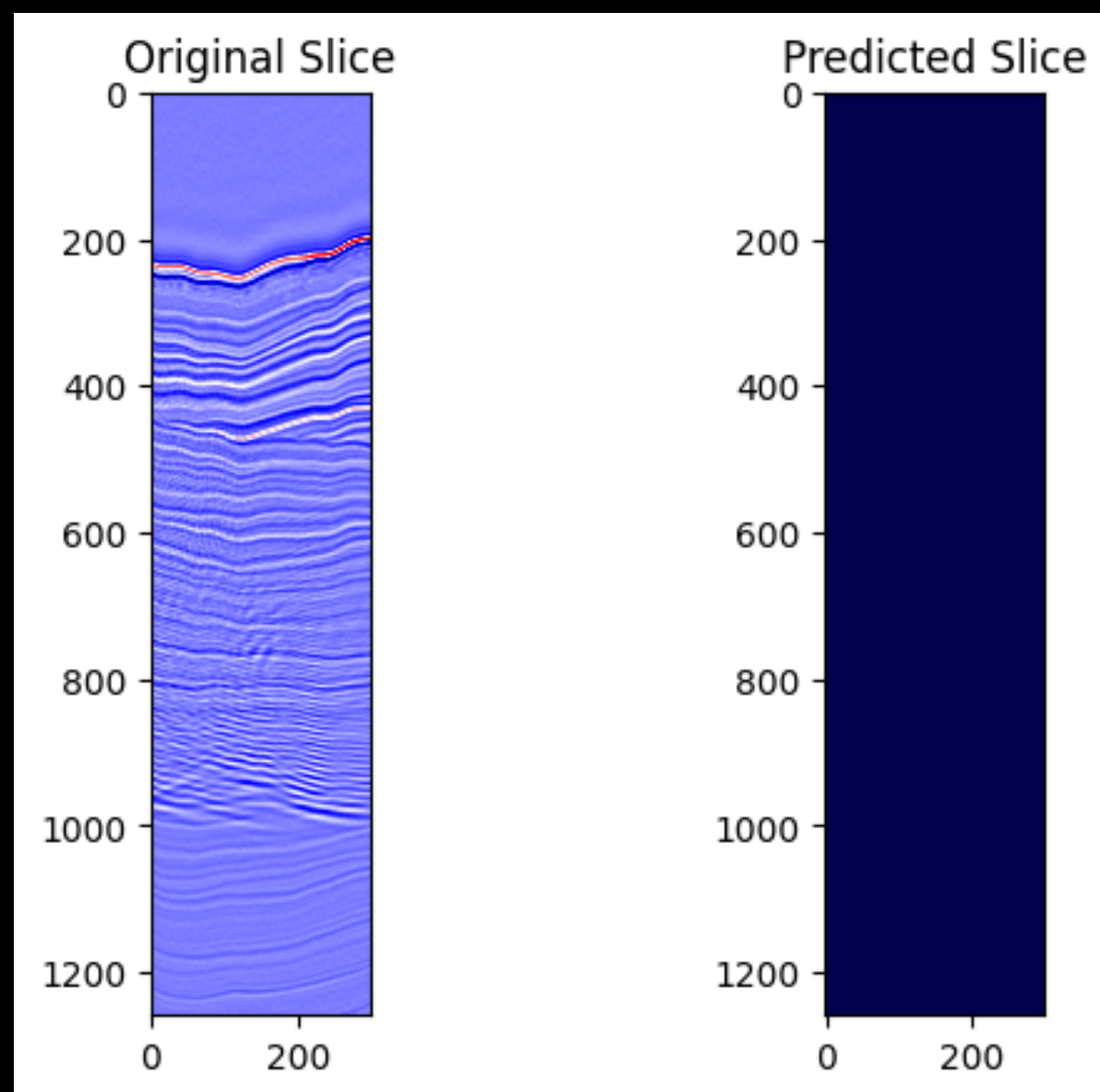
Model Design



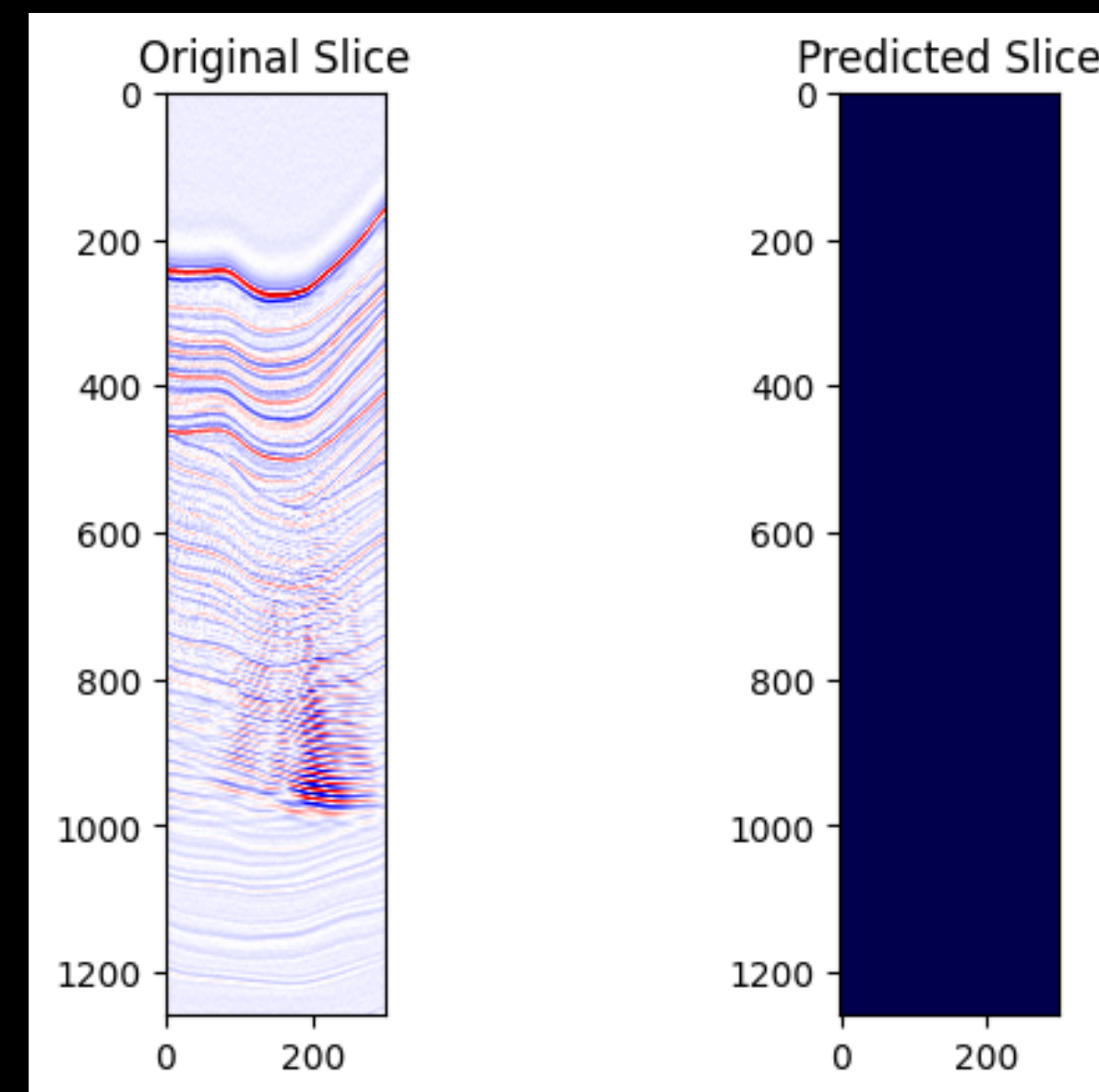
Model Design



Model Performance



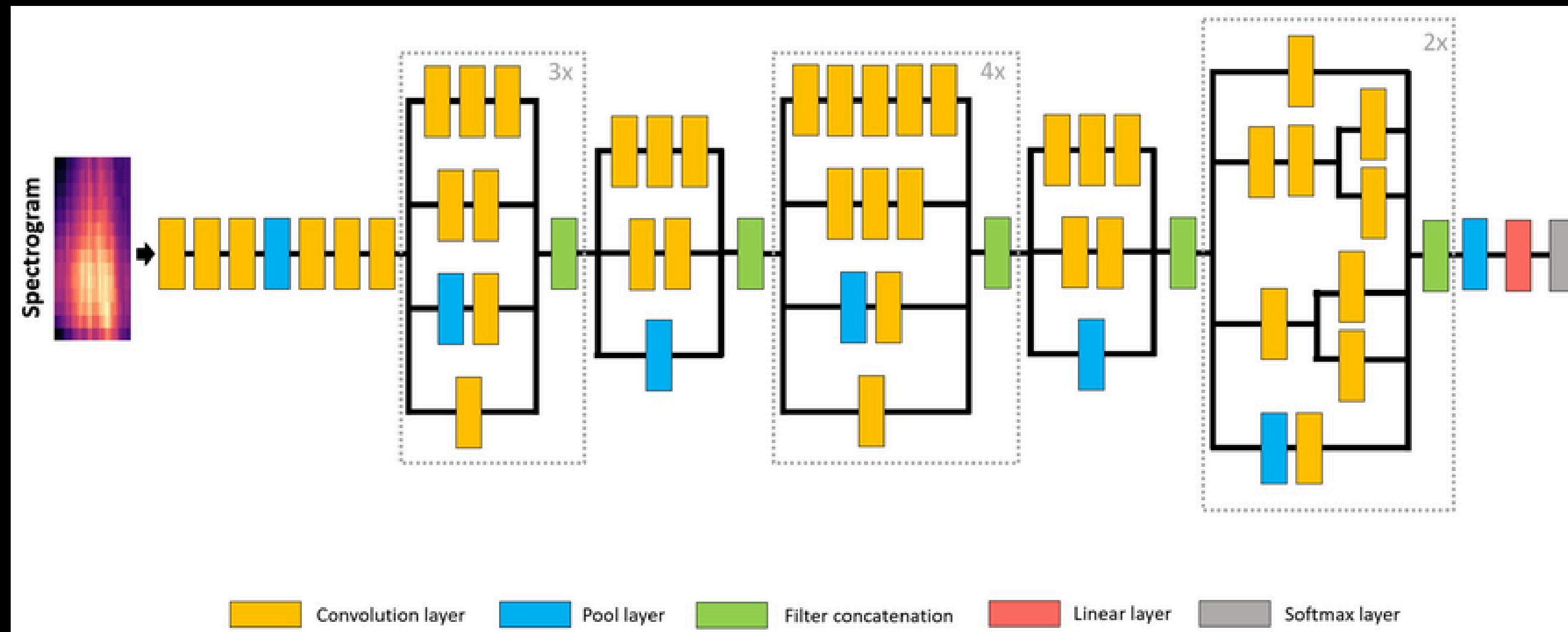
Epoch [144/200],
Train Loss: 10062.4538, Train SSIM: 0.4738,
Val Loss: 10088.1909, Val SSIM: 0.4163



Epoch [145/200],
Train Loss: 10066.3513, Train SSIM: 0.4724,
Val Loss: 10088.1914, Val SSIM: 0.4163

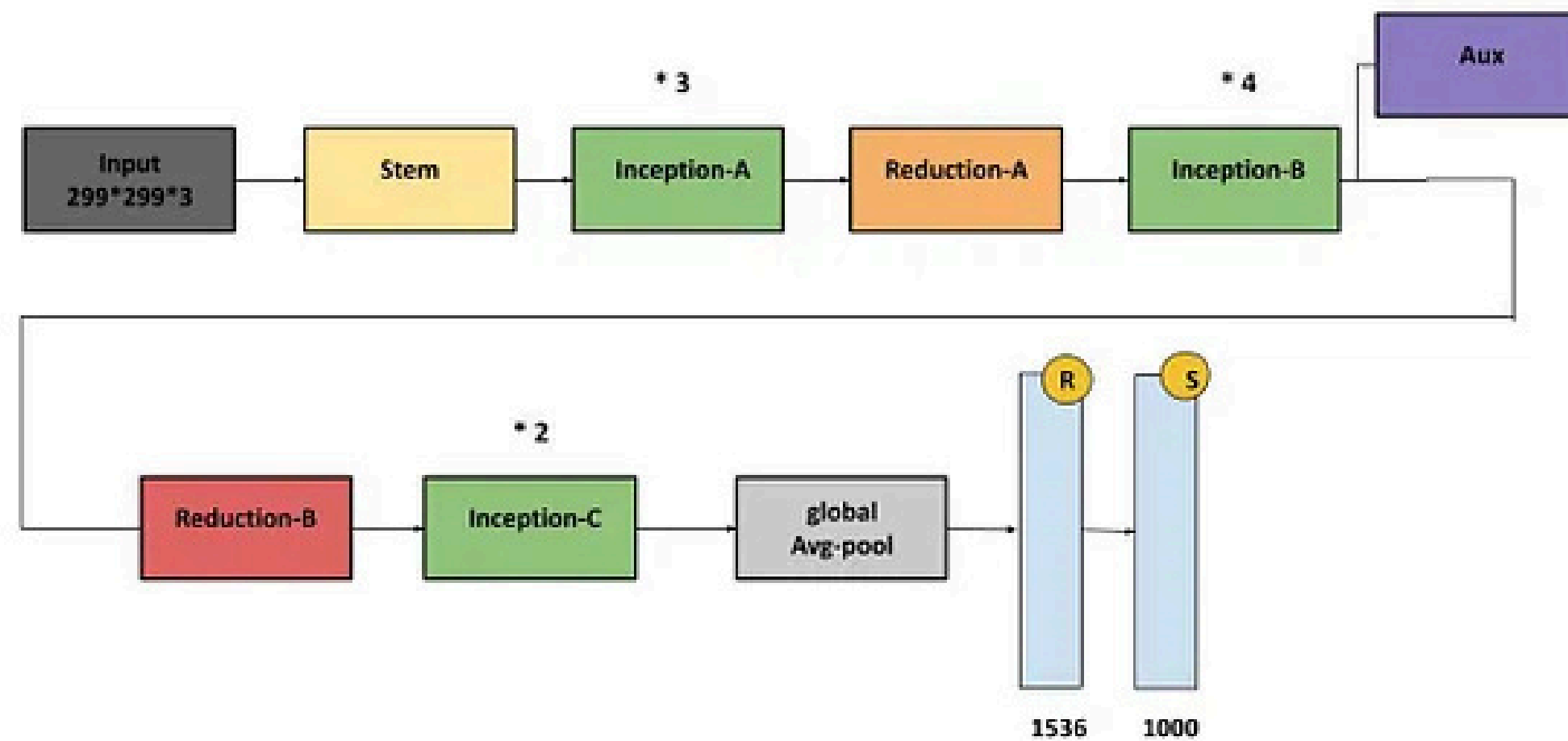
Inception v3

Model Design

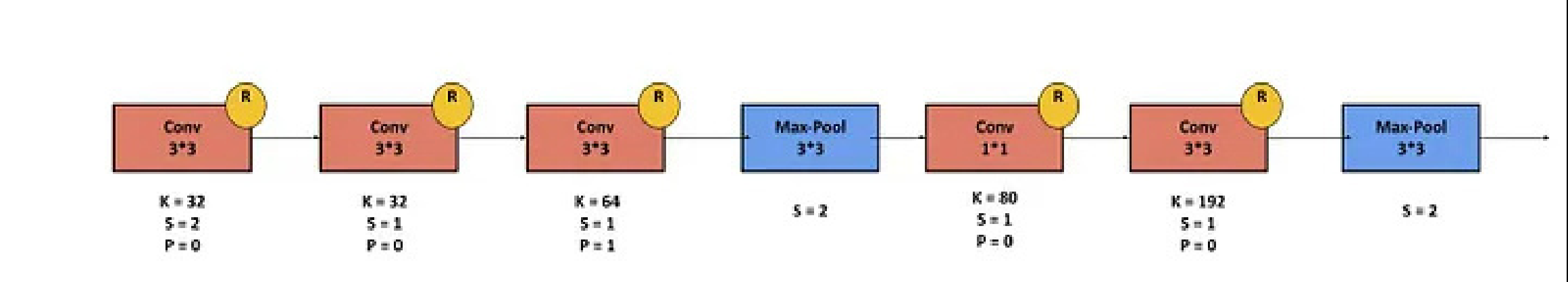


Model Design

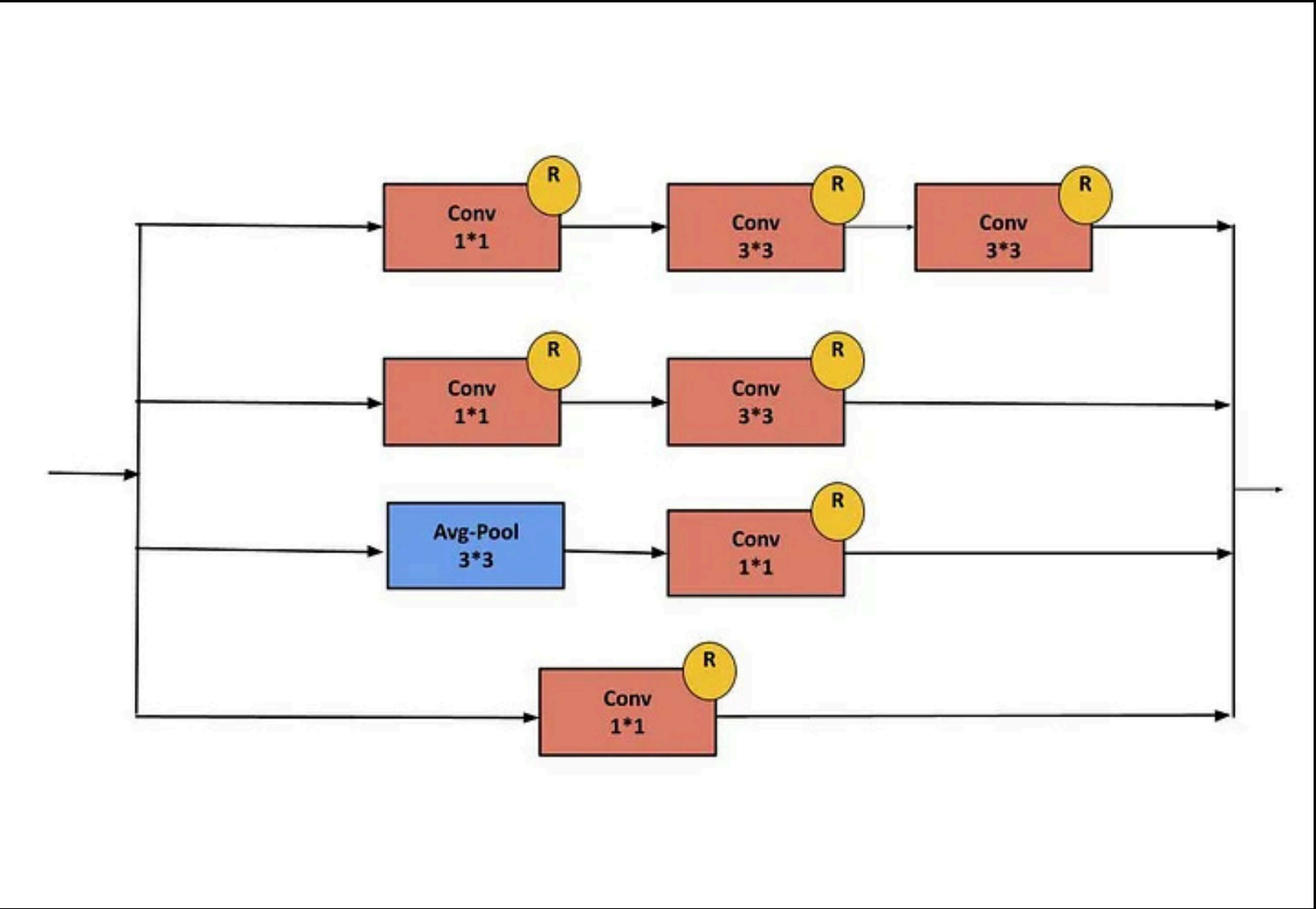
Inception V3



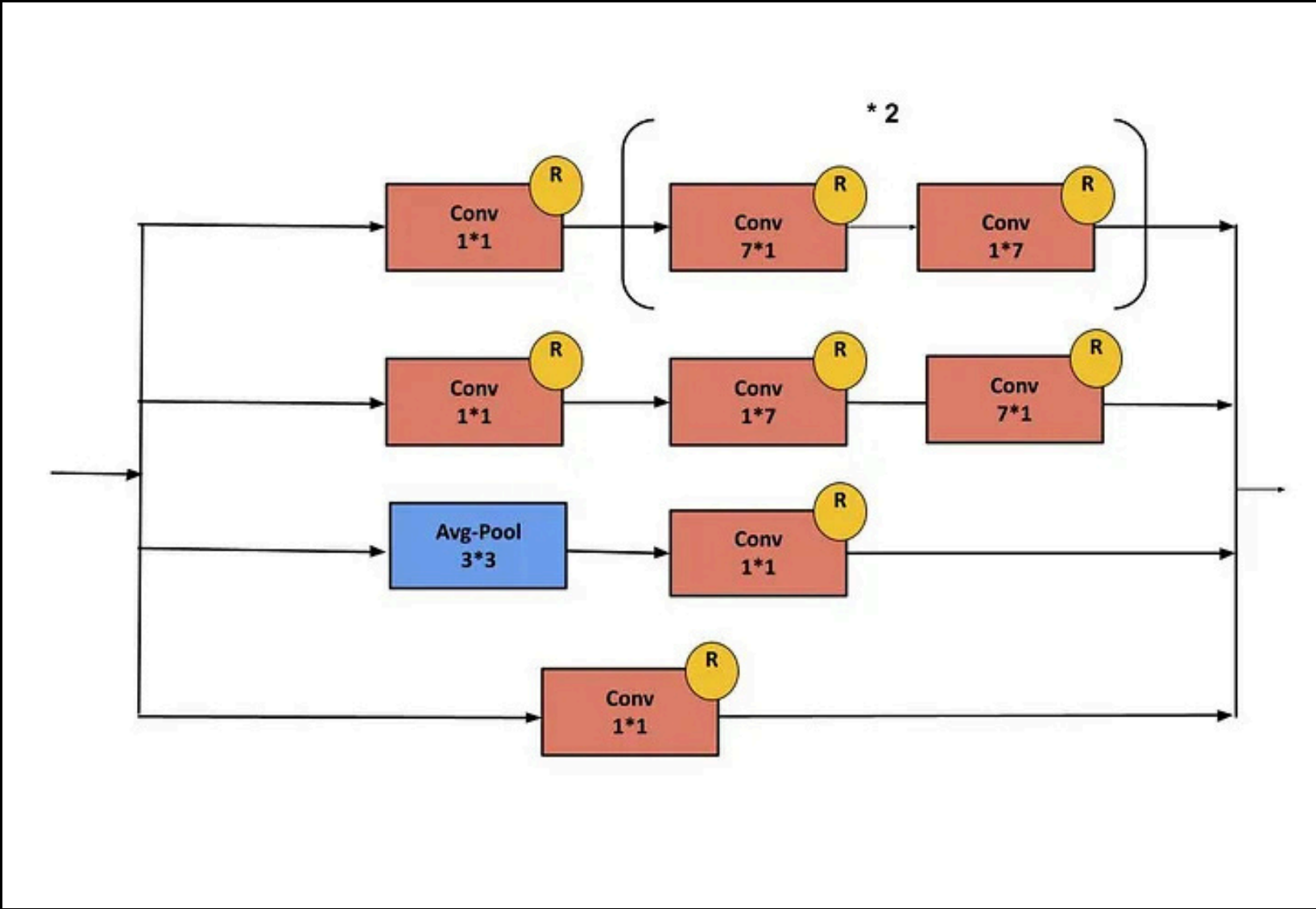
Model Design



(Stem Block)

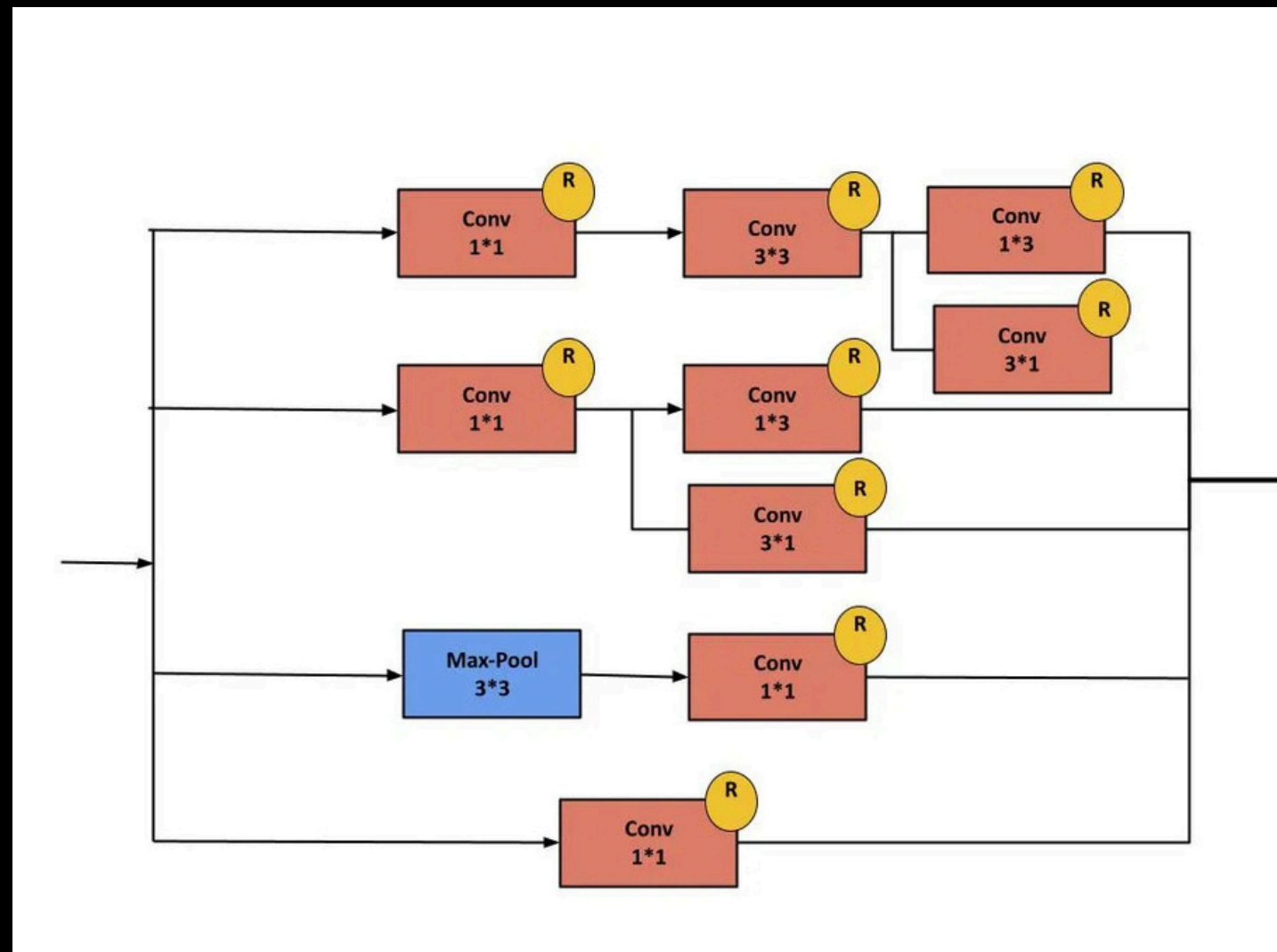


(Inception-A Block)

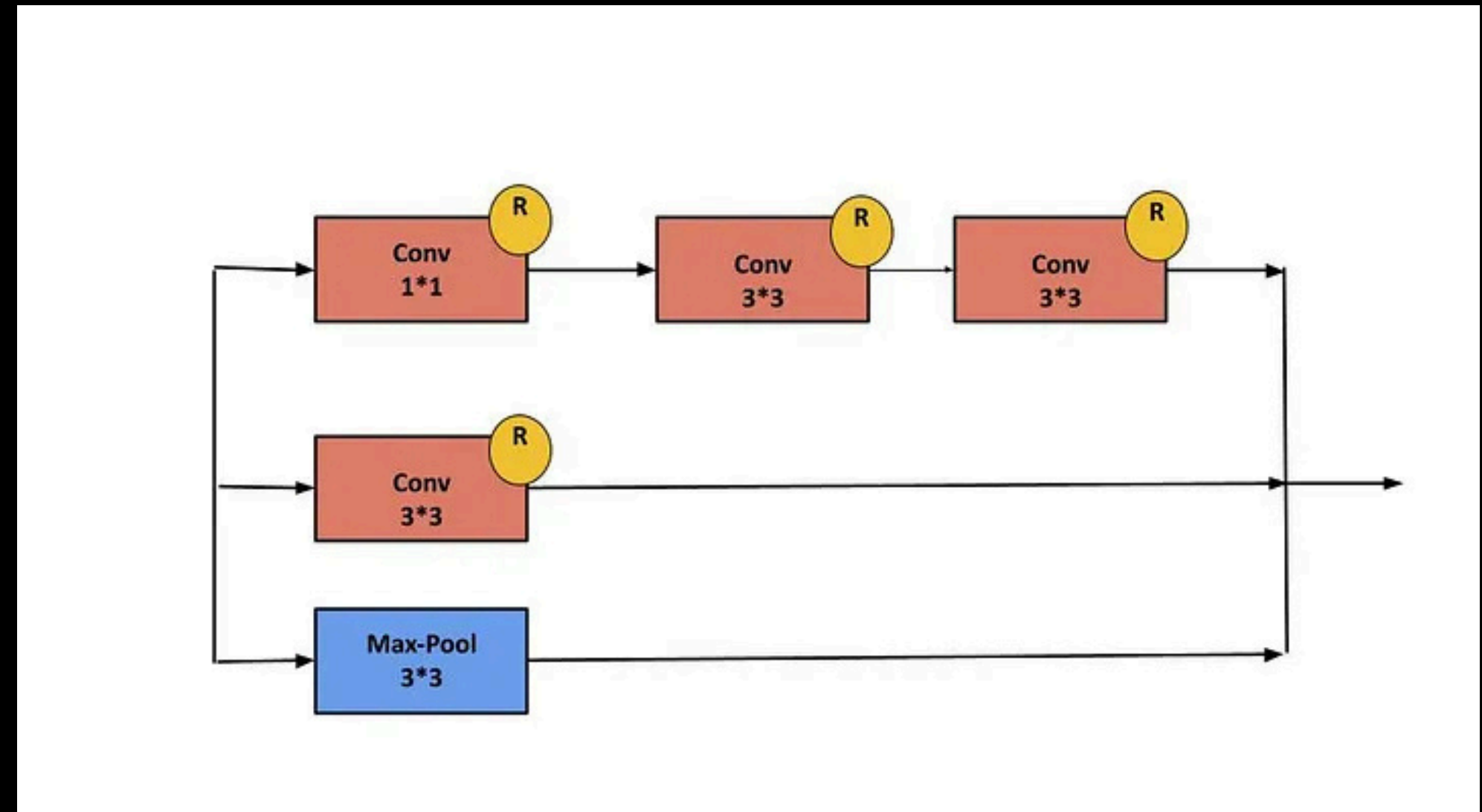


(Inception-B Block)

Model Design

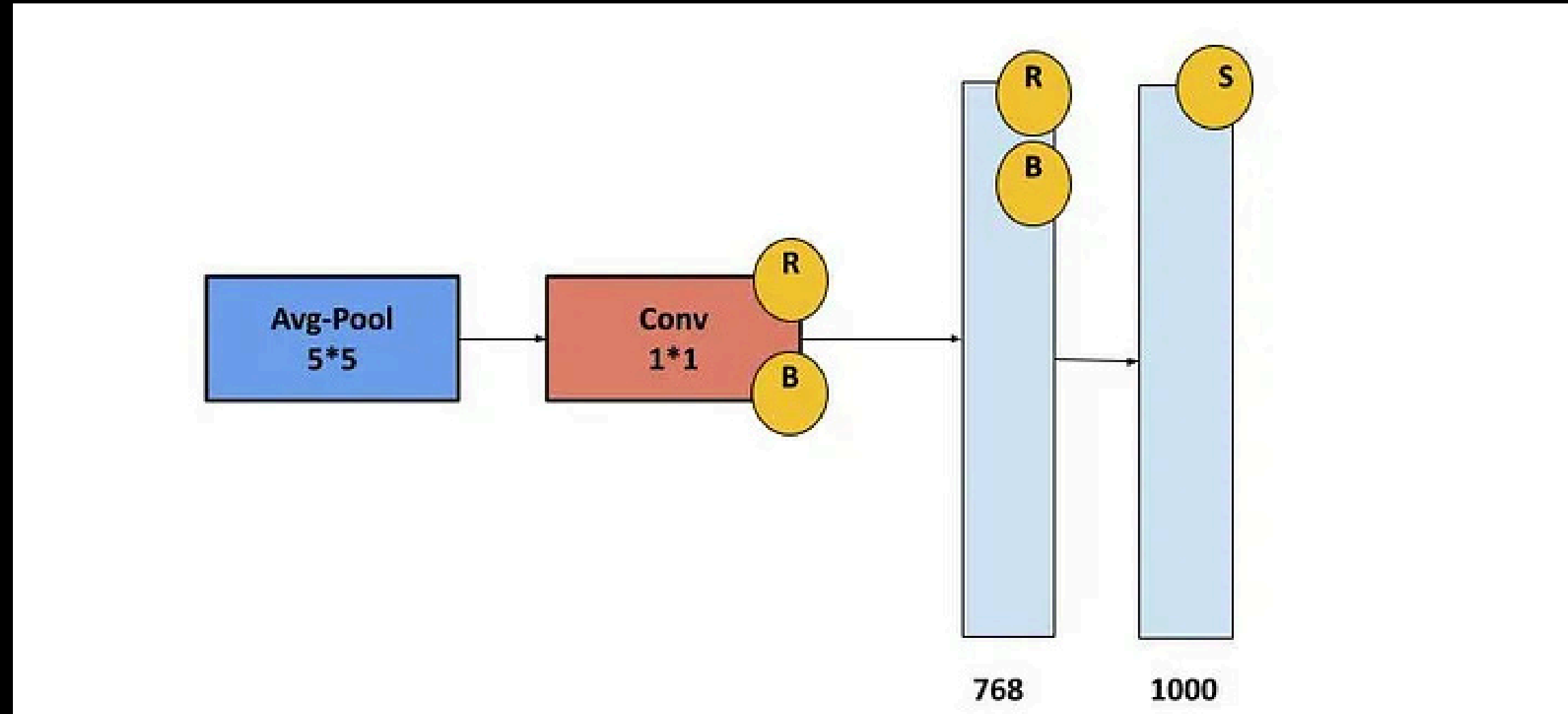


(Inception-C Block)



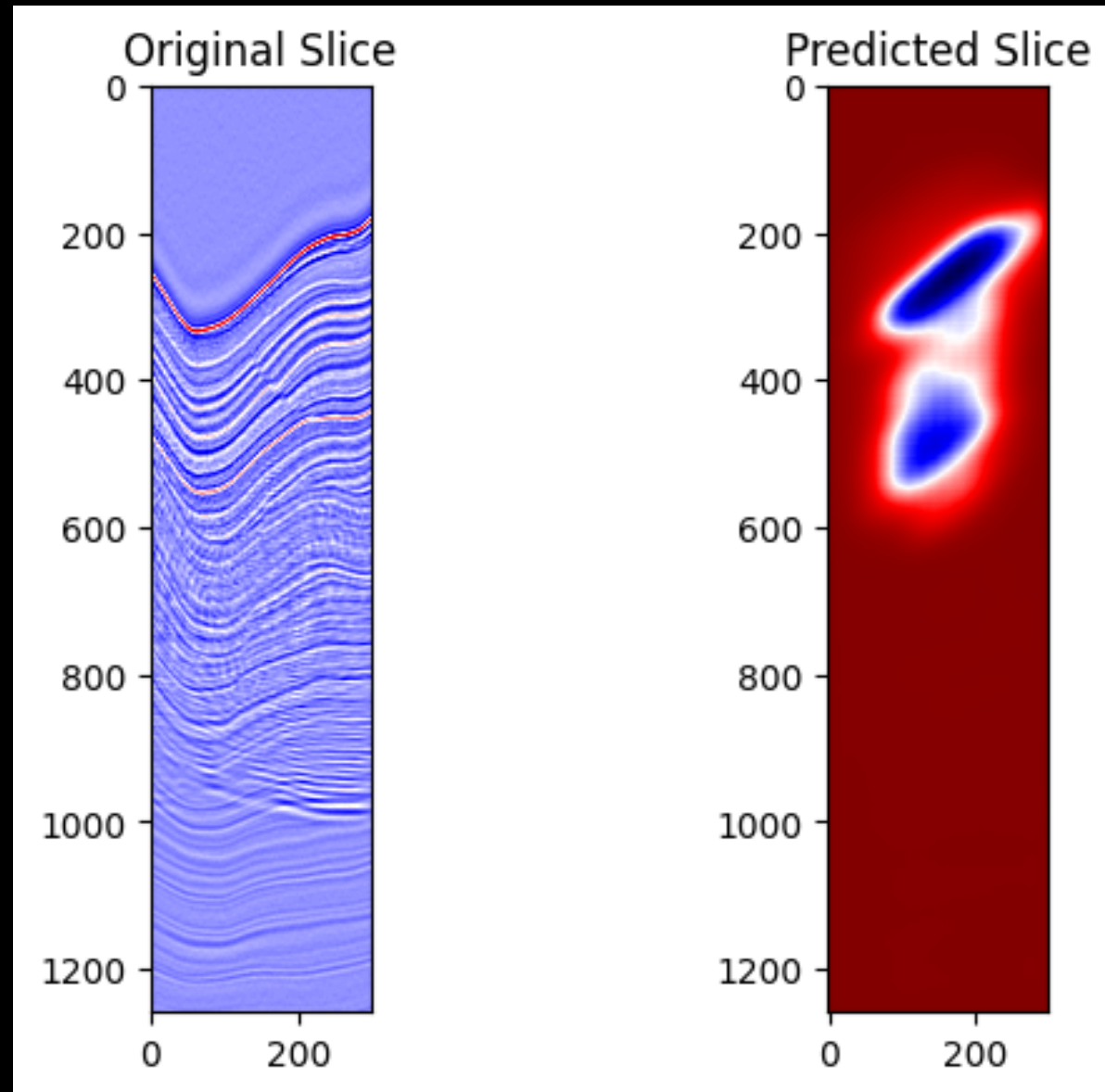
(Reduction-B Block)

Model Design

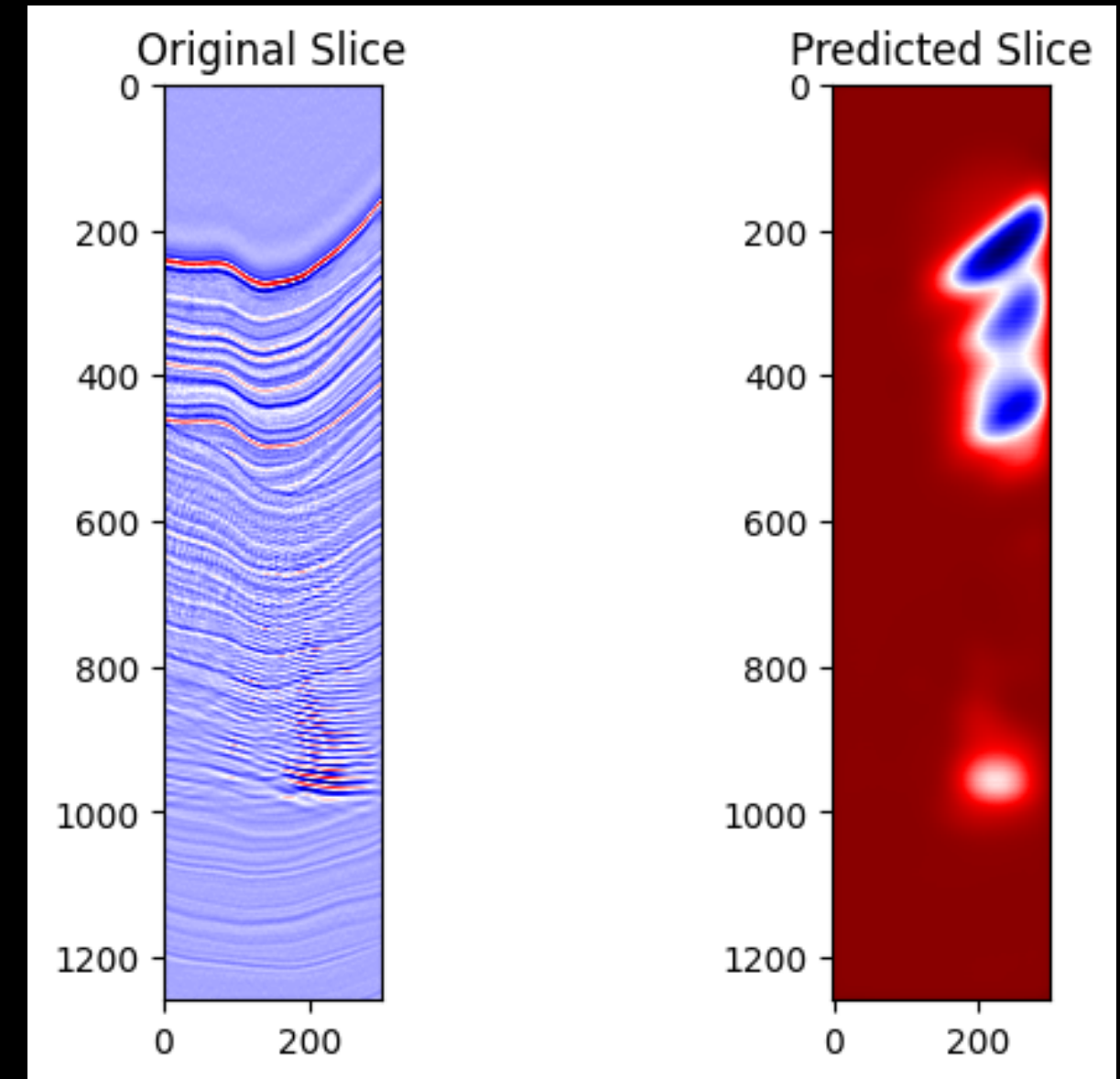


(Auxiliary Classifier Block)

Model Performance



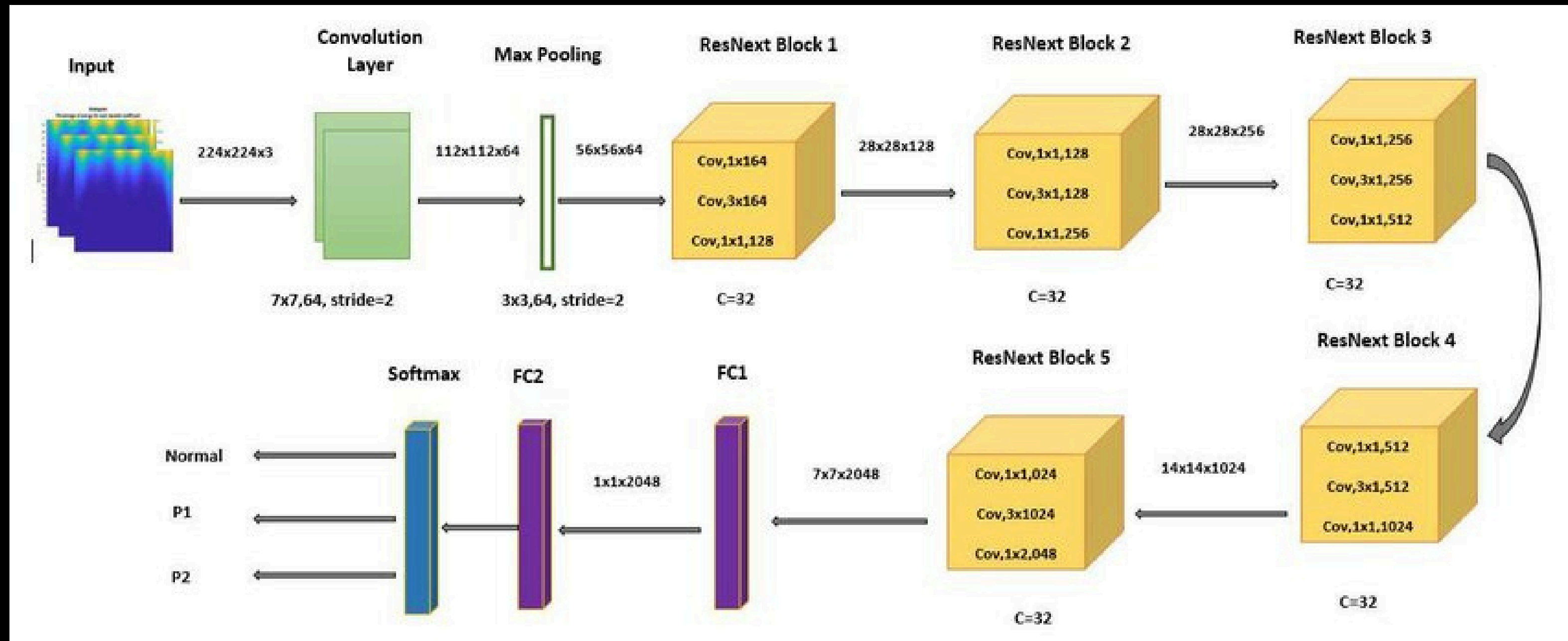
Epoch [1/200],
Train Loss: 2049449862938519560152678400.0000, Train SSIM: nan,
Val Loss: 2329127968478420220554248192.0000, Val SSIM: nan



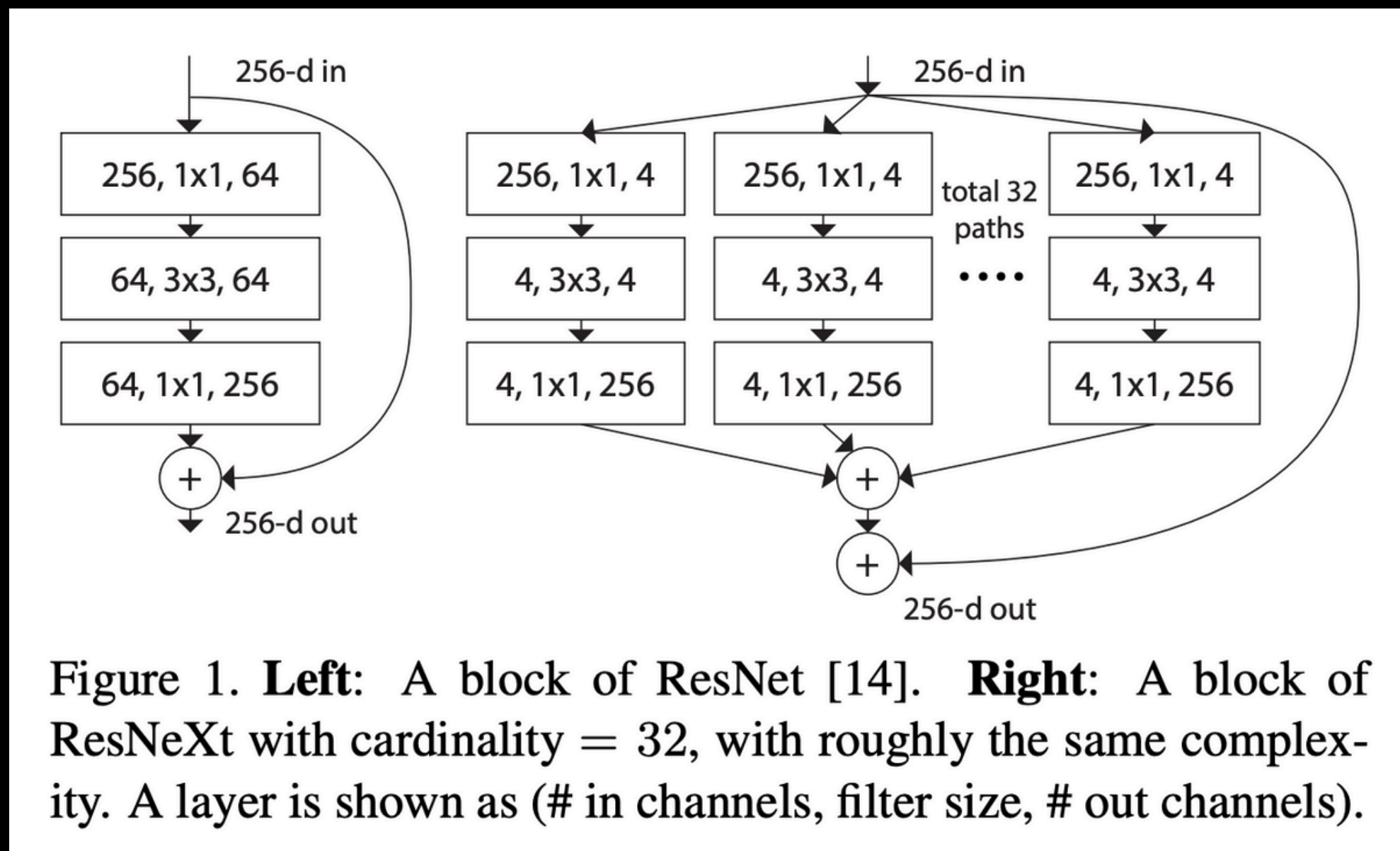
Epoch [2/200],
Train Loss: 1695706040153834187497930752.0000, Train SSIM: nan,
Val Loss: 1236440763291727545252184064.0000, Val SSIM: nan

ResNeXt

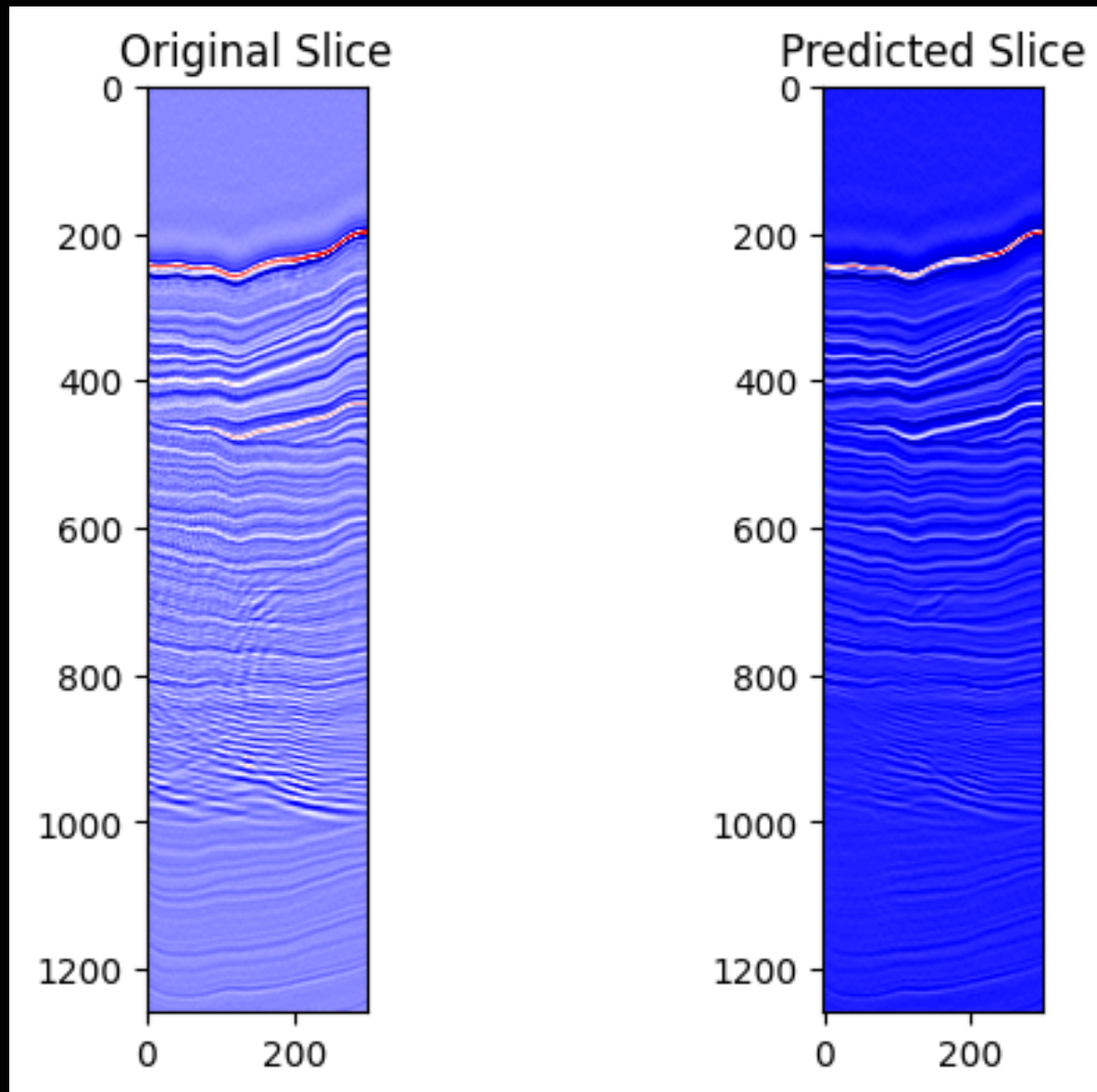
Model Design



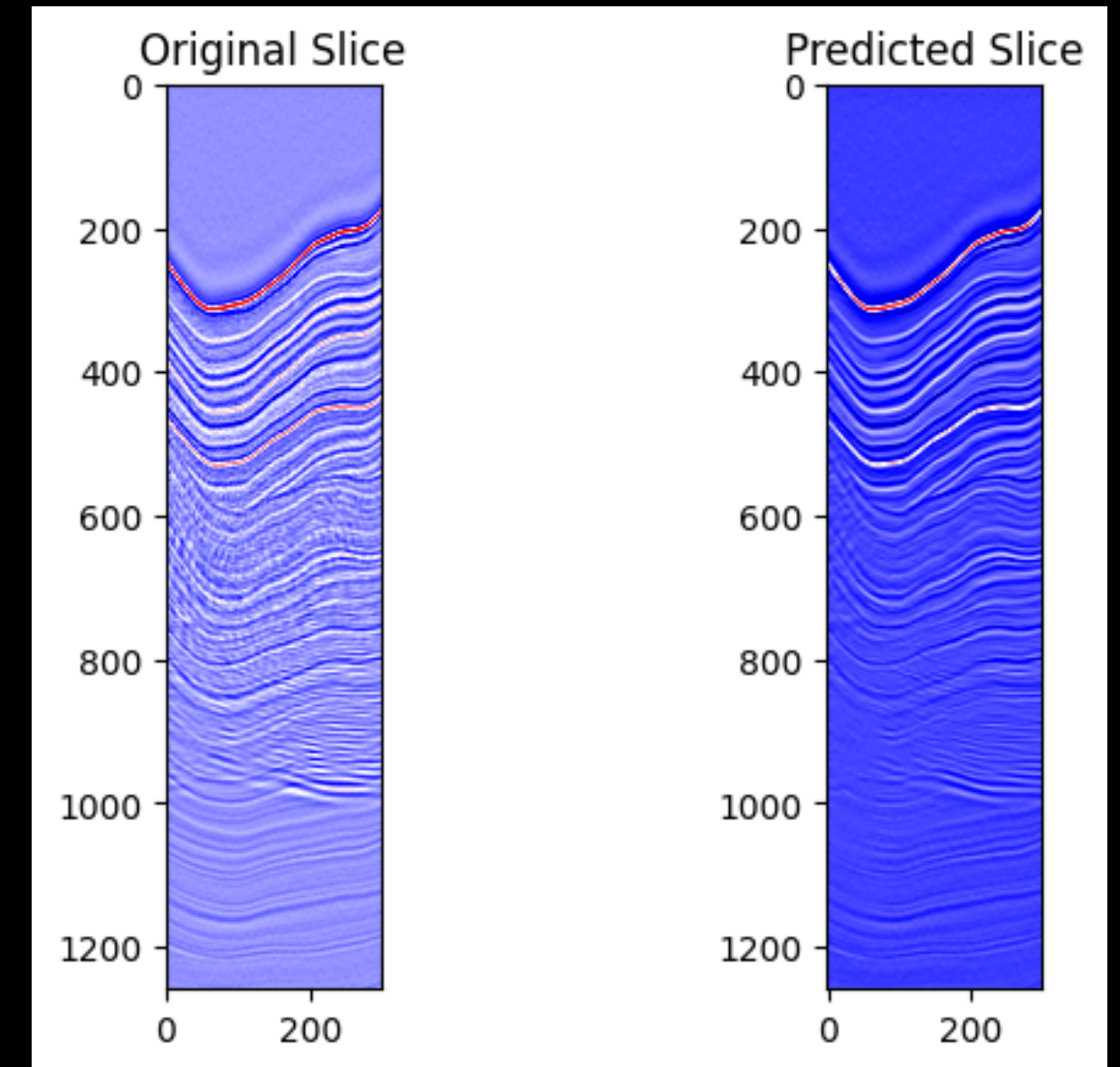
Model Design



Model Performance



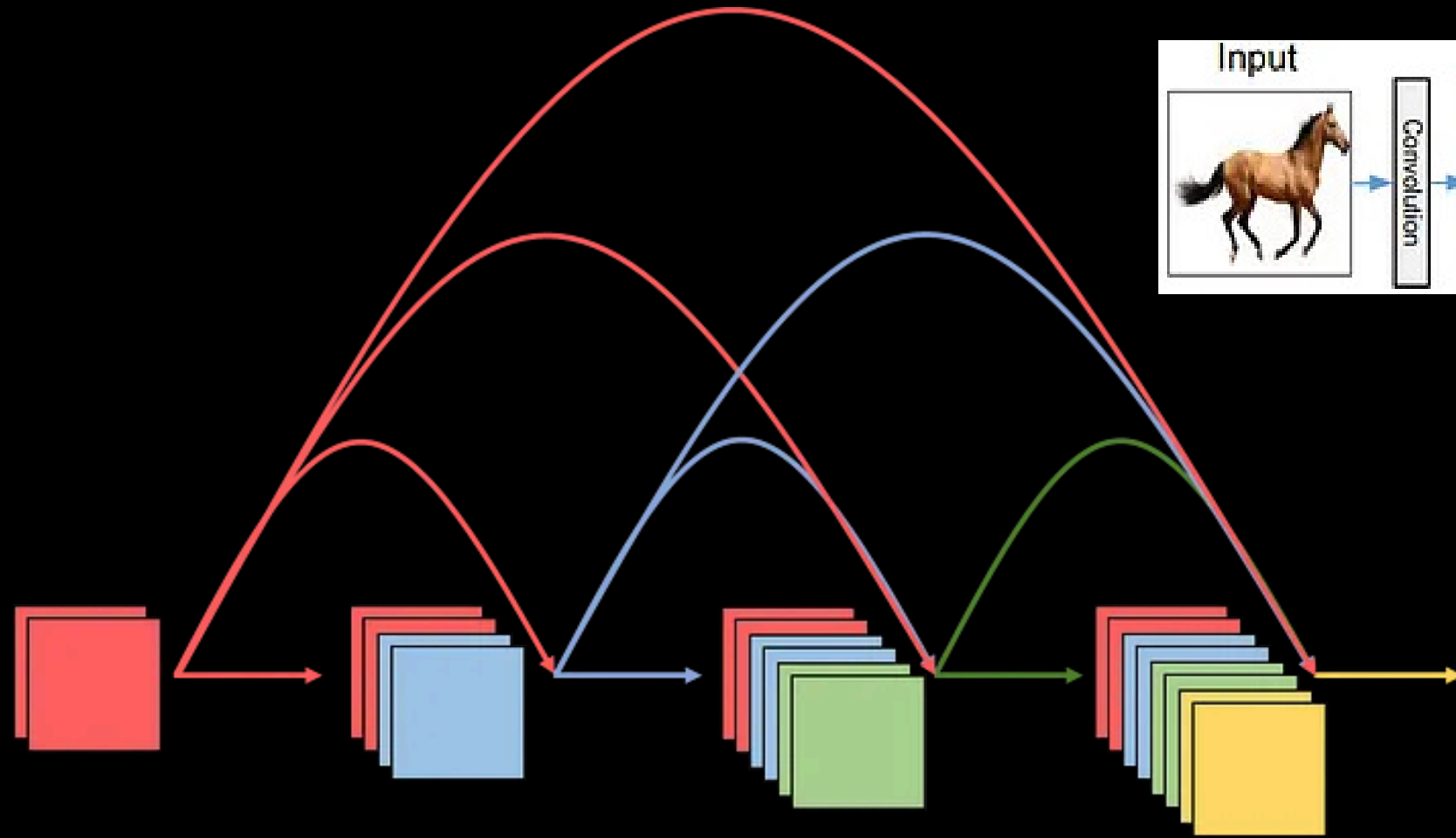
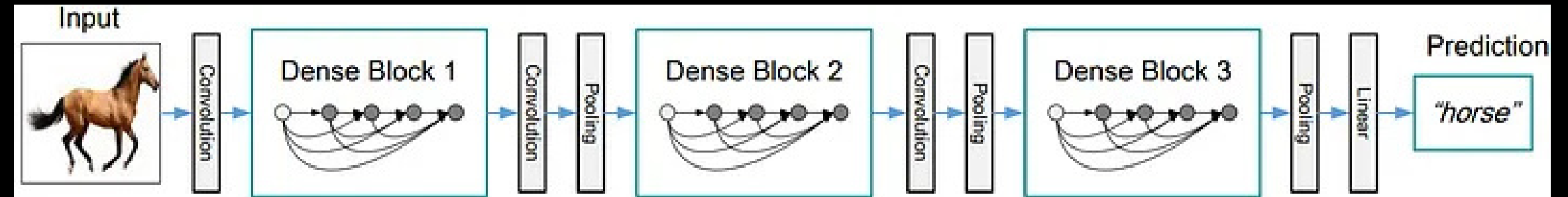
Epoch [120/200],
Train Loss: 1171.0722, Train SSIM: 0.9243,
Val Loss: 1500.4455, Val SSIM: 0.8553



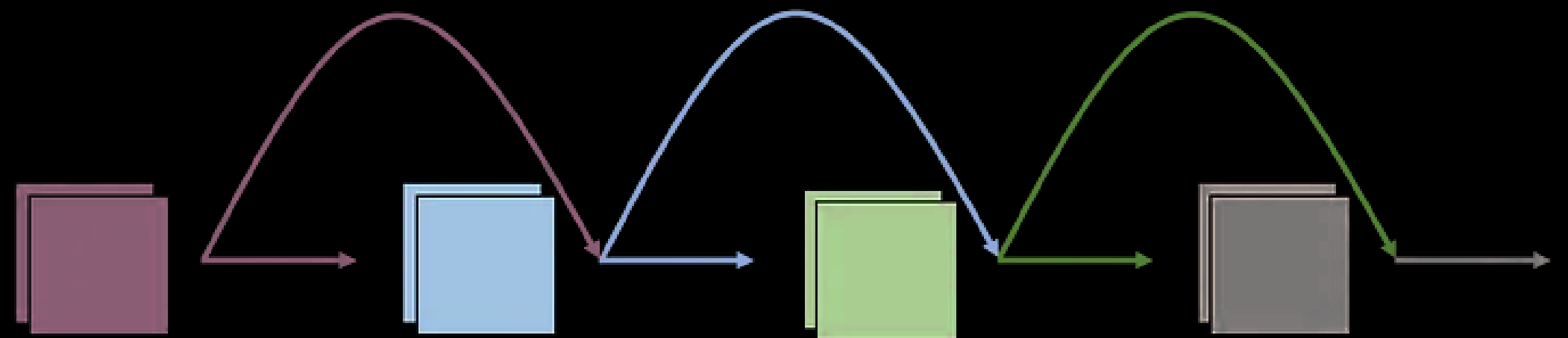
Epoch [121/200],
Train Loss: 1170.8711, Train SSIM: 0.9243,
Val Loss: 1486.7113, Val SSIM: 0.8616

DenseNet

Model Design

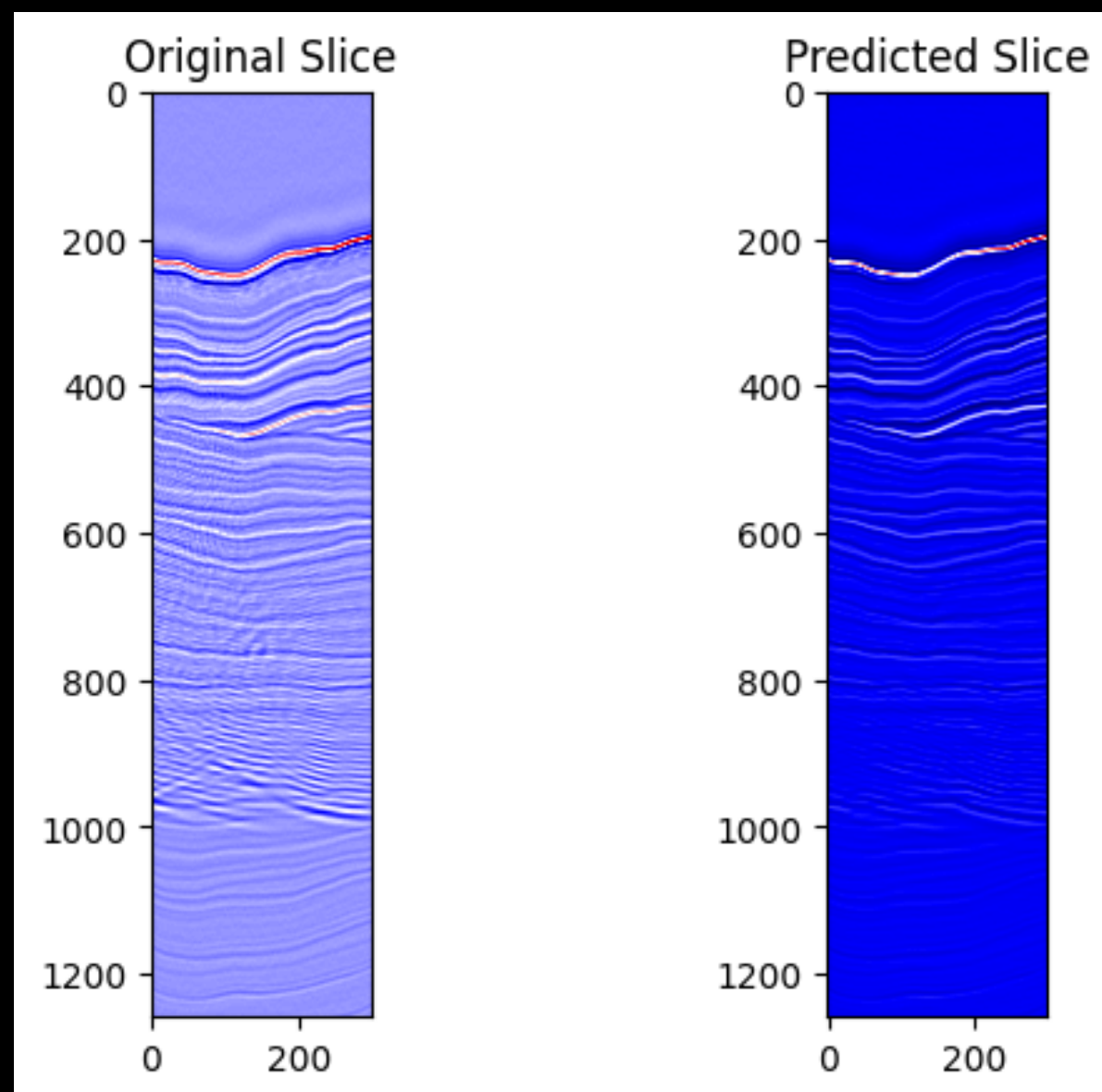


(Dense Net)

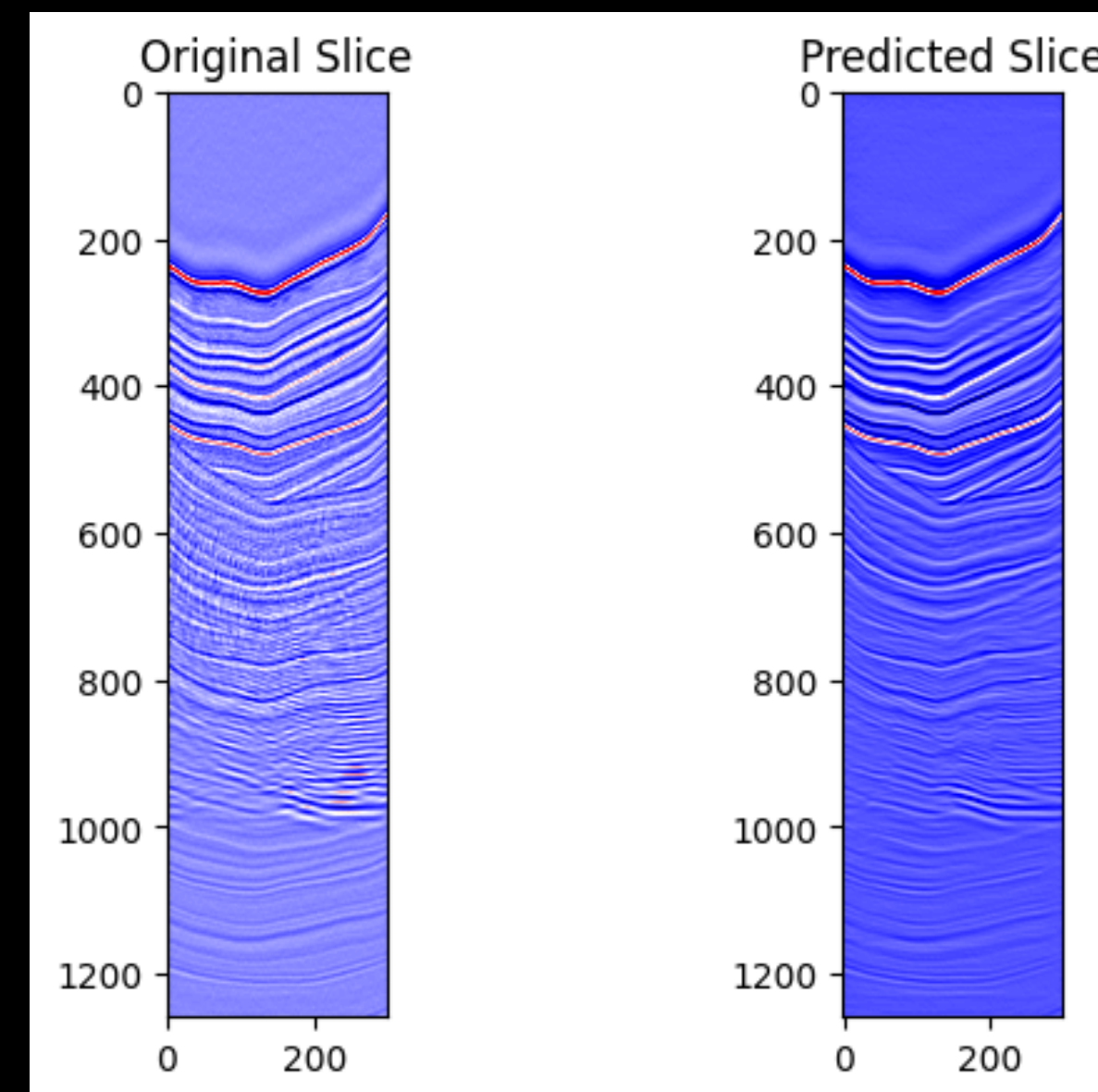


(ResNet)

Model Performance



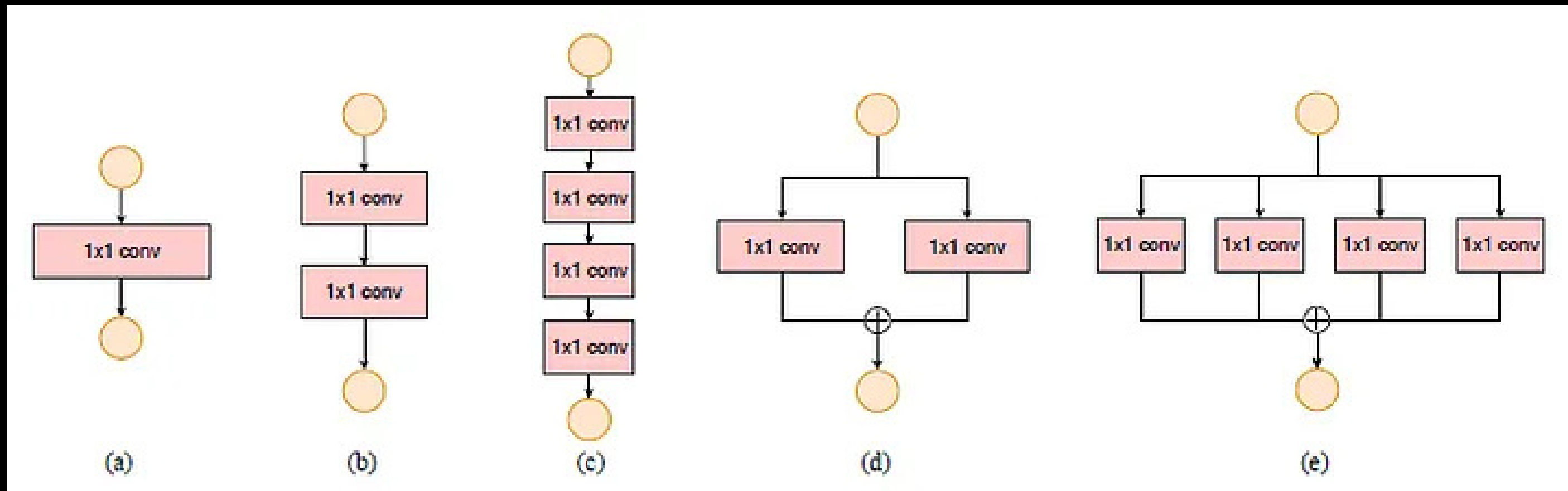
Epoch [20/200],
Train Loss: 73.2345, Train SSIM: 0.9787,
Val Loss: 1266.5717, Val SSIM: 0.8570



Epoch [21/200],
Train Loss: 73.0801, Train SSIM: 0.9786,
Val Loss: 1185.0675, Val SSIM: 0.8630

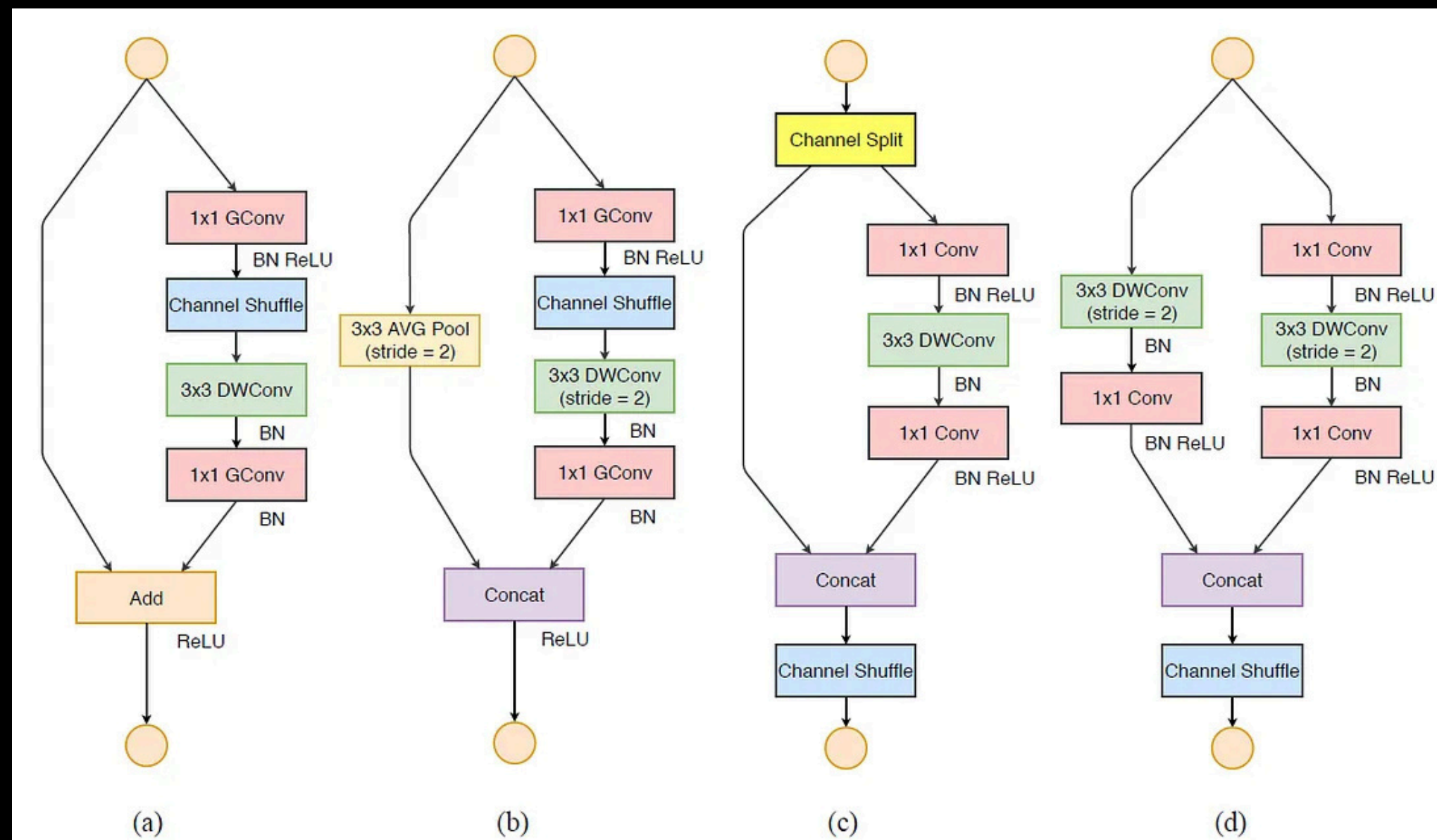
ShuffleNet V2

Model Design



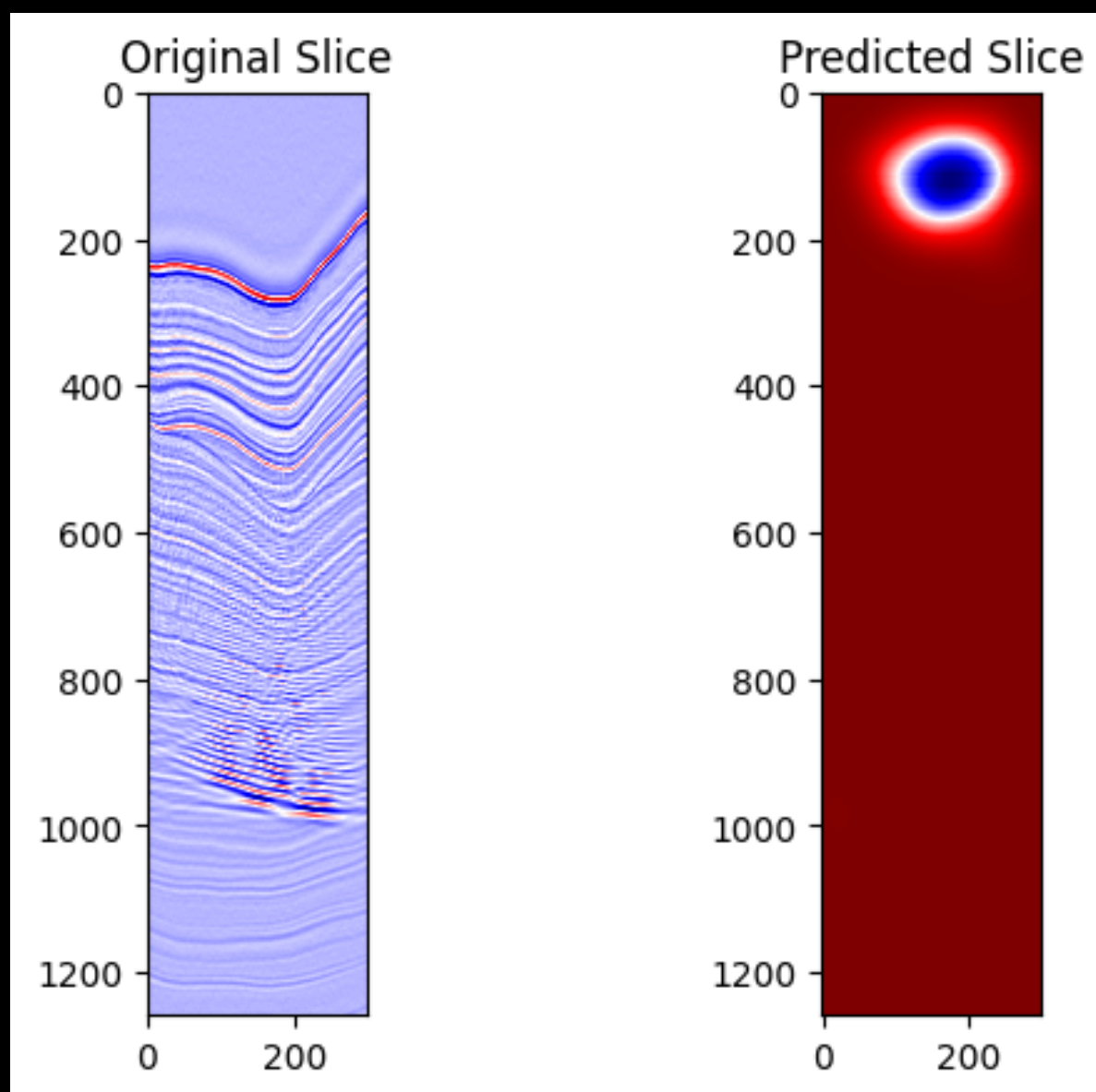
(a) 1-fragment. (b) 2-fragment-series. (c) 4-fragment-series. (d) 2-fragment-parallel. (e) 4-fragment-parallel.

Model Design

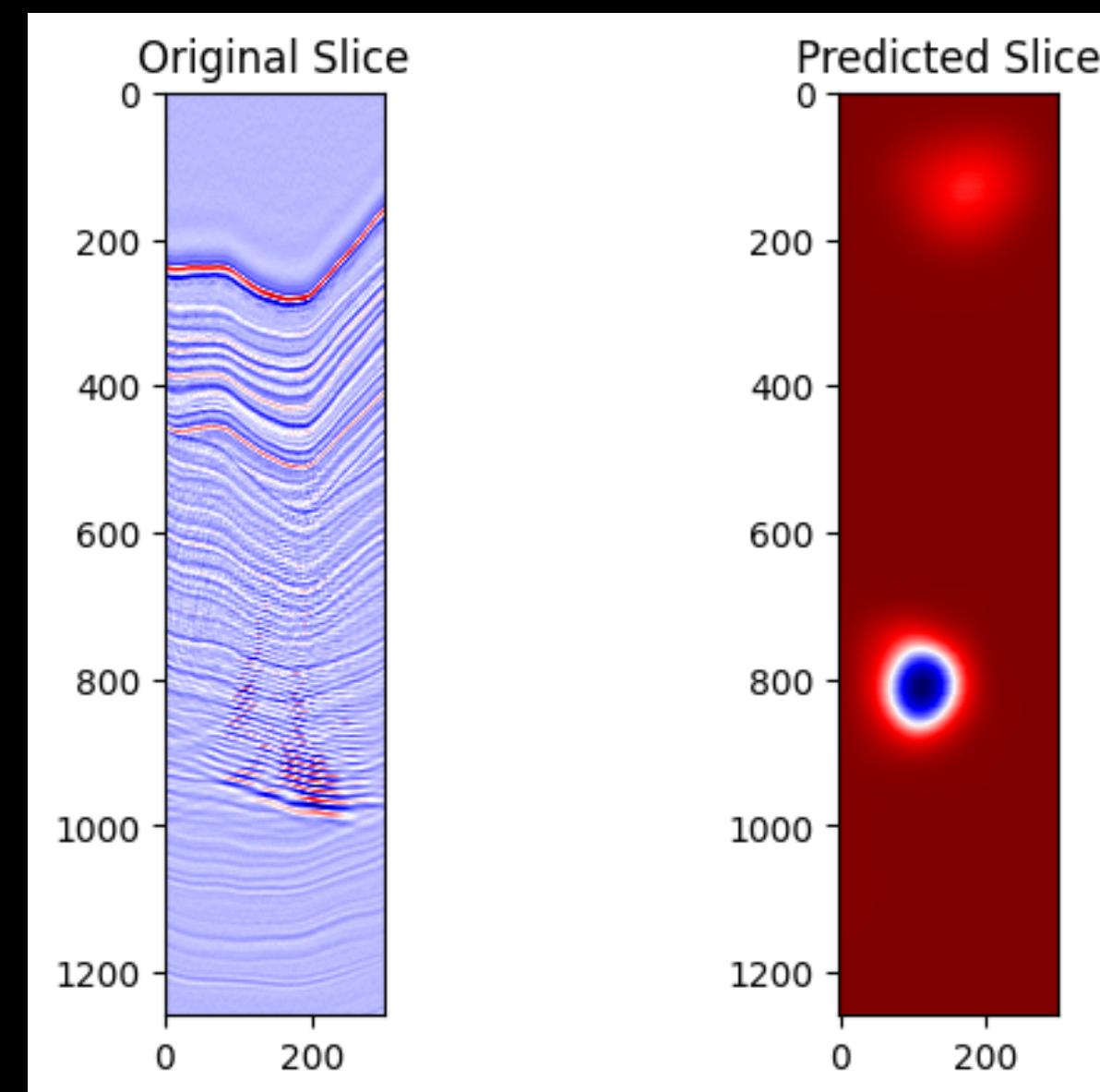


Building blocks of ShuffleNet V1 and this work. (a): the basic ShuffleNet V1 unit; (b) the ShuffleNet V1 unit for spatial down sampling (2x); (c) the ShuffleNet V2 basic unit; (d) the ShuffleNet V2 unit for spatial down sampling (2x). DWConv: depthwise convolution. GConv: group convolution.

Model Performance



Epoch [8/200],
Train Loss: inf, Train SSIM: nan,
Val Loss: inf, Val SSIM: nan



Epoch [9/200],
Train Loss: inf, Train SSIM: nan,
Val Loss: inf, Val SSIM: nan

SqueezeNet

Model Design

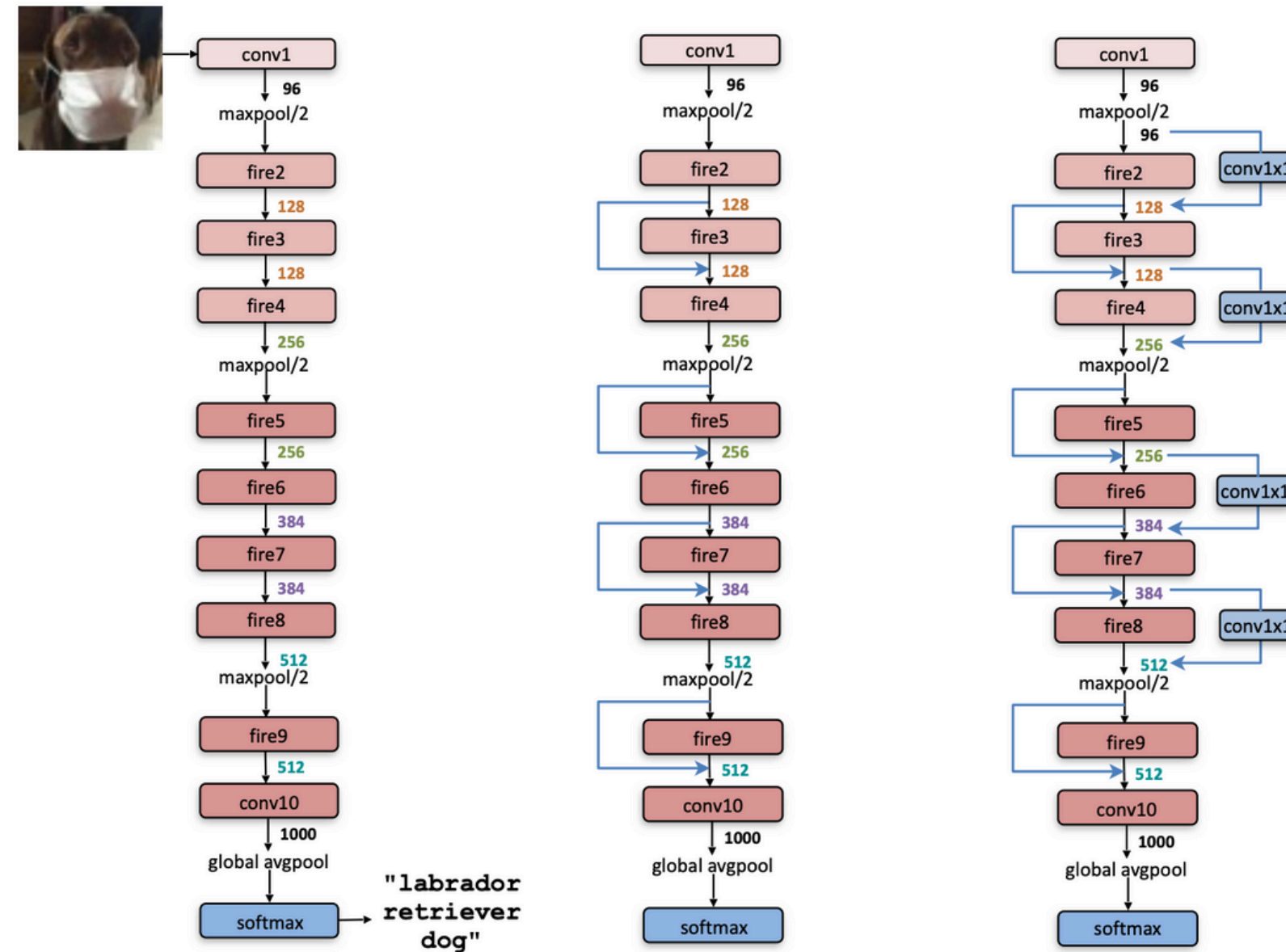
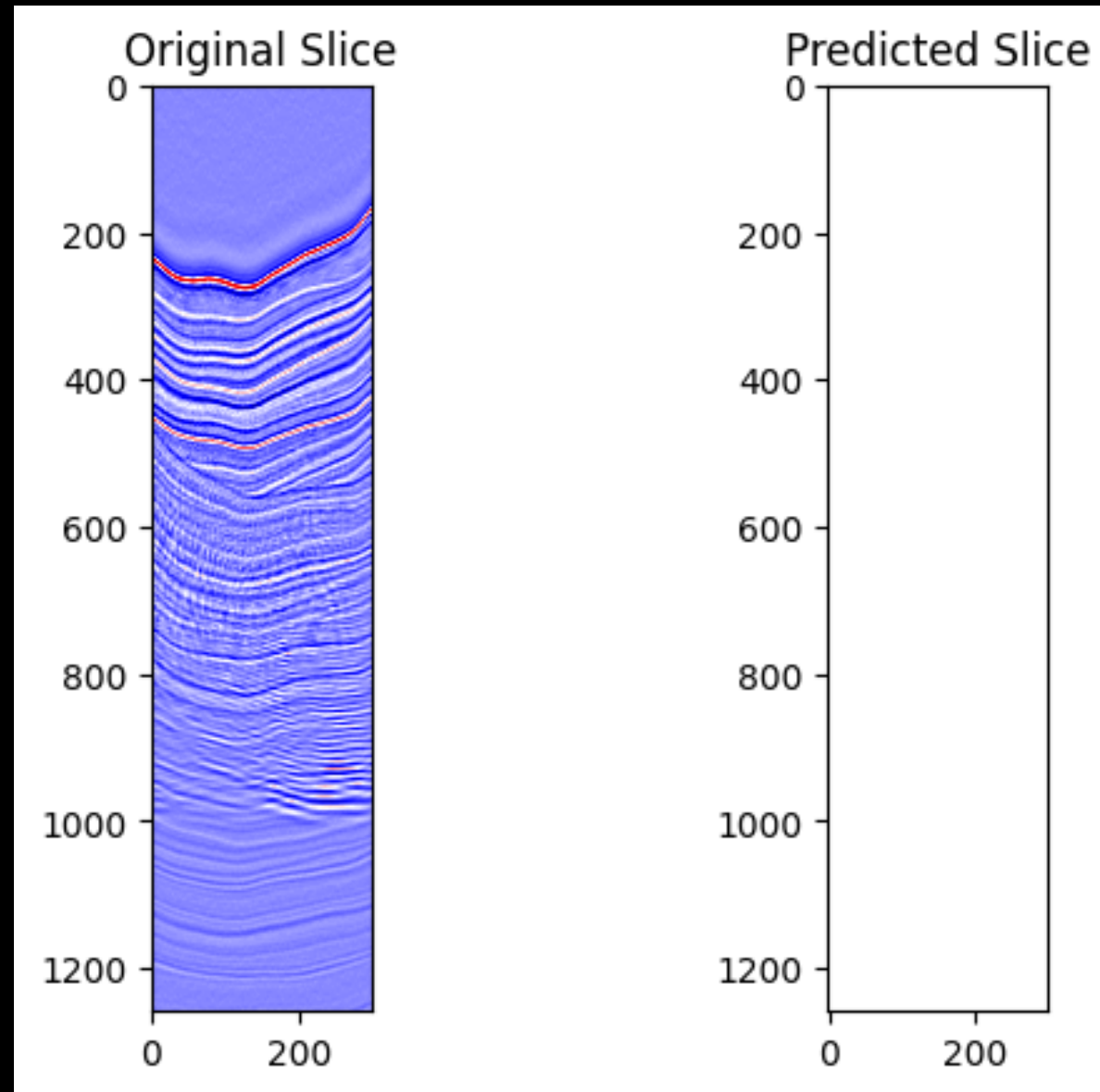
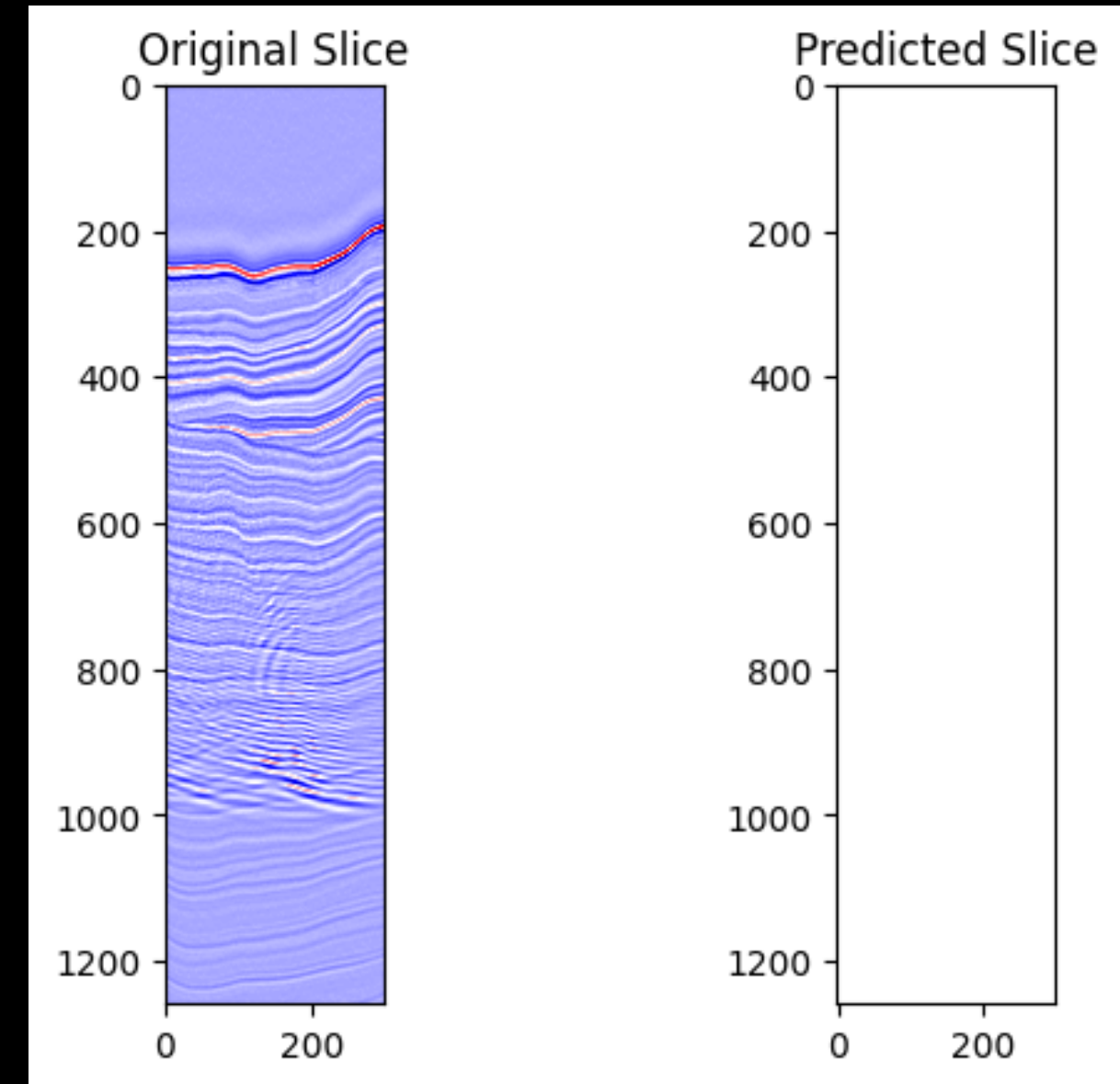


Figure 2: Macroarchitectural view of our SqueezeNet architecture. Left: SqueezeNet (Section 3.3); Middle: SqueezeNet with simple bypass (Section 6); Right: SqueezeNet with complex bypass (Section 6).

Model Performance



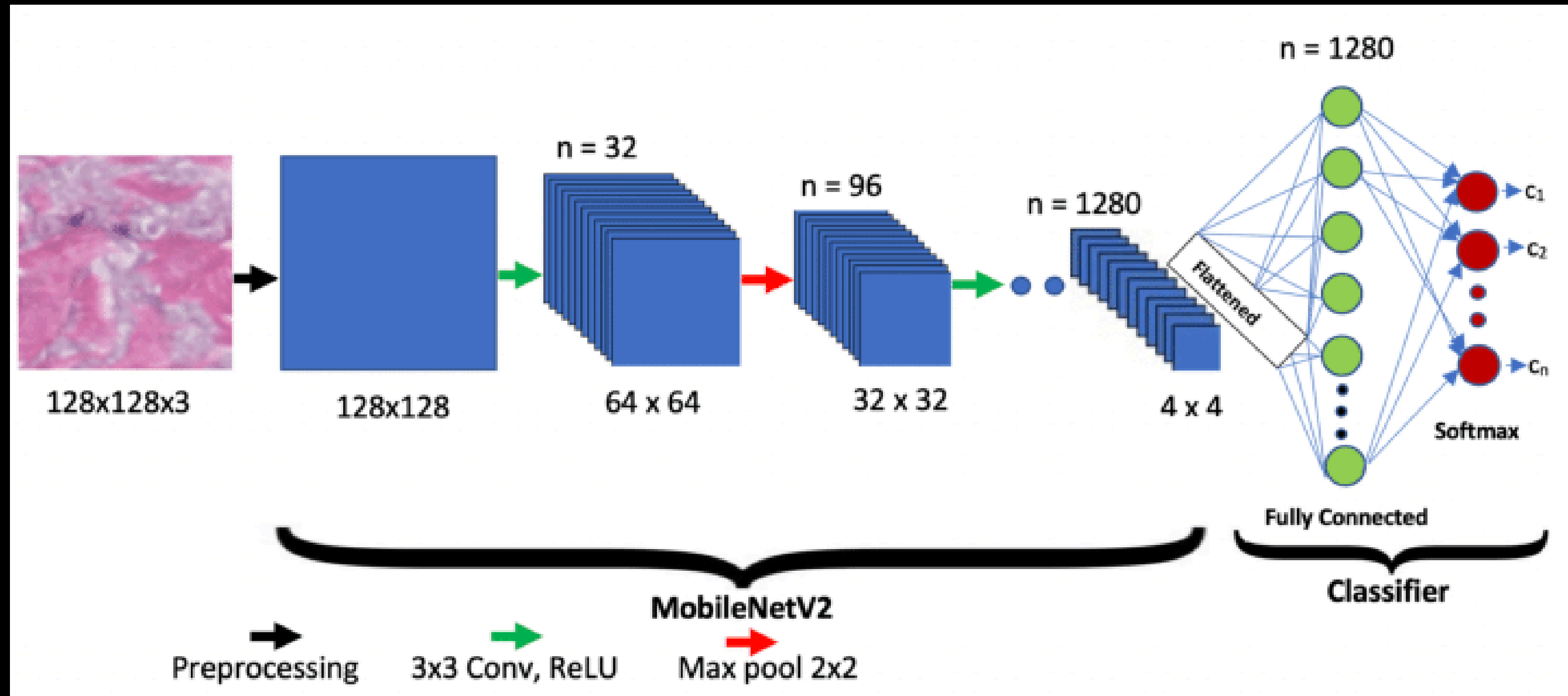
Epoch [1/200],
Train Loss: nan, Train SSIM: nan,
Val Loss: nan, Val SSIM: nan



Epoch [2/200],
Train Loss: nan, Train SSIM: nan,
Val Loss: nan, Val SSIM: nan

MobileNet V2

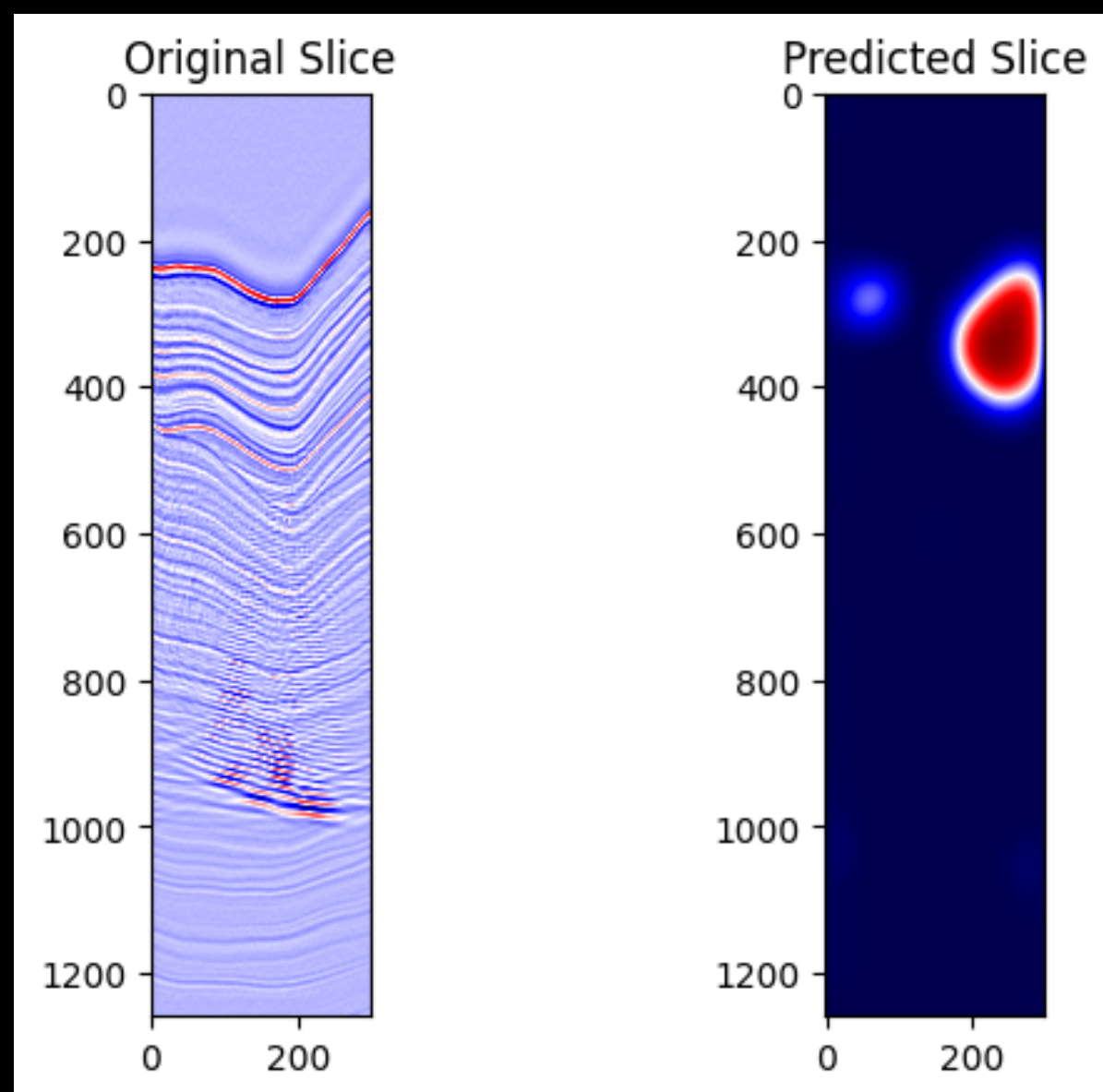
Model Design



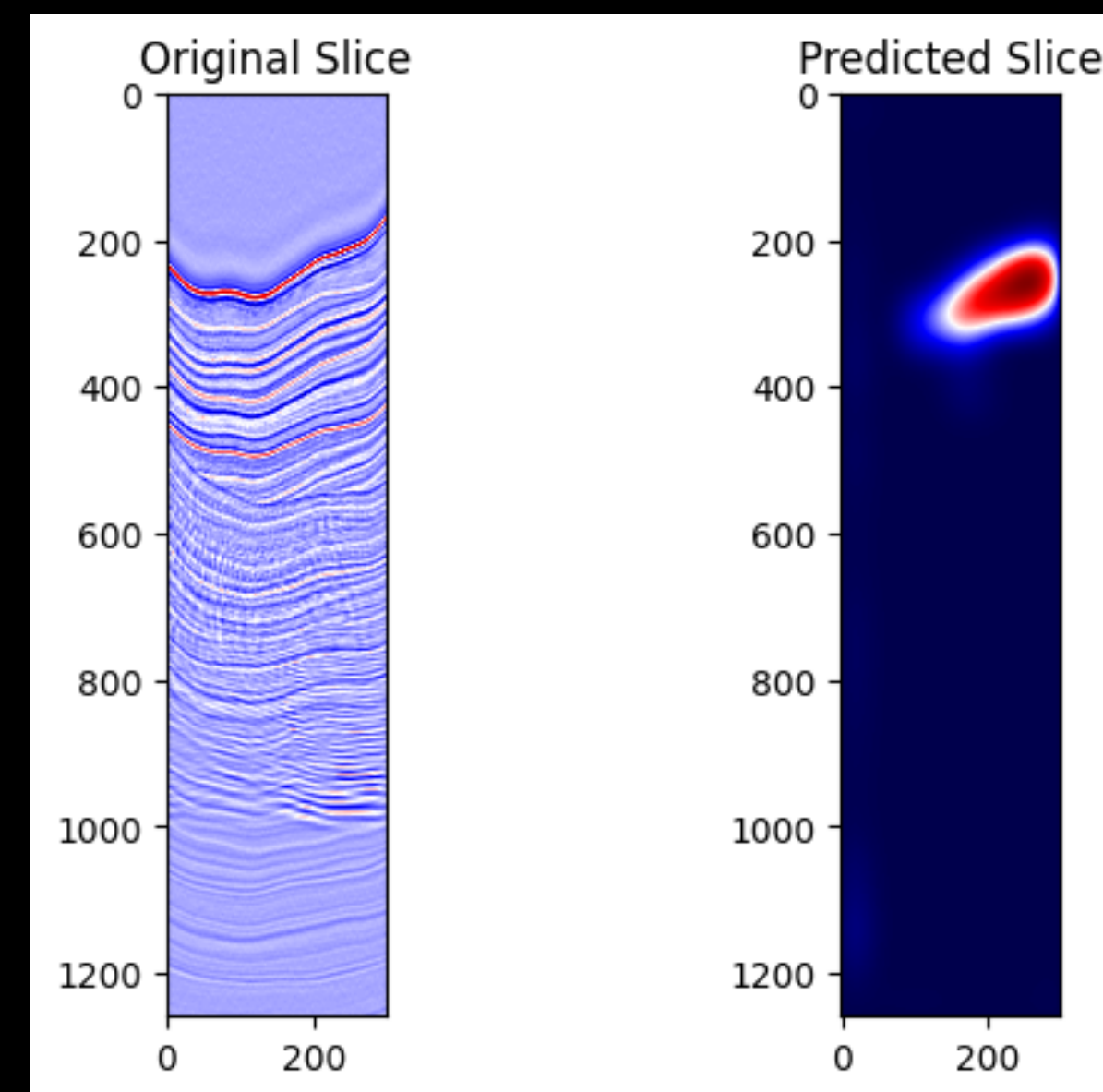
Model Design

Input	Operator	t	c	n	s
$224^2 \times 3$	conv2d	-	32	1	2
$112^2 \times 32$	bottleneck	1	16	1	1
$112^2 \times 16$	bottleneck	6	24	2	2
$56^2 \times 24$	bottleneck	6	32	3	2
$28^2 \times 32$	bottleneck	6	64	4	2
$14^2 \times 64$	bottleneck	6	96	3	1
$14^2 \times 96$	bottleneck	6	160	3	2
$7^2 \times 160$	bottleneck	6	320	1	1
$7^2 \times 320$	conv2d 1x1	-	1280	1	1
$7^2 \times 1280$	avgpool 7x7	-	-	1	-
$1 \times 1 \times 1280$	conv2d 1x1	-	k	-	-

Model Performance



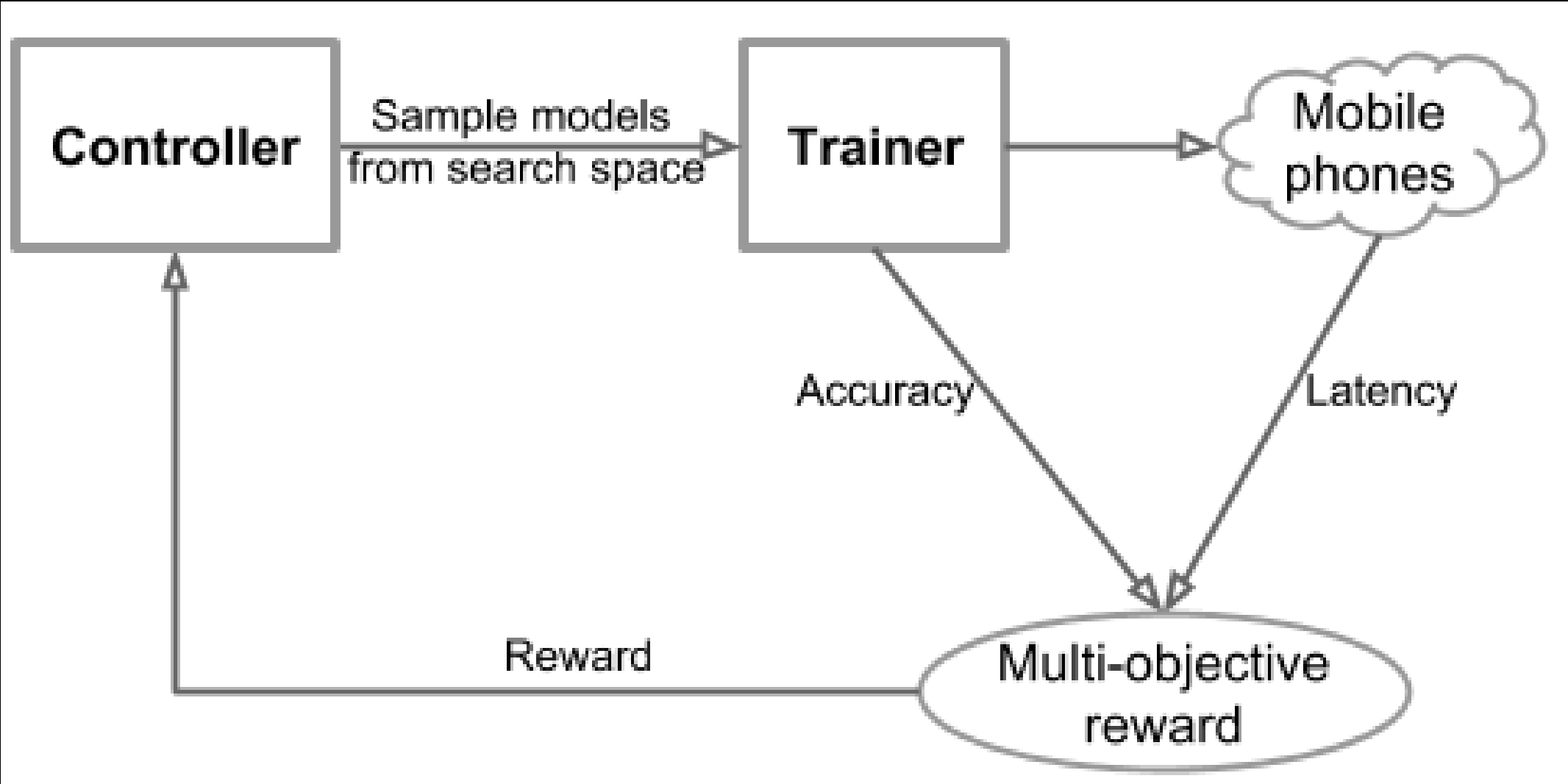
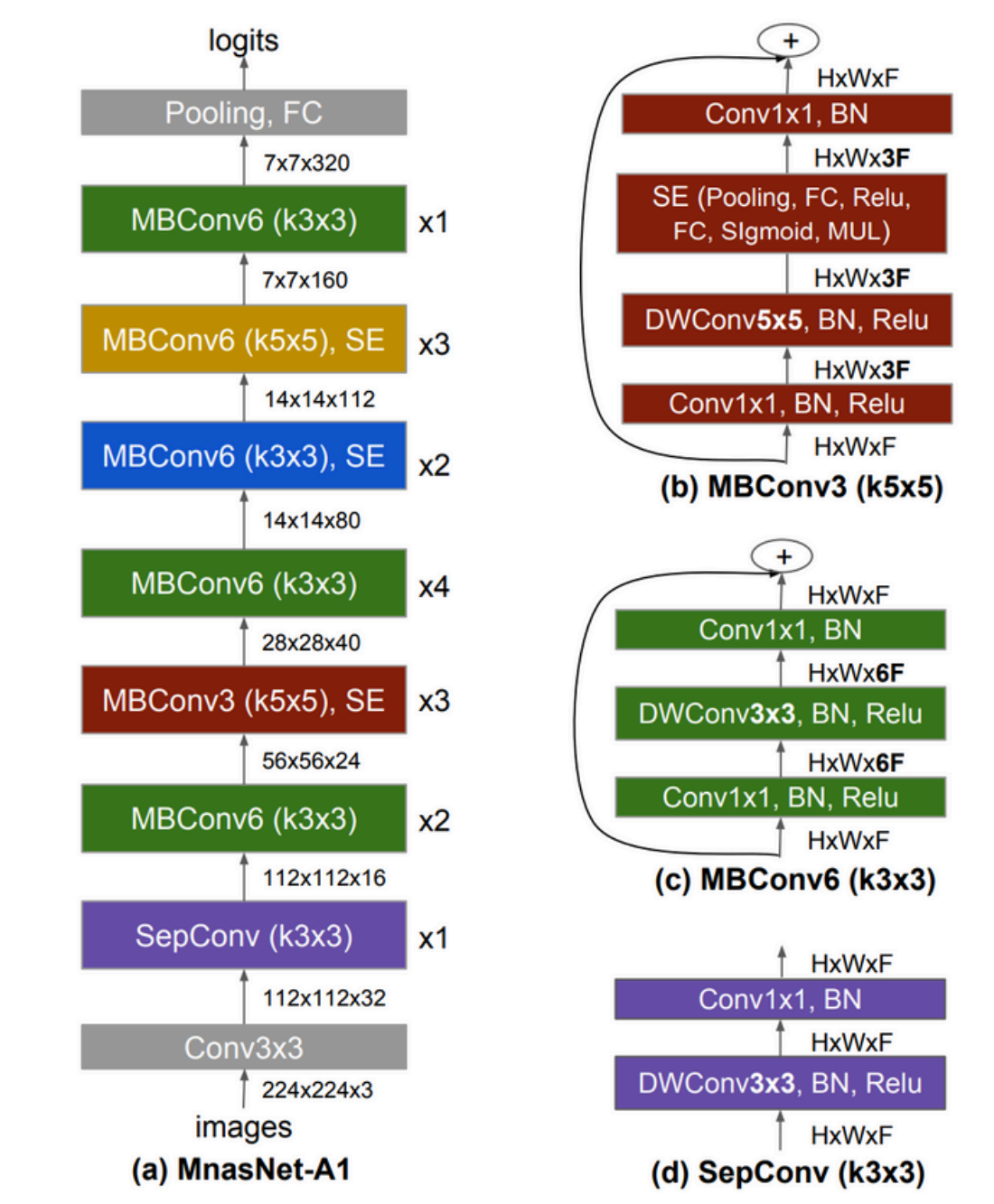
Epoch [8/200],
Train Loss: inf, Train SSIM: nan,
Val Loss: inf, Val SSIM: nan



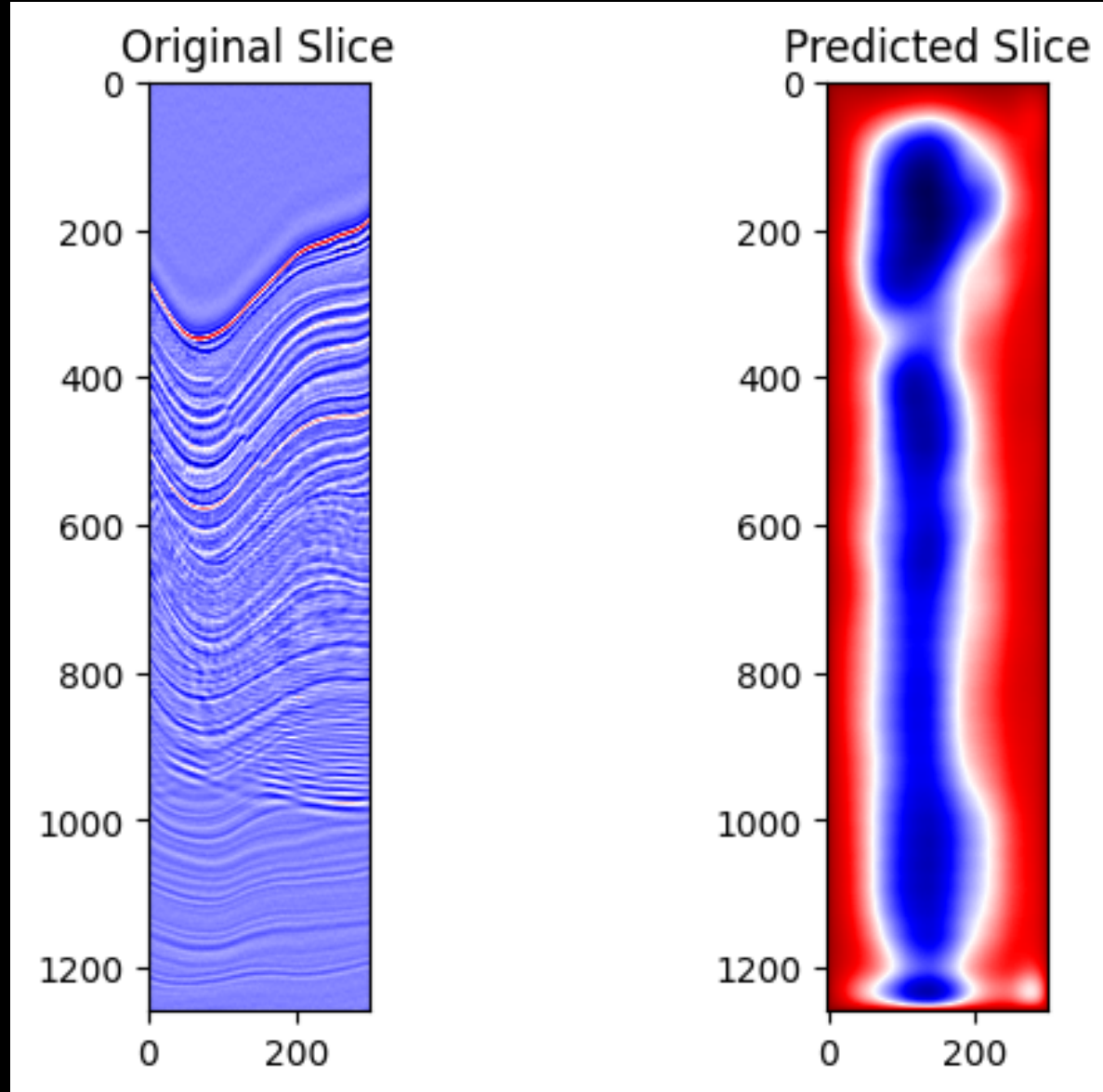
Epoch [9/200],
Train Loss: inf, Train SSIM: nan,
Val Loss: inf, Val SSIM: nan

MNASNet

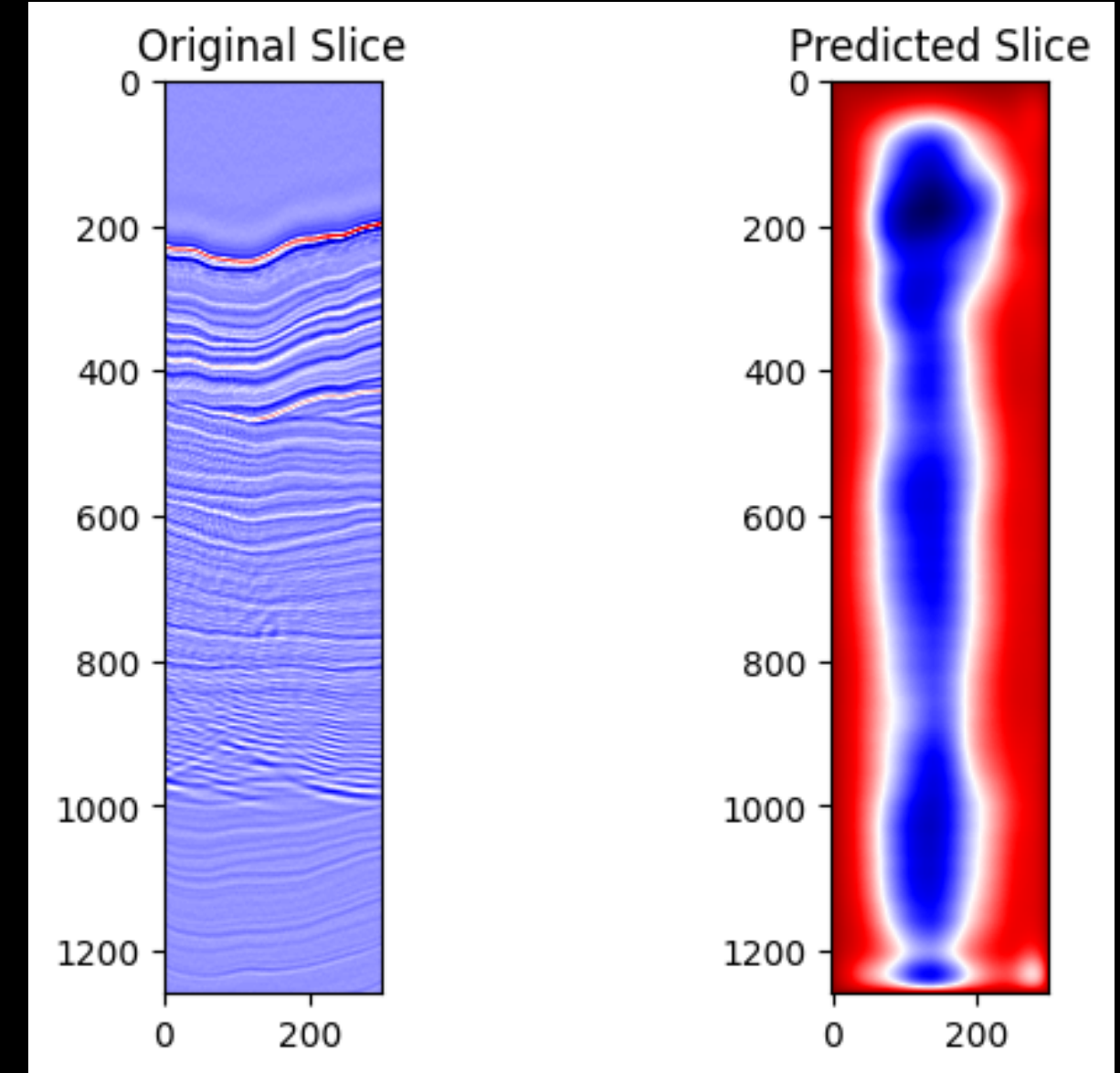
Model Design



Model Performance



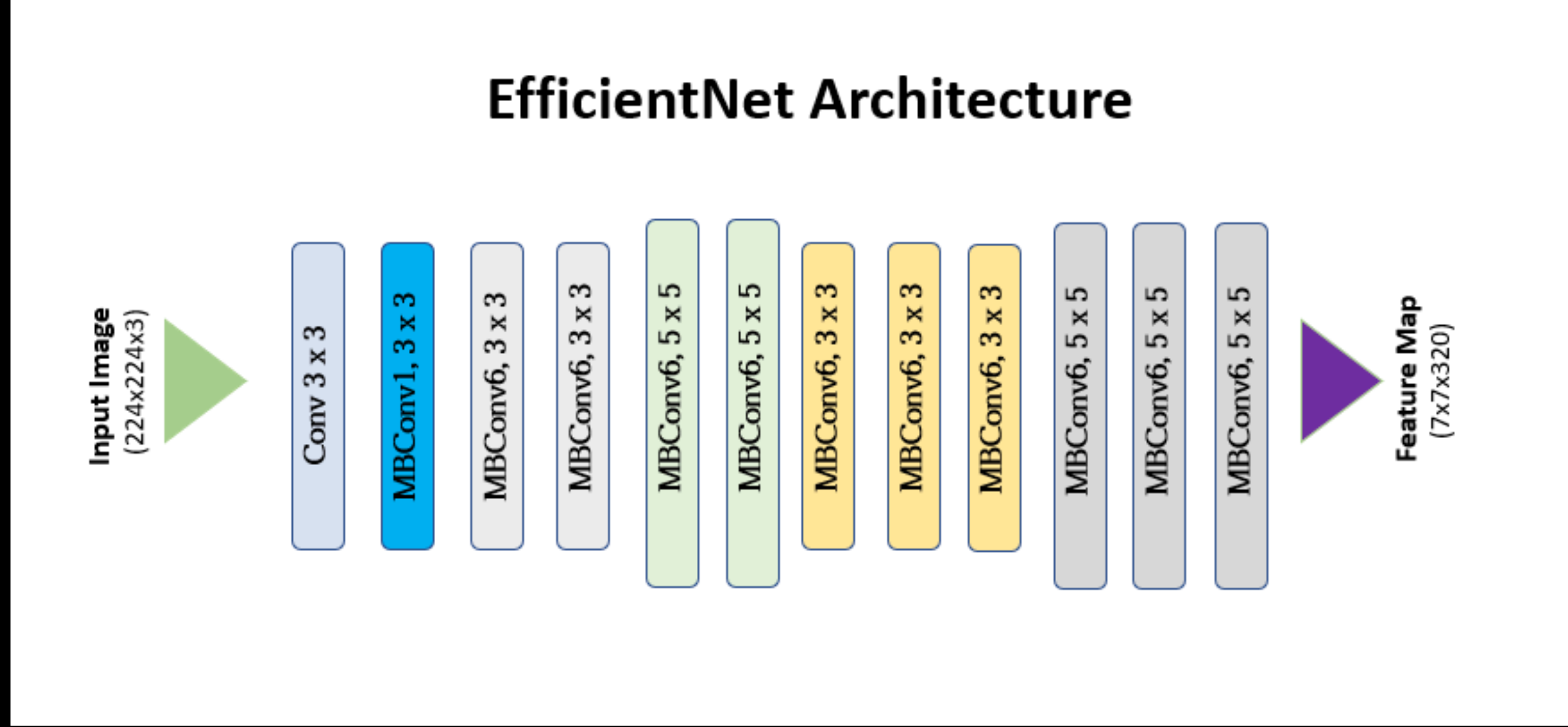
Epoch [10/200],
Train Loss: inf, Train SSIM: nan,
Val Loss: inf, Val SSIM: nan



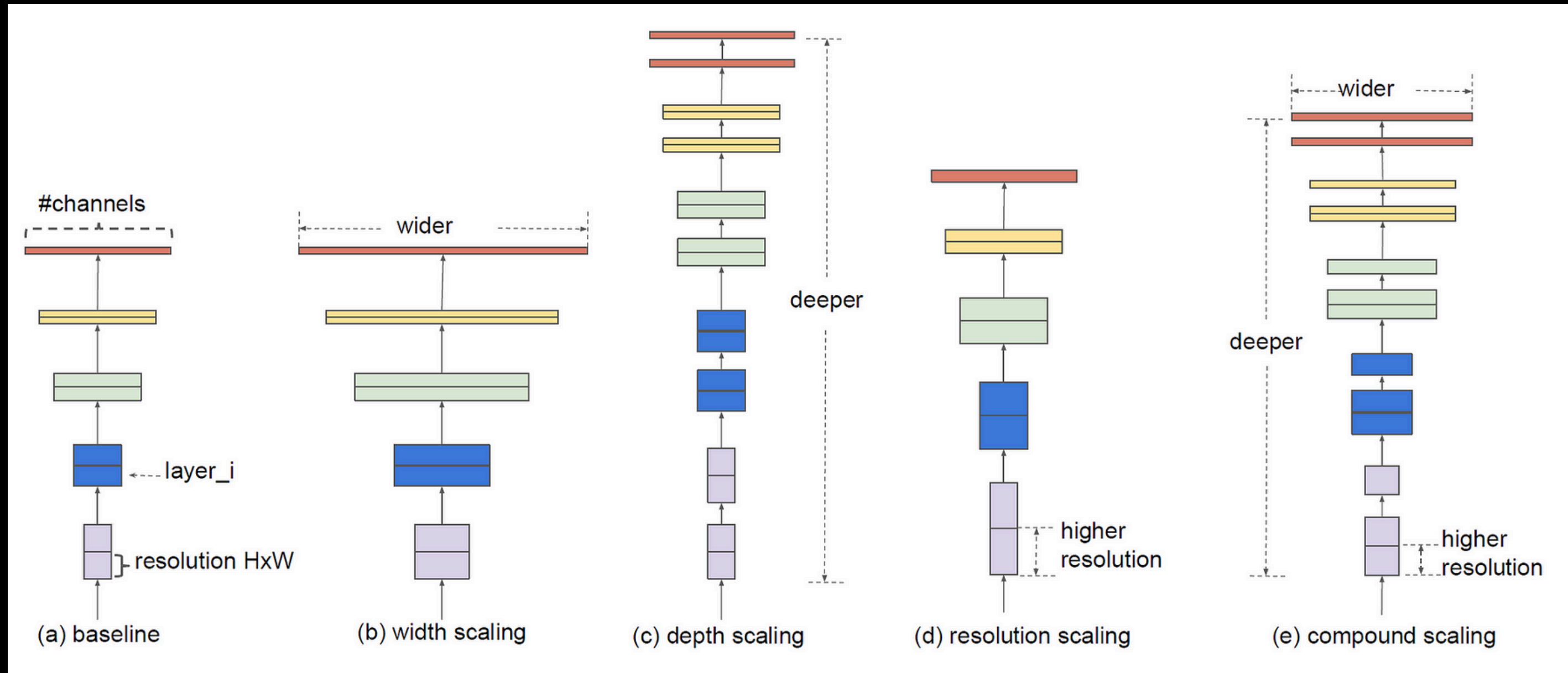
Epoch [9/200],
Train Loss: inf, Train SSIM: nan,
Val Loss: inf, Val SSIM: nan

EfficientNetV2

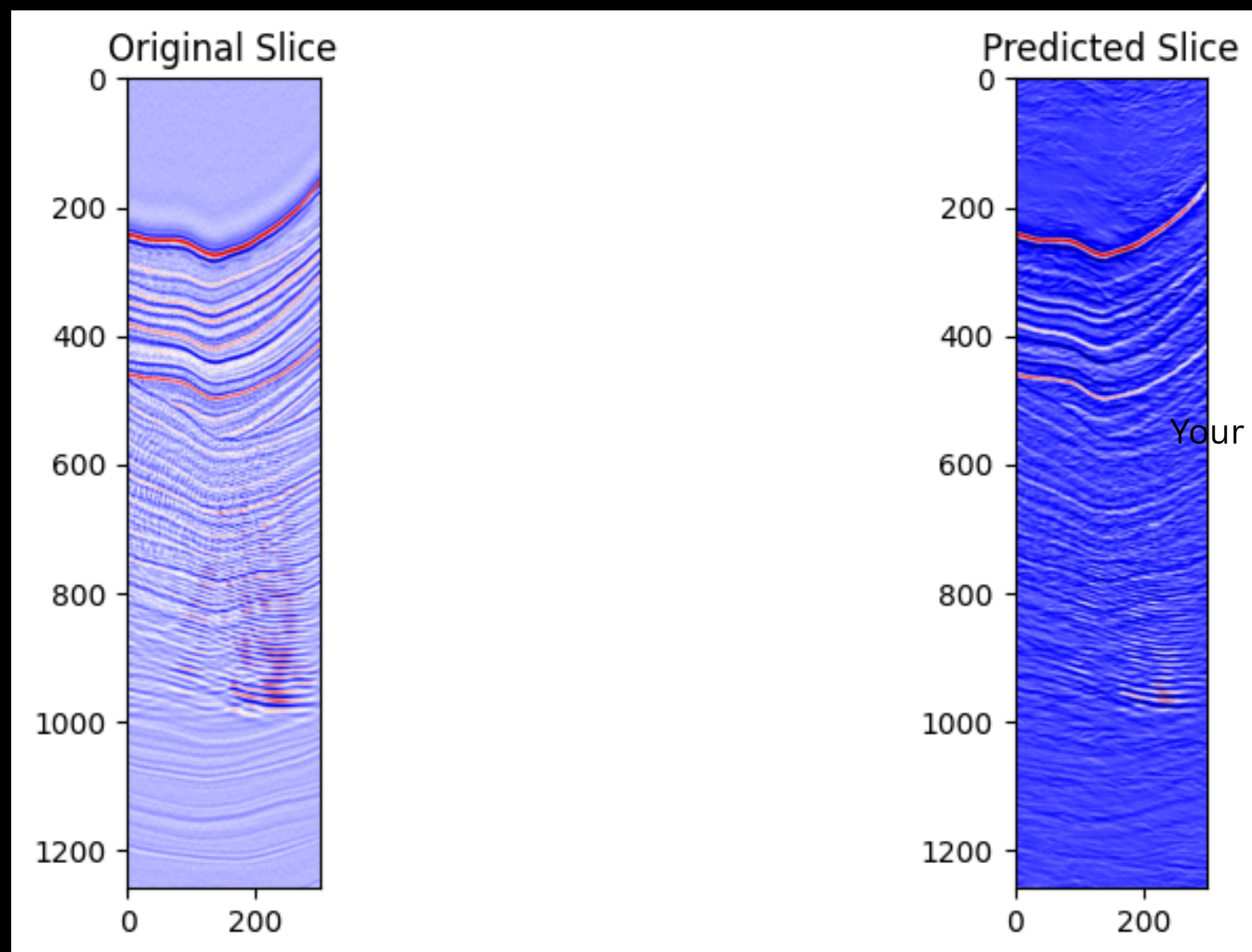
Model Design



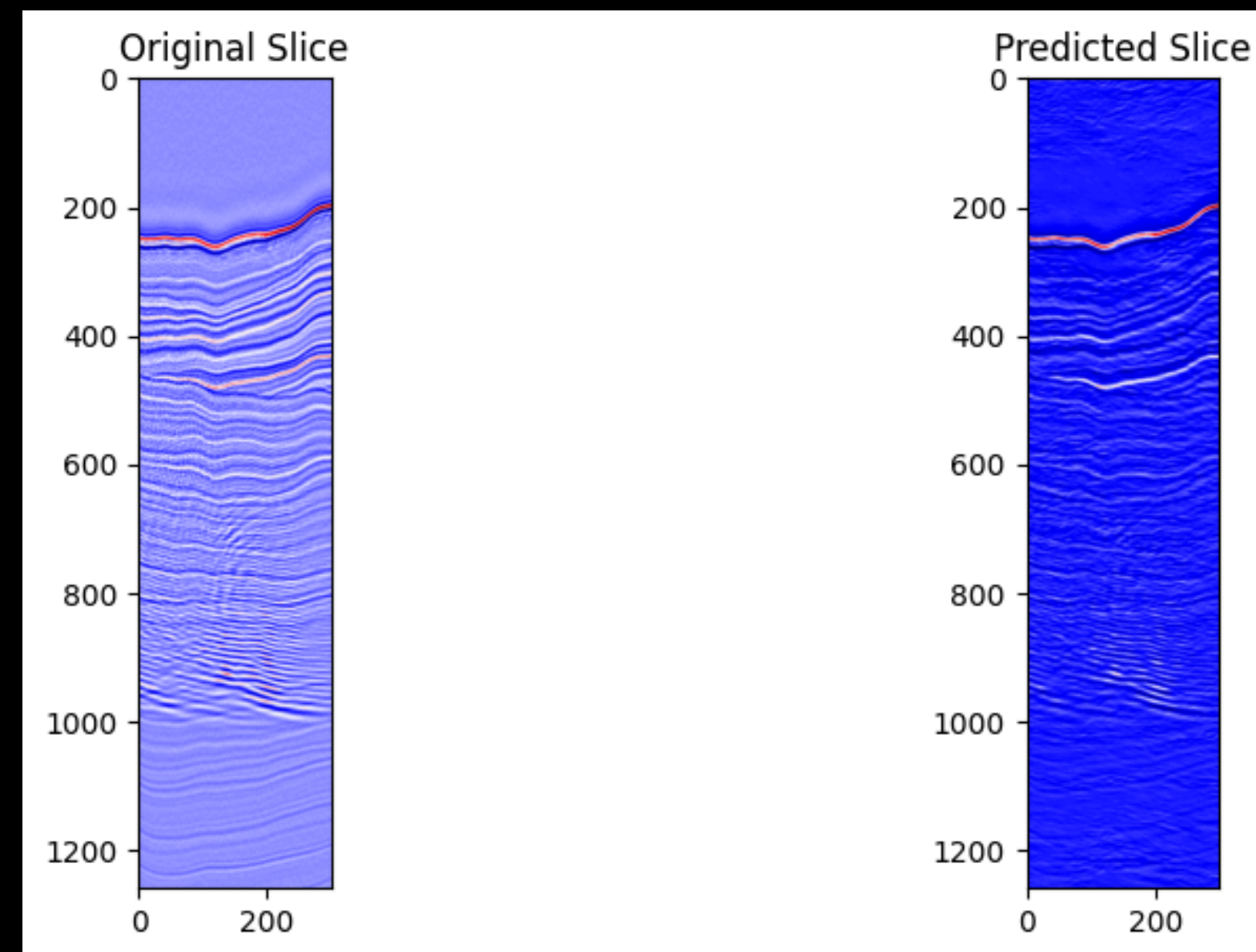
Model Design



Model Performance



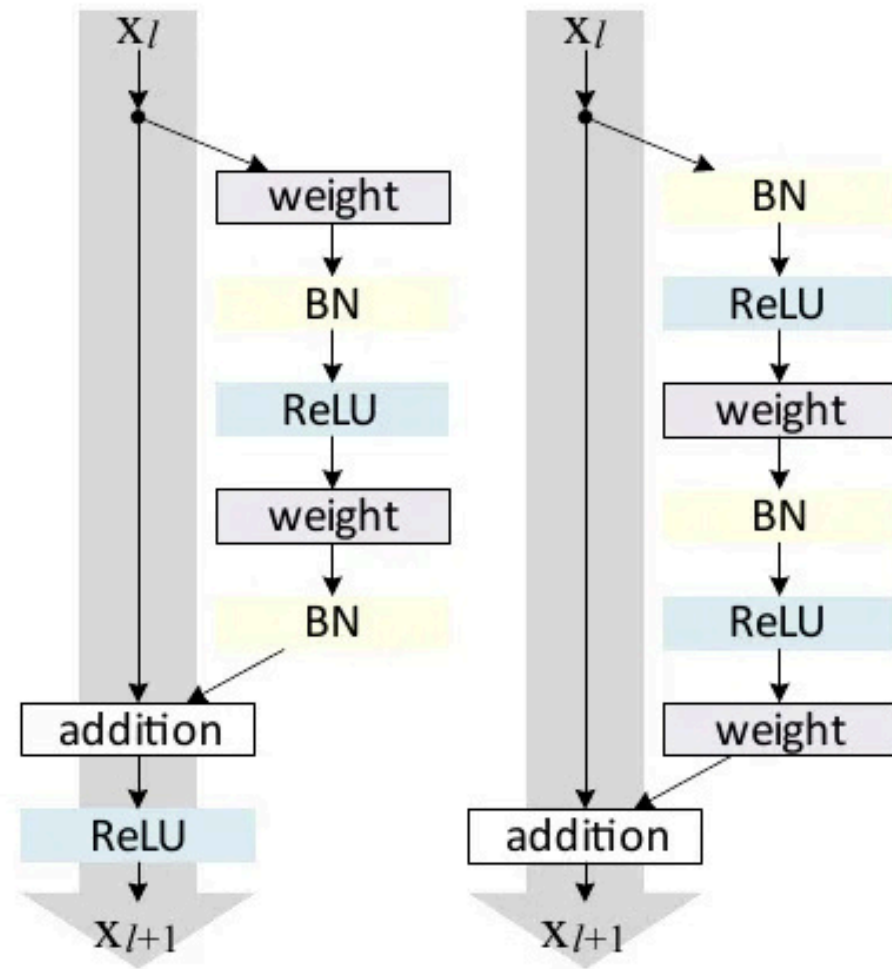
Epoch [194/500],
Train Loss: 1260.2117, Train SSIM: 0.9230,
Val Loss: 4489.4179, Val SSIM: 0.8176



Epoch [195/500],
Train Loss: 1236.3432, Train SSIM: 0.9238,
Val Loss: 4431.8253, Val SSIM: 0.8230

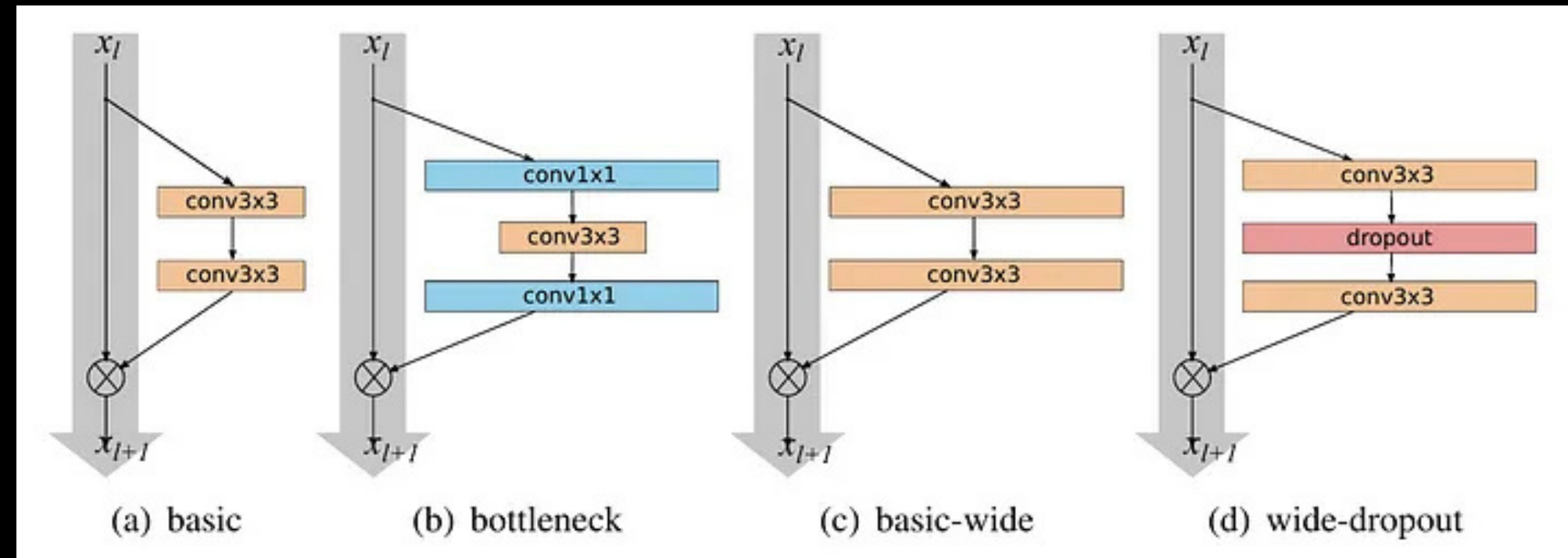
Wide ResNet

Model Design



(a) original

(b) proposed



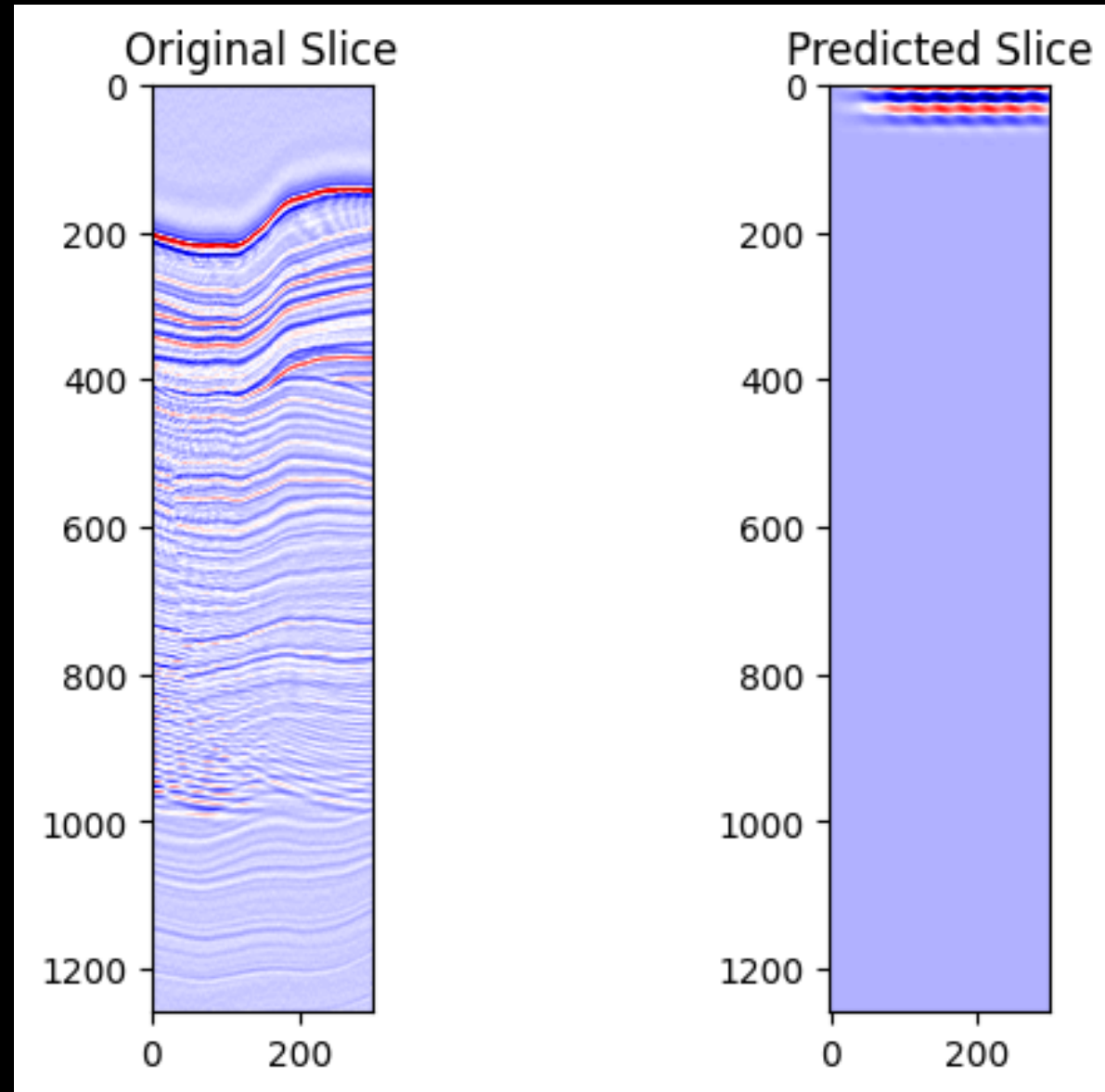
(a) basic

(b) bottleneck

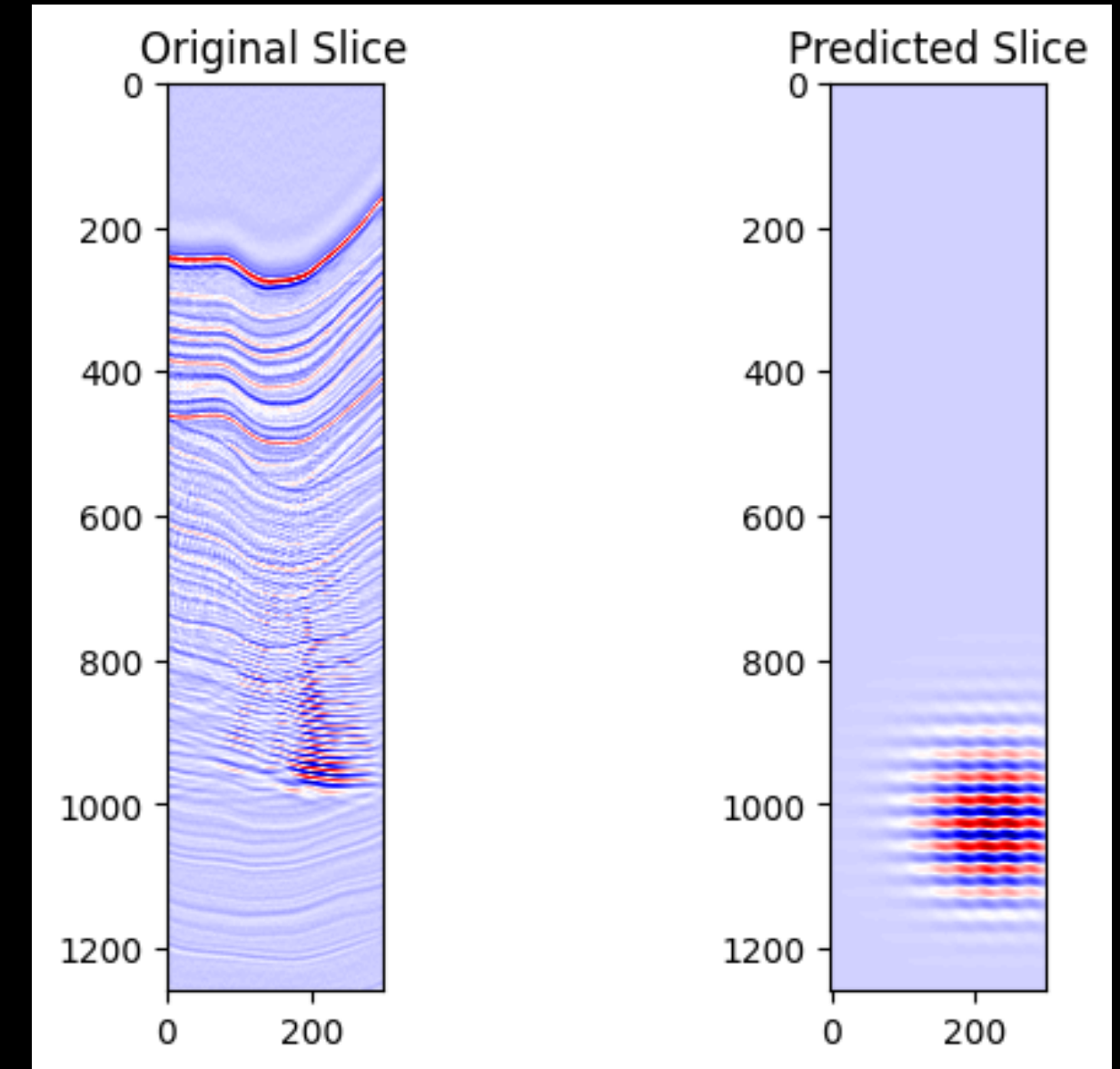
(c) basic-wide

(d) wide-dropout

Model Performance



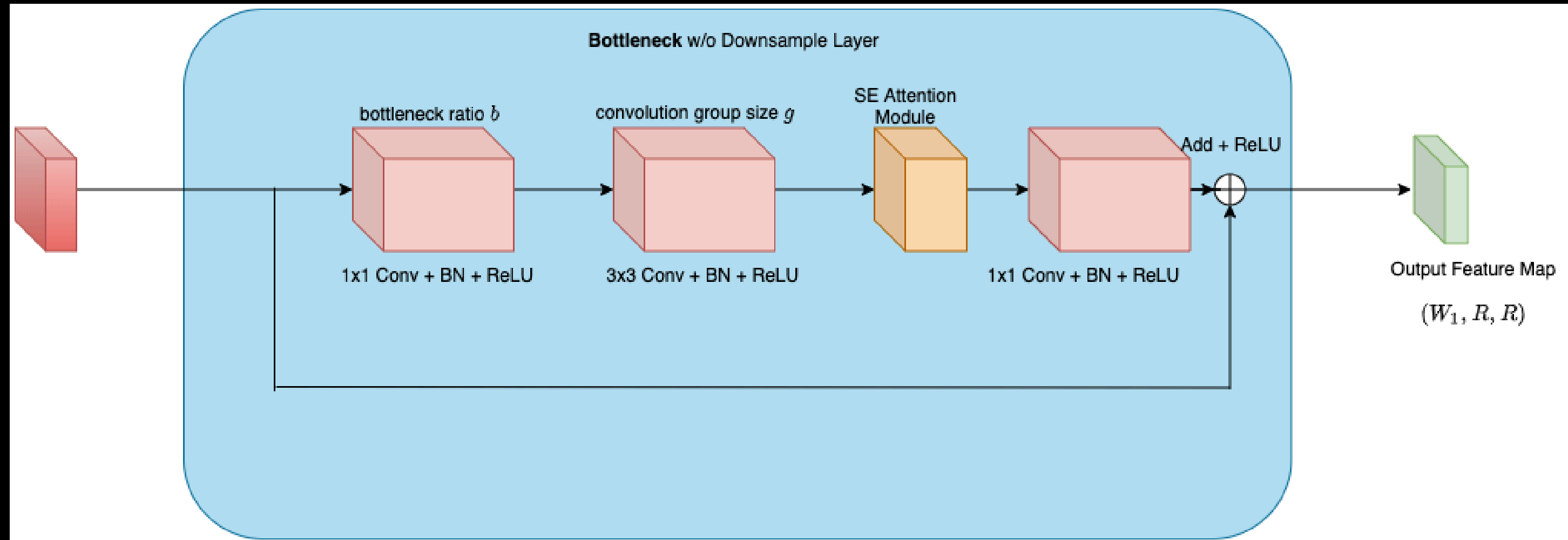
Epoch [6/200],
Train Loss: inf, Train SSIM: nan,
Val Loss: inf, Val SSIM: nan



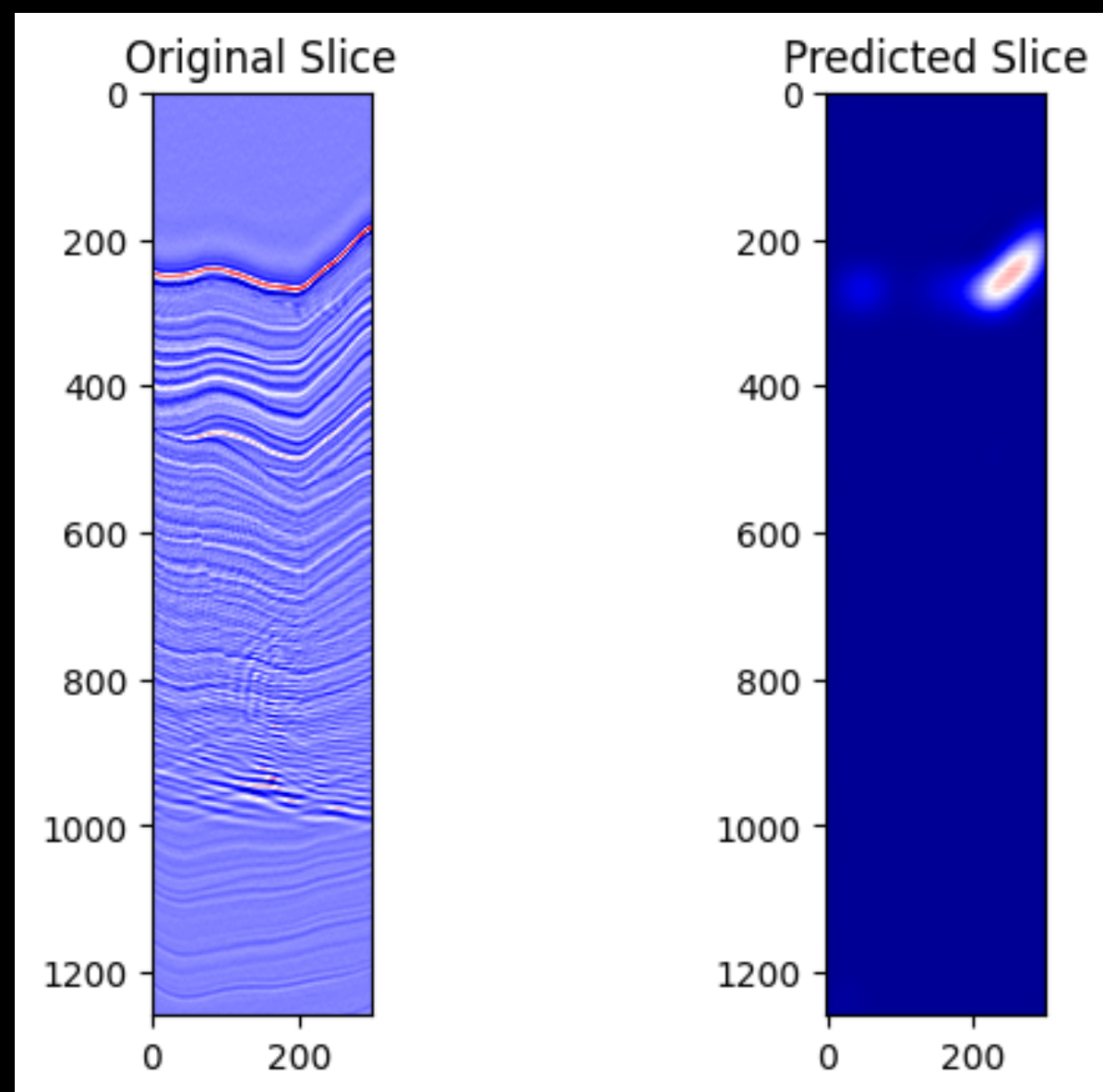
Epoch [7/200],
Train Loss: inf, Train SSIM: nan,
Val Loss: inf, Val SSIM: nan

RegNet

Model Design



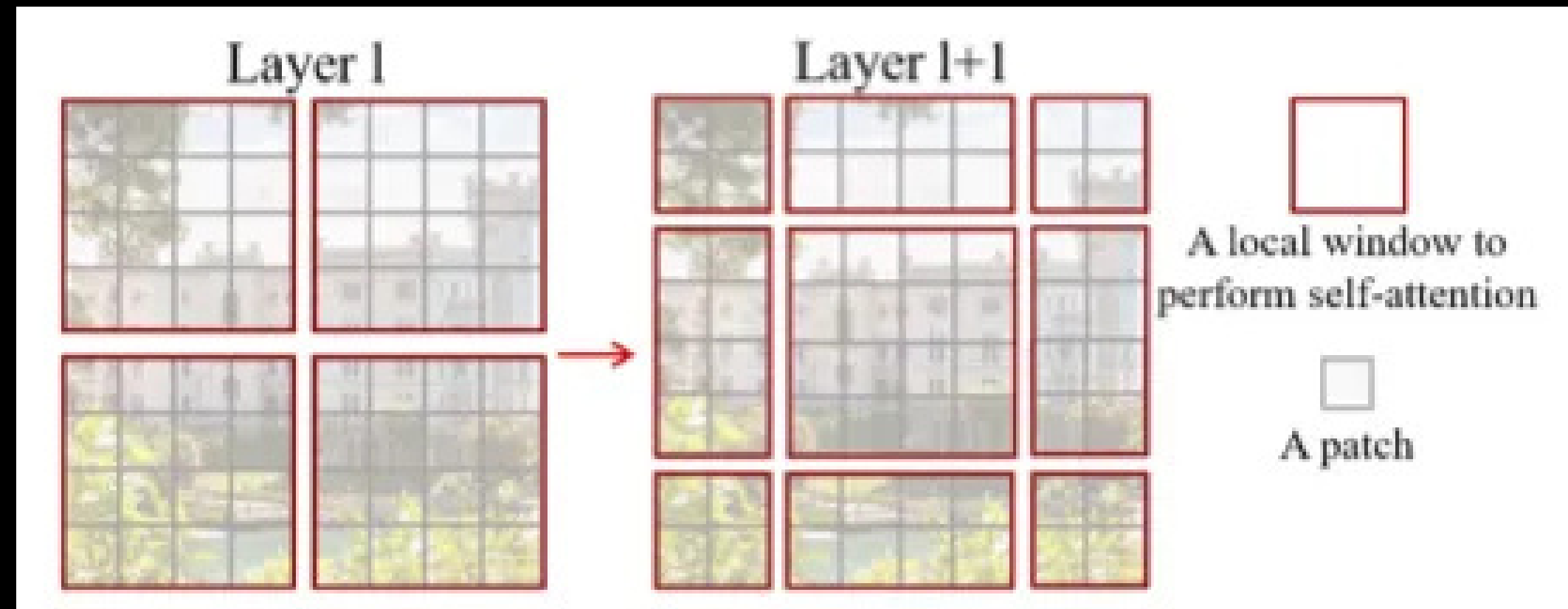
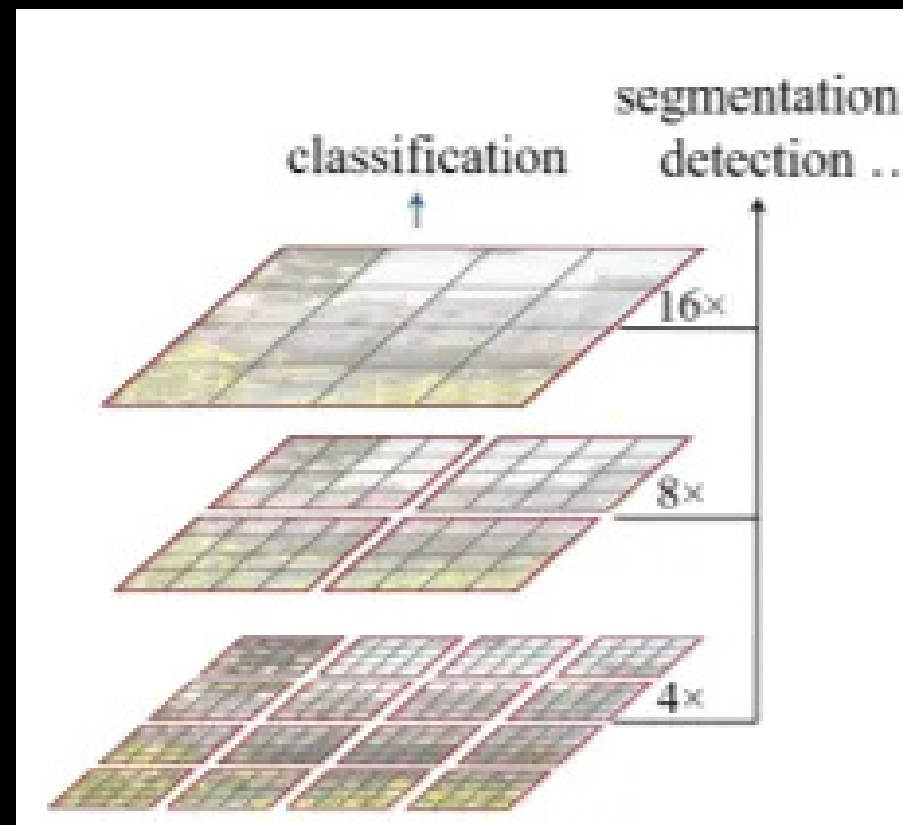
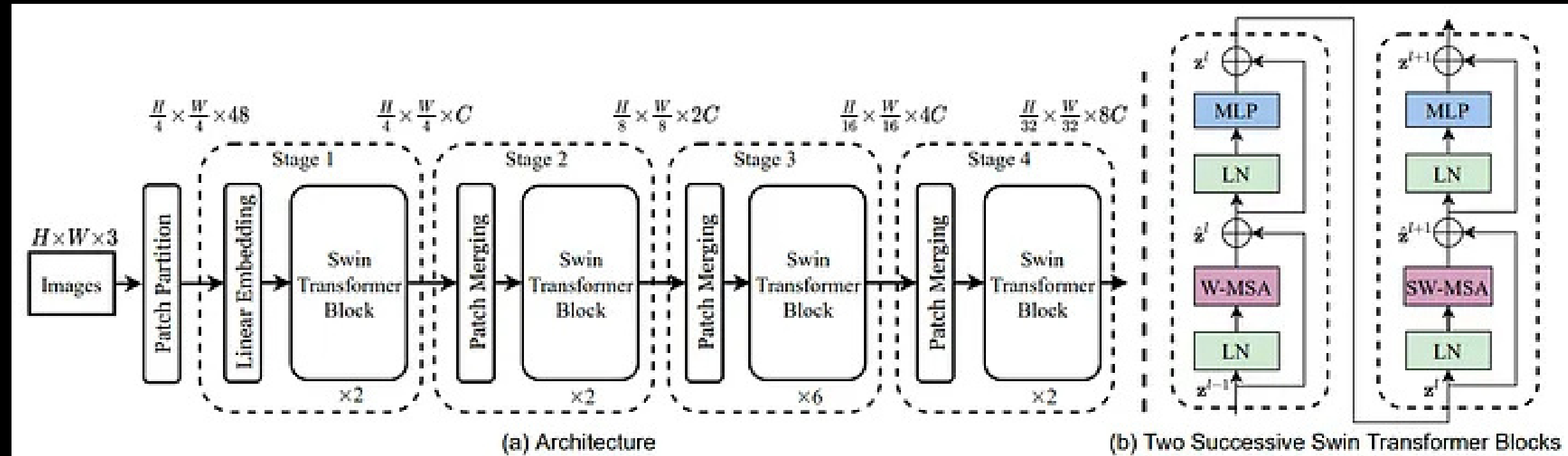
Model Performance



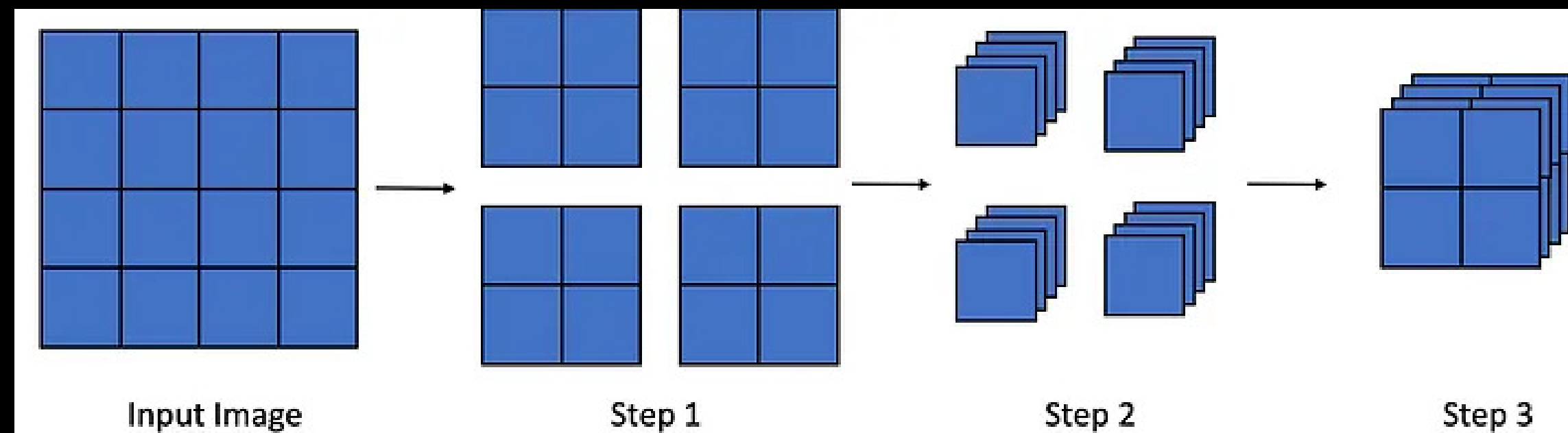
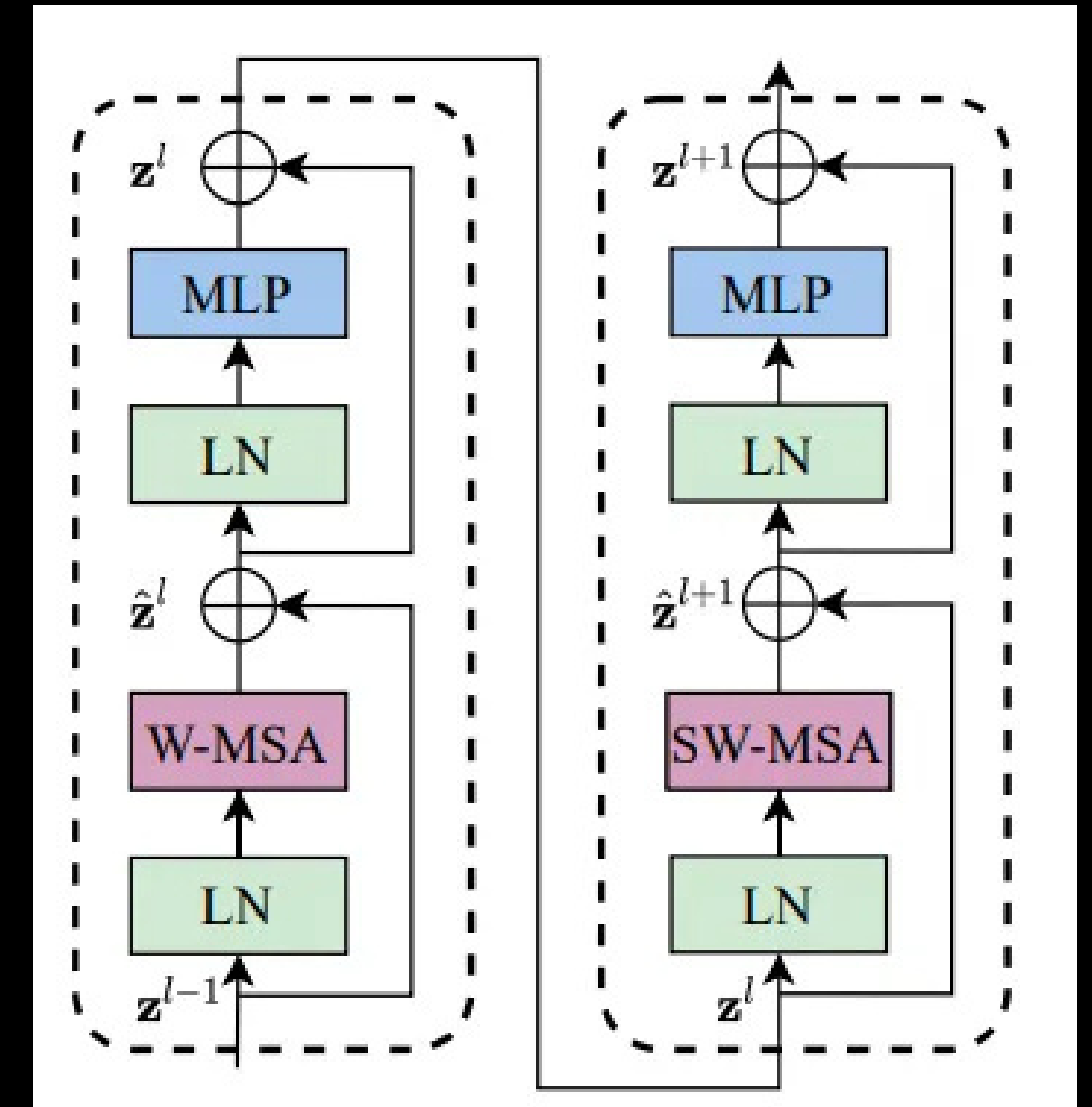
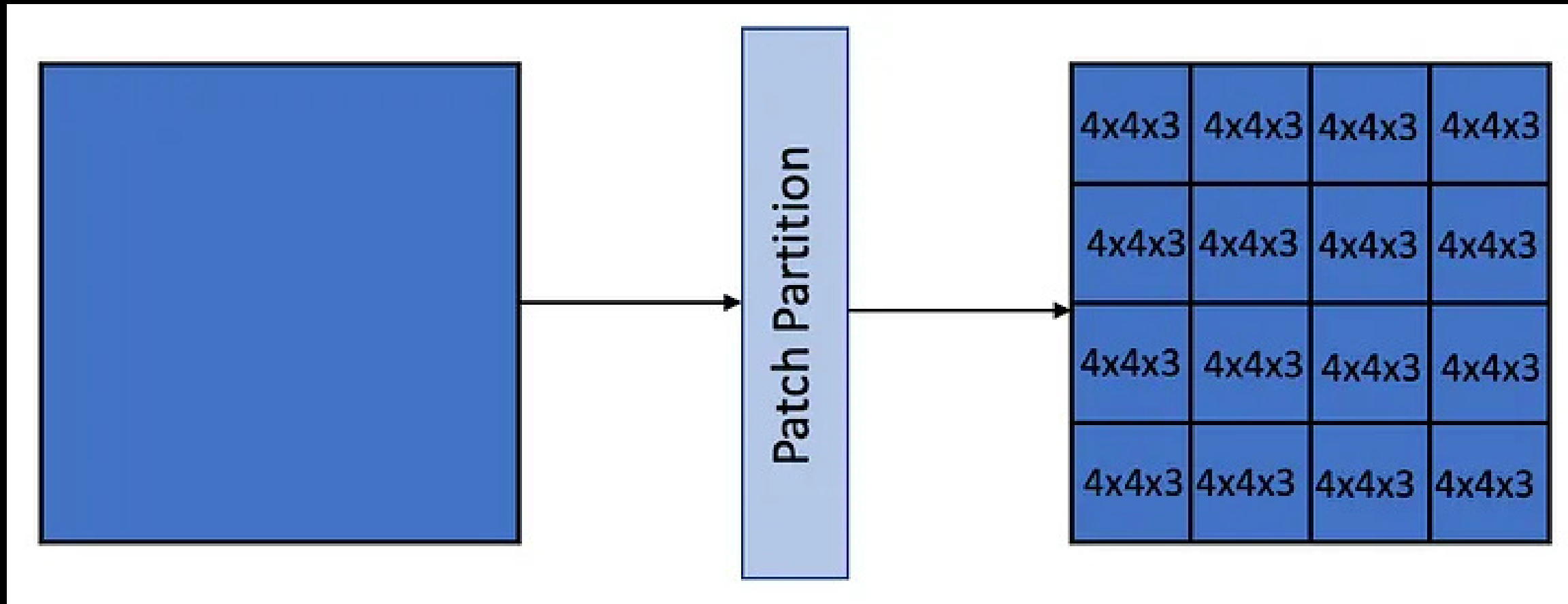
Epoch [1/200],
Train Loss: inf, Train SSIM: nan,
Val Loss: inf, Val SSIM: nan

Swin Transformer

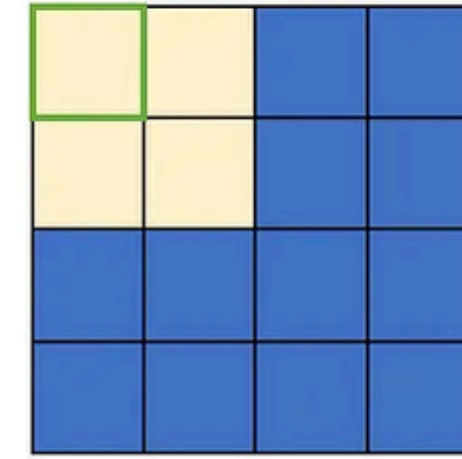
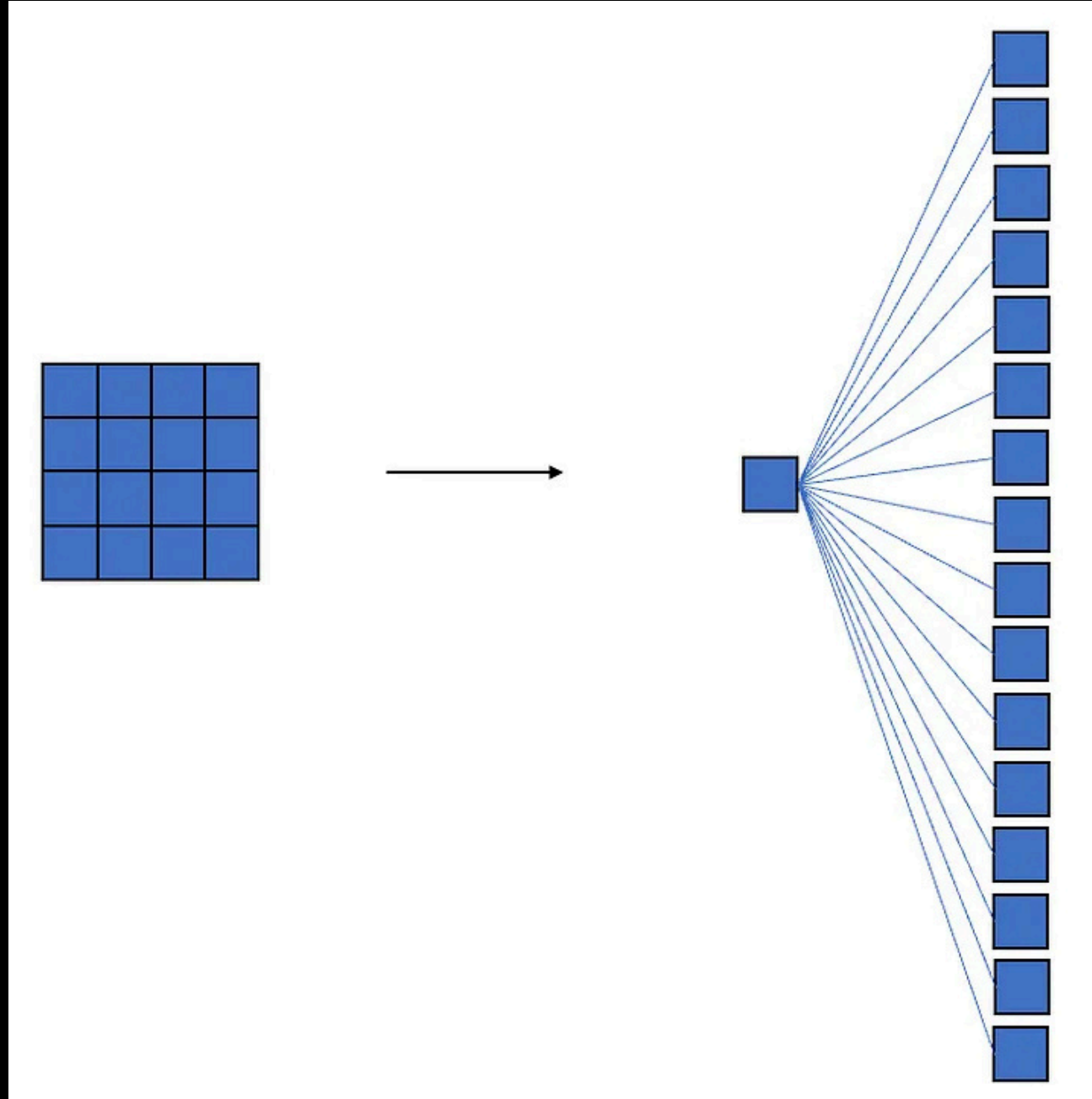
Model Design



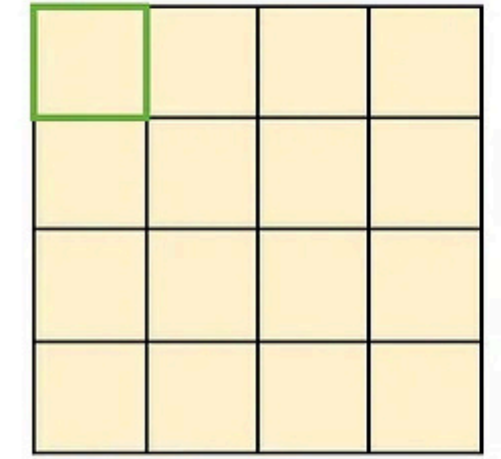
Model Design



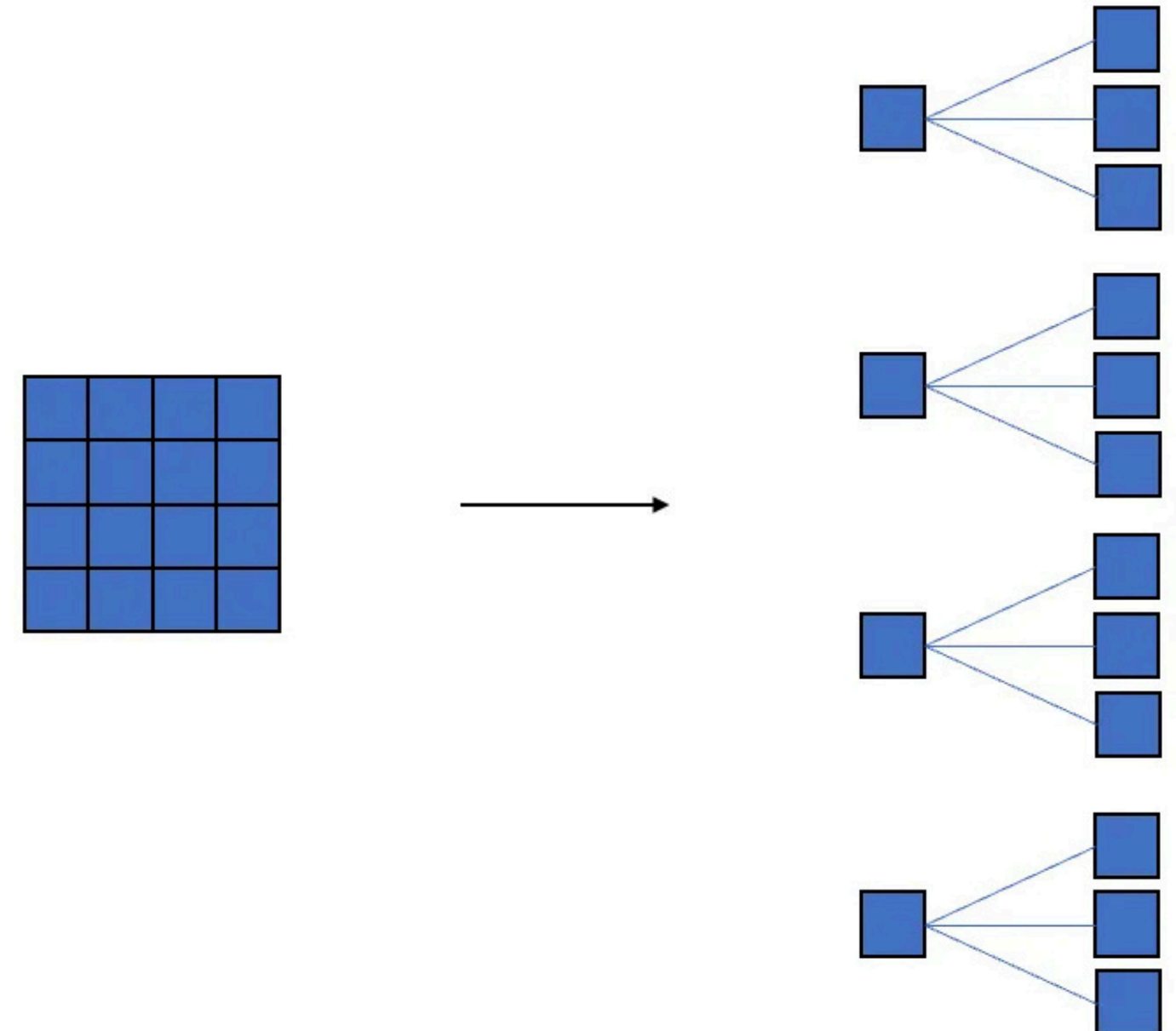
Model Design



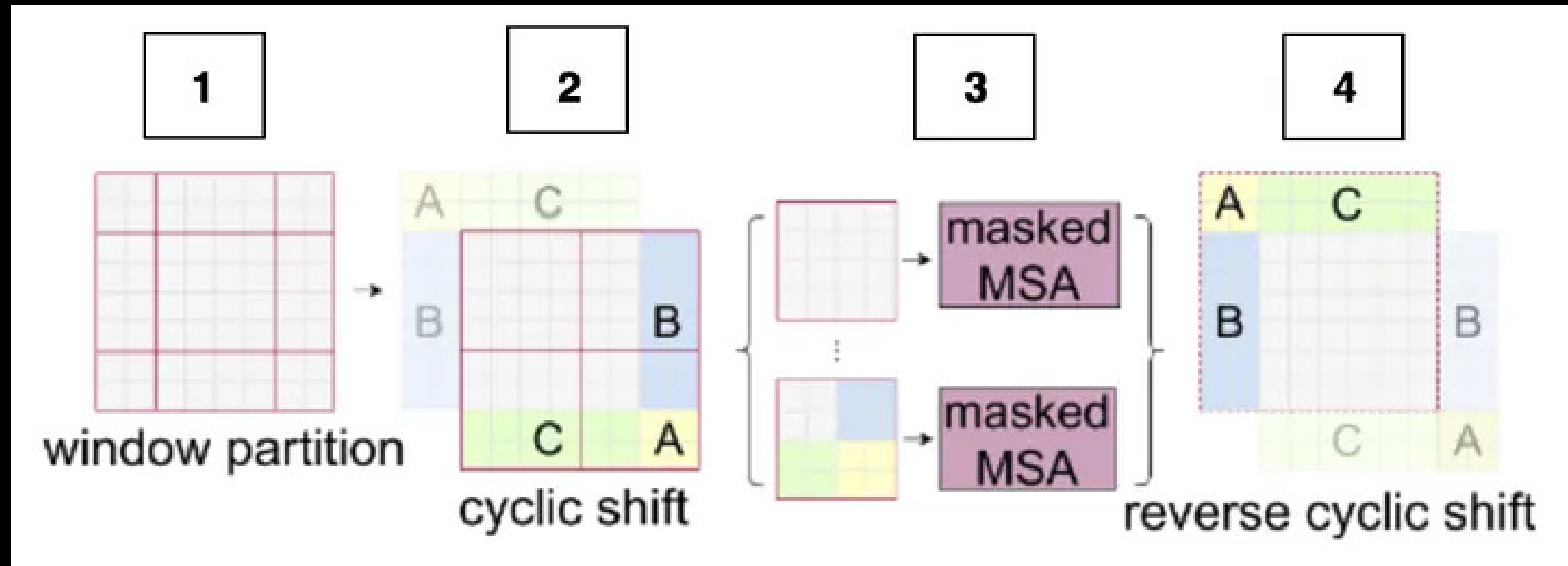
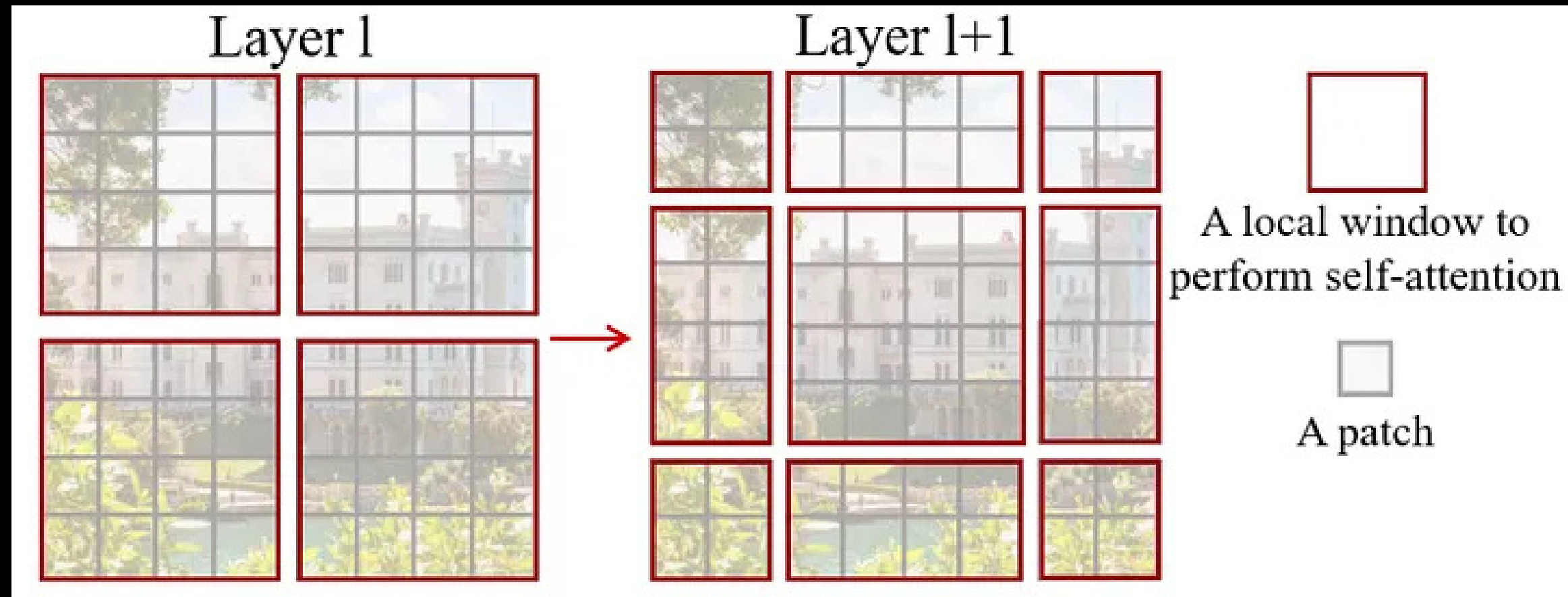
Window Based Self-Attention - SWIN



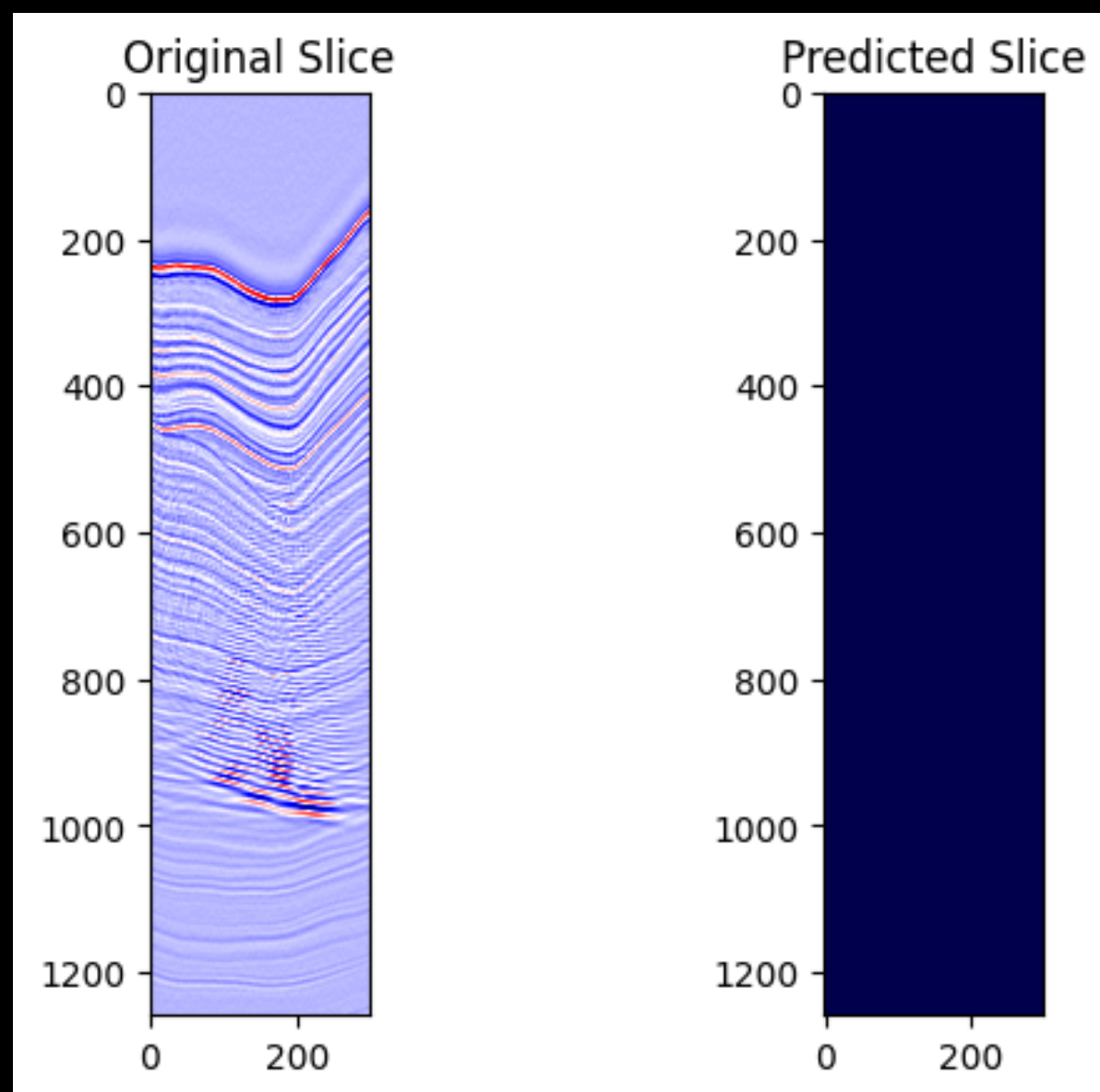
Global Self-Attention - ViT



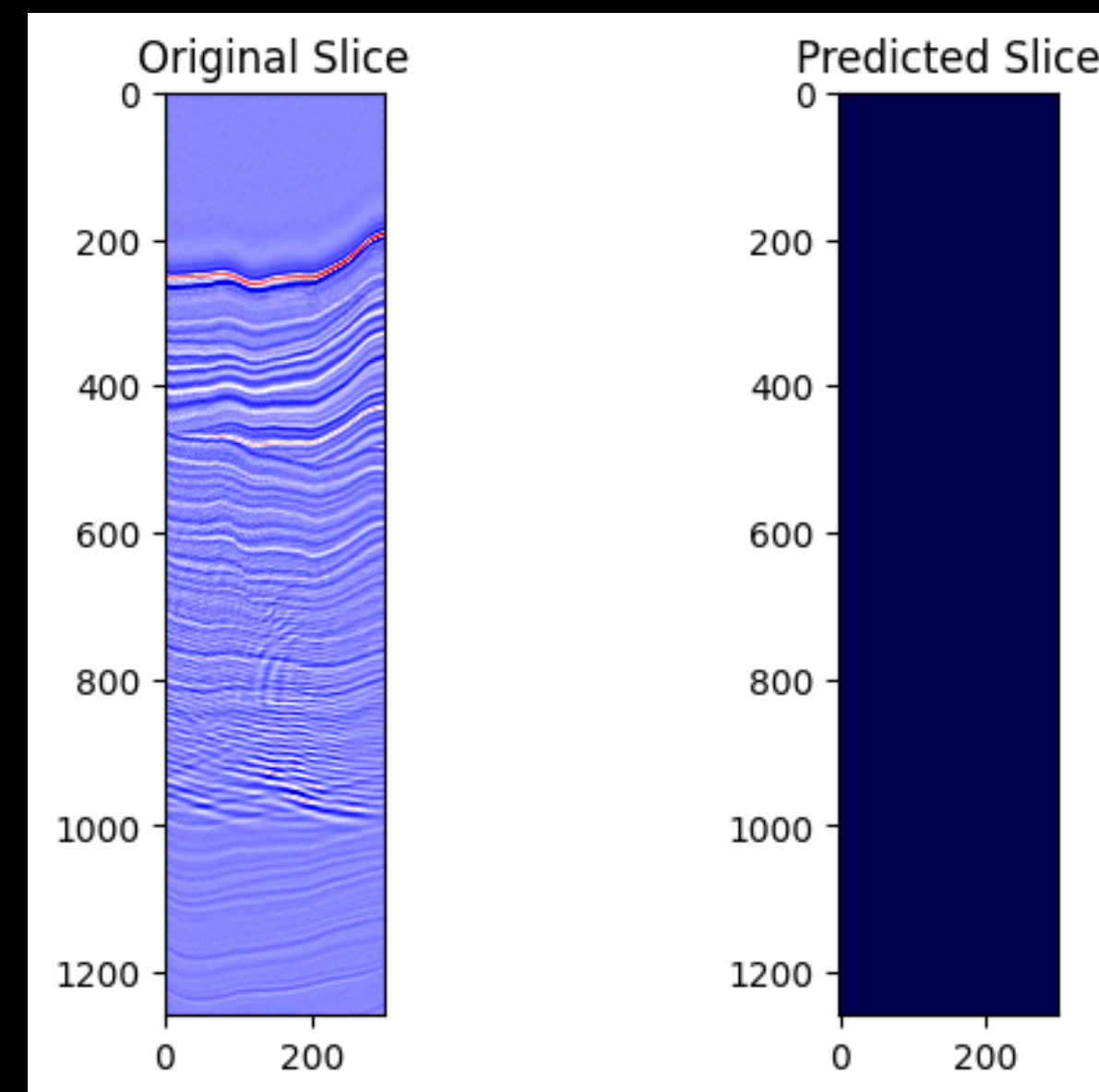
Model Design



Model Performance



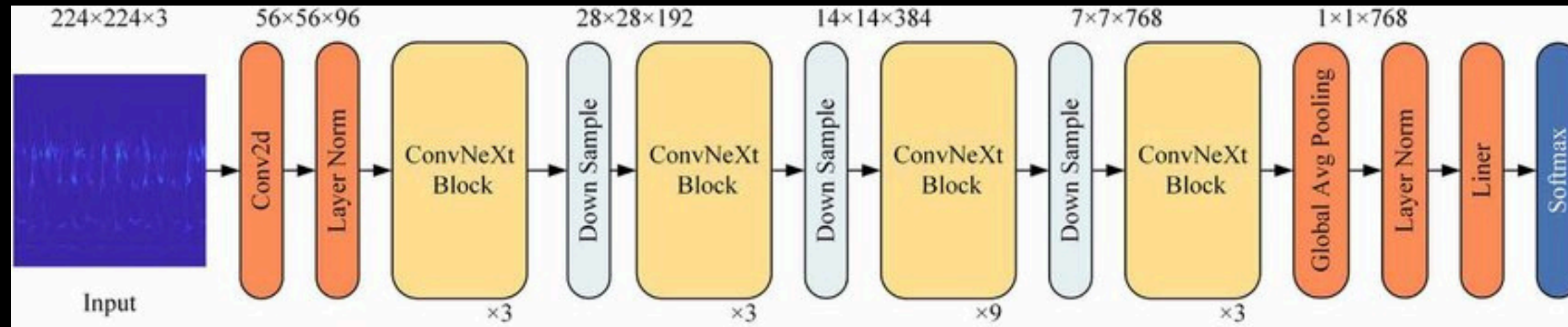
Epoch [136/200],
Train Loss: 10075.4040, Train SSIM: 0.4732,
Val Loss: 10088.1913, Val SSIM: 0.4163



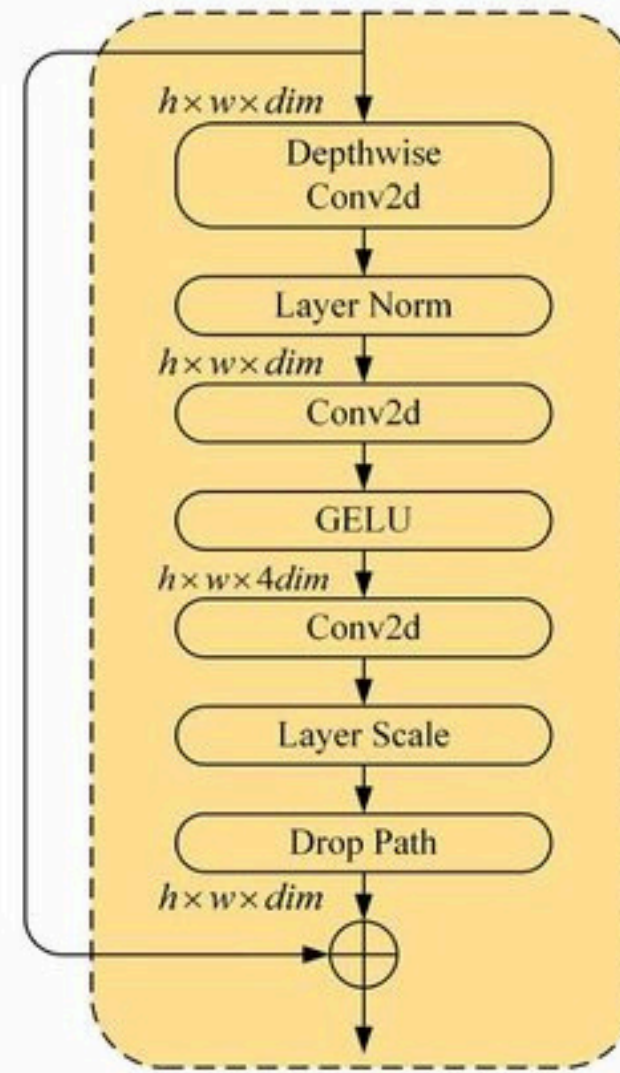
Epoch [138/200],
Train Loss: 10078.8154, Train SSIM: 0.4753,
Val Loss: 10088.1913, Val SSIM: 0.4163

ConvNeXt

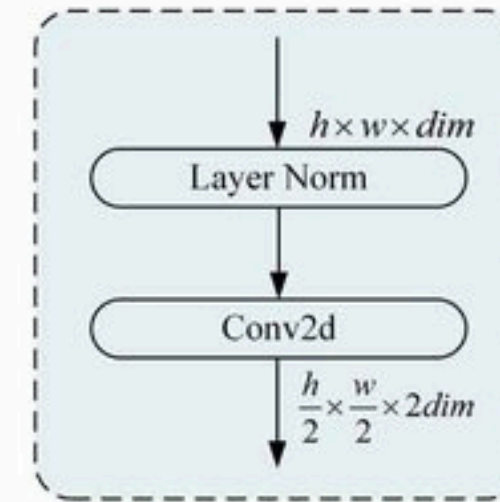
Model Design



(a)

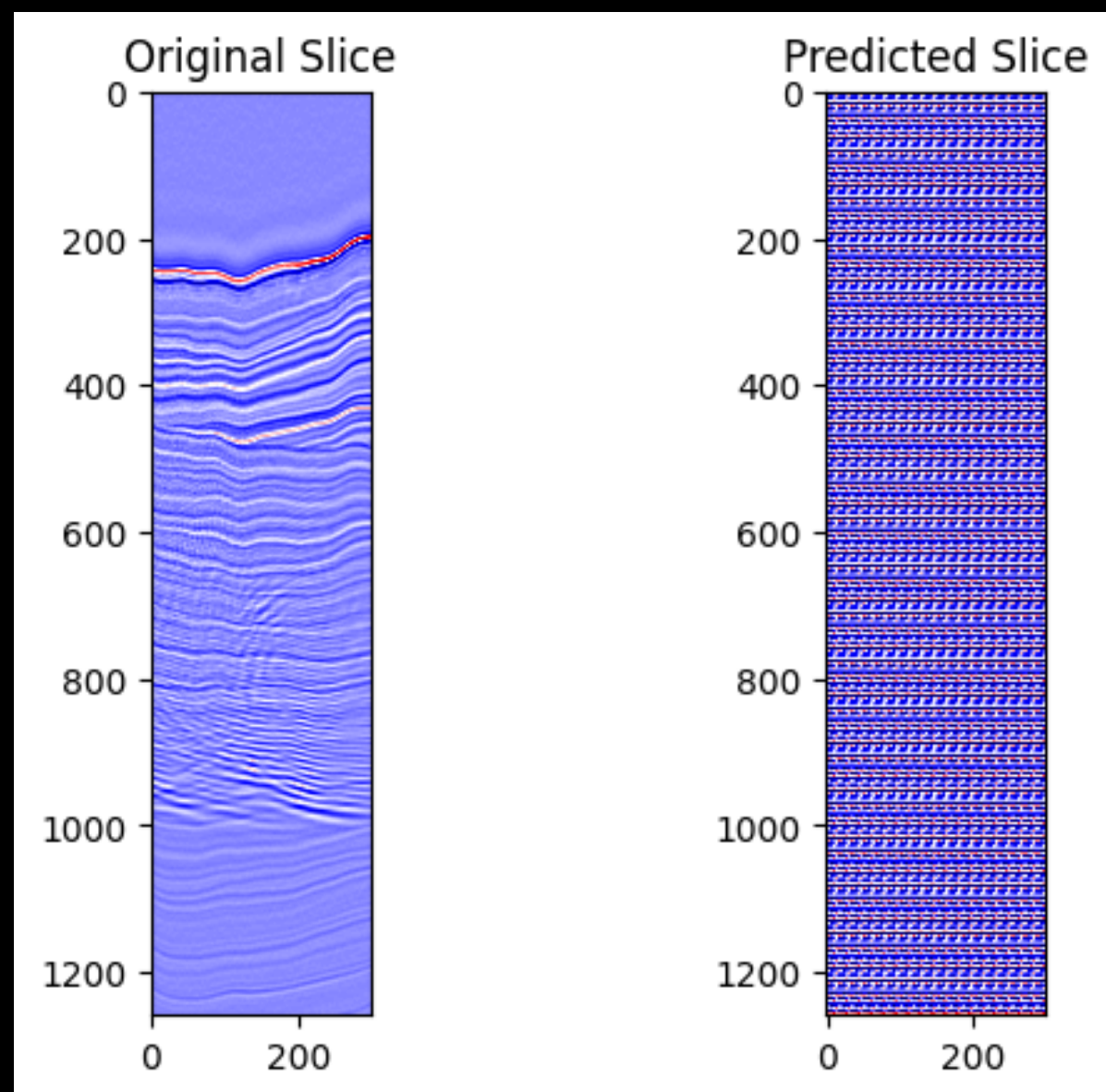


(b)

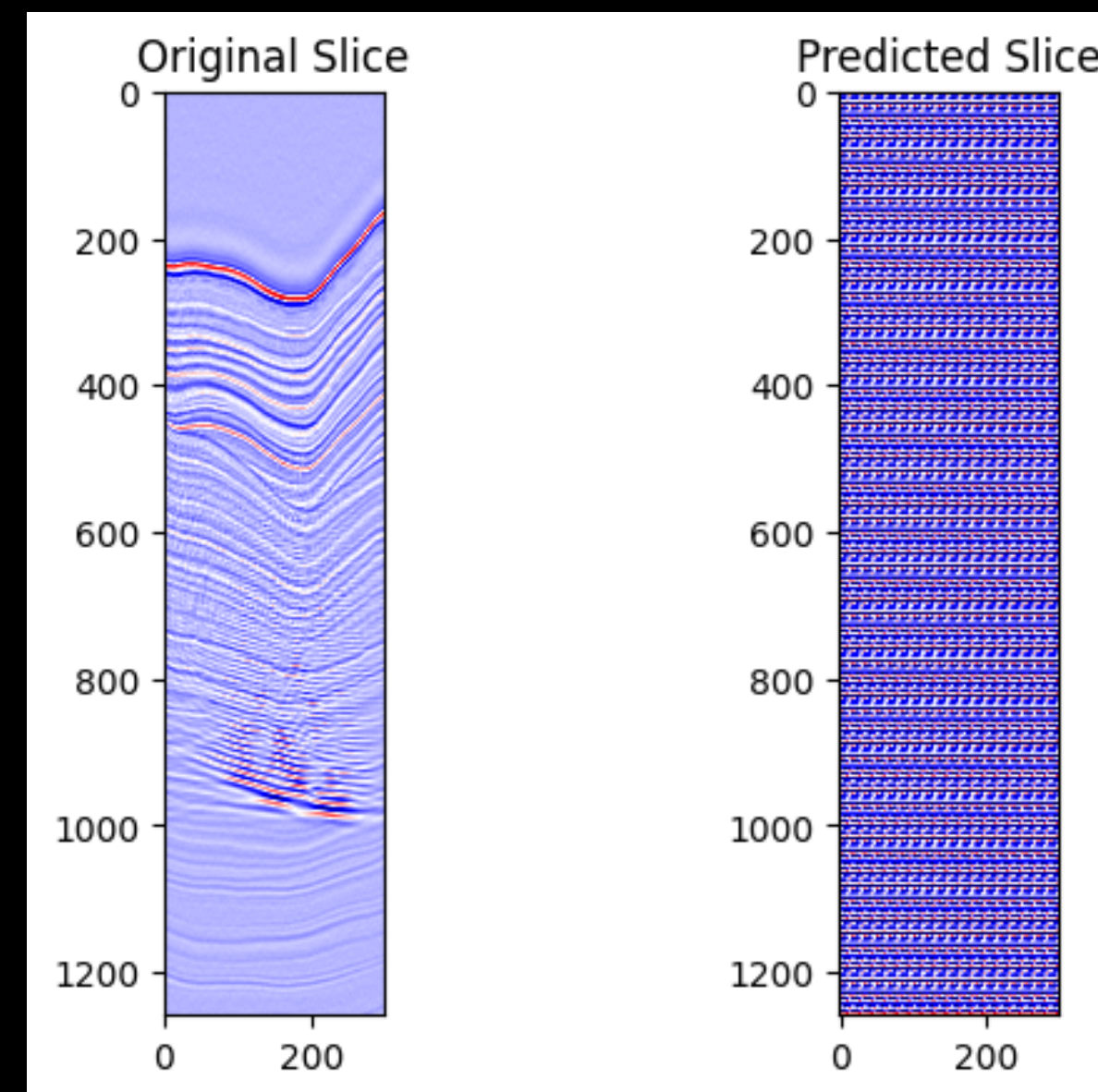


(c)

Model Performance



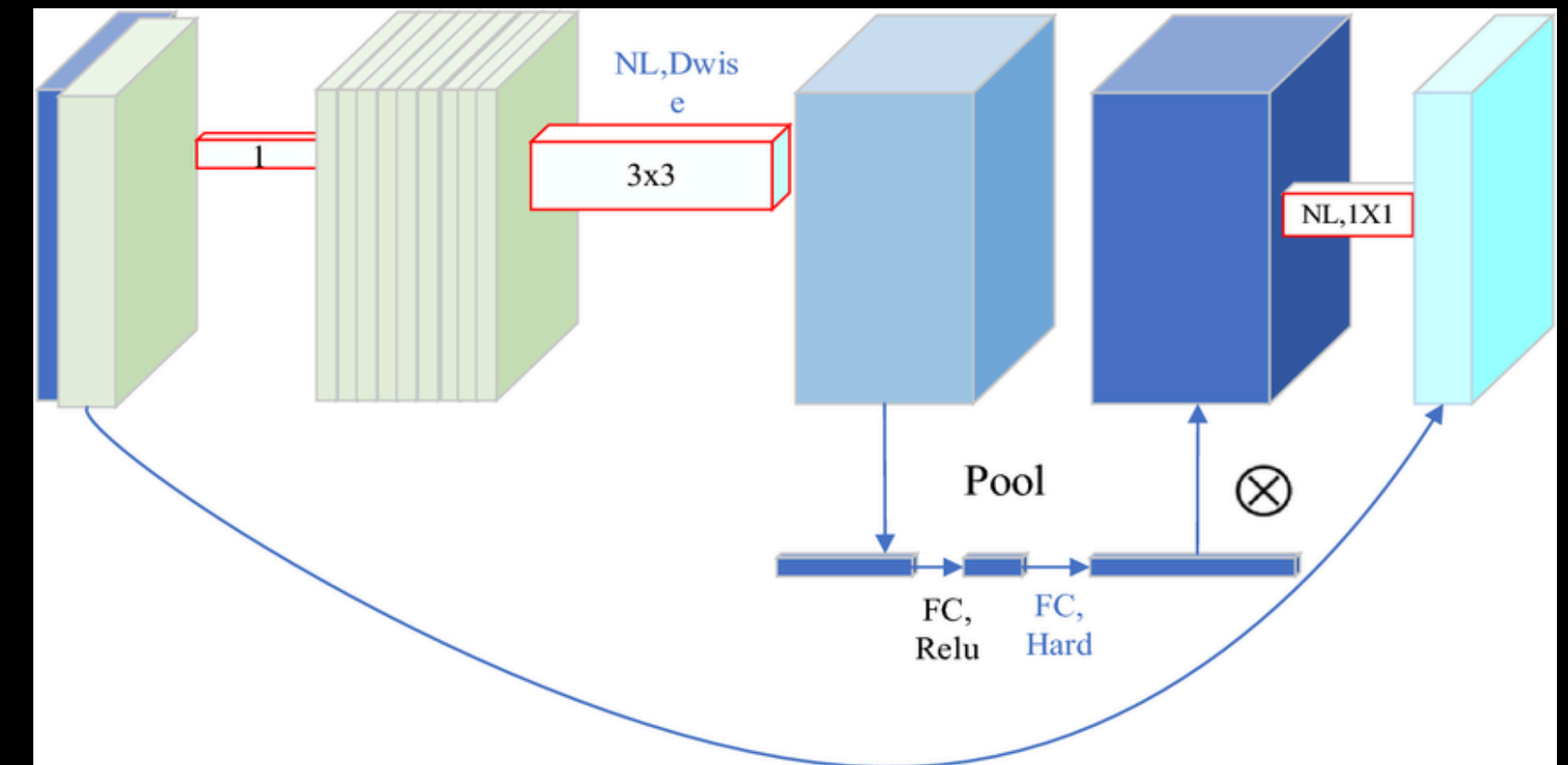
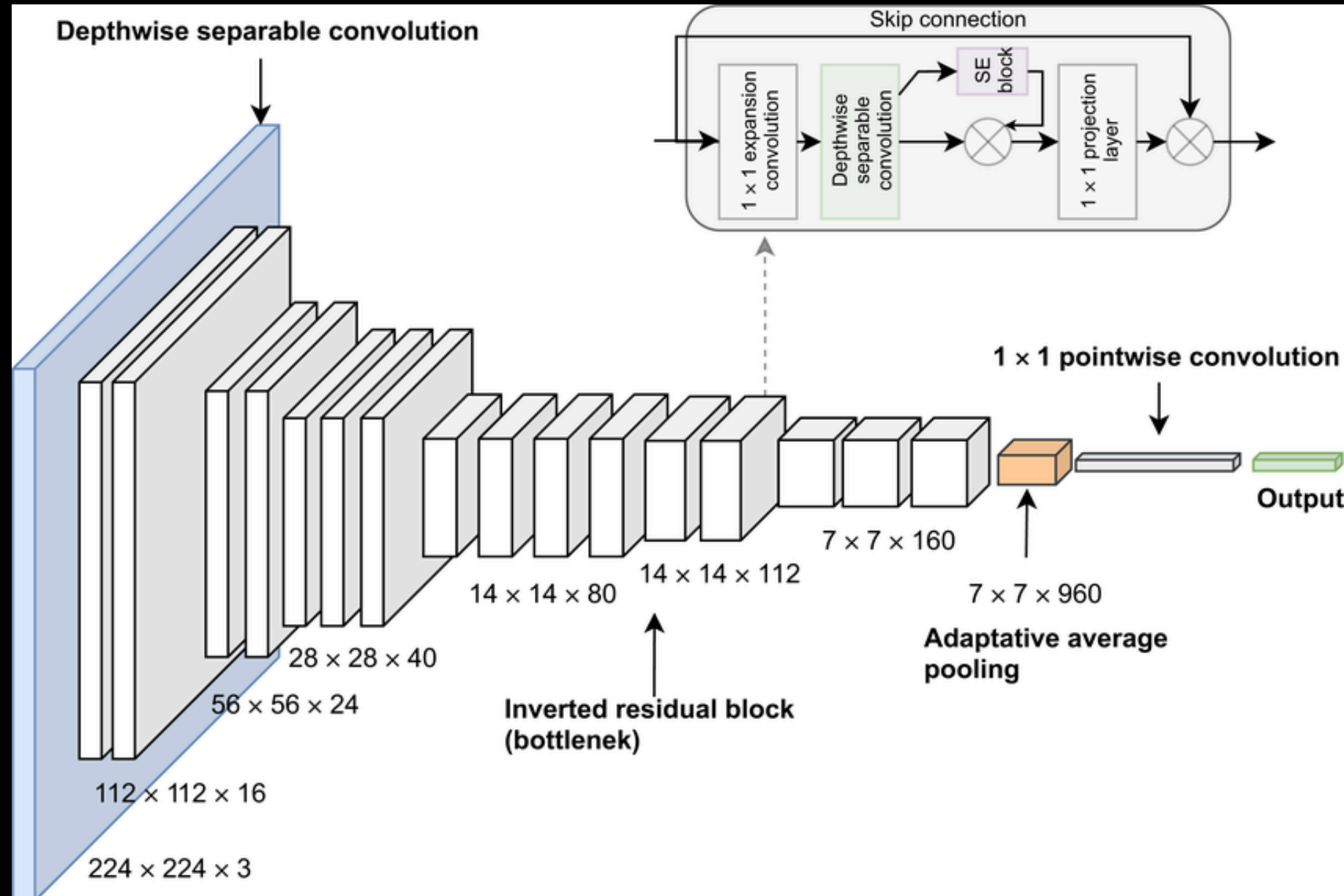
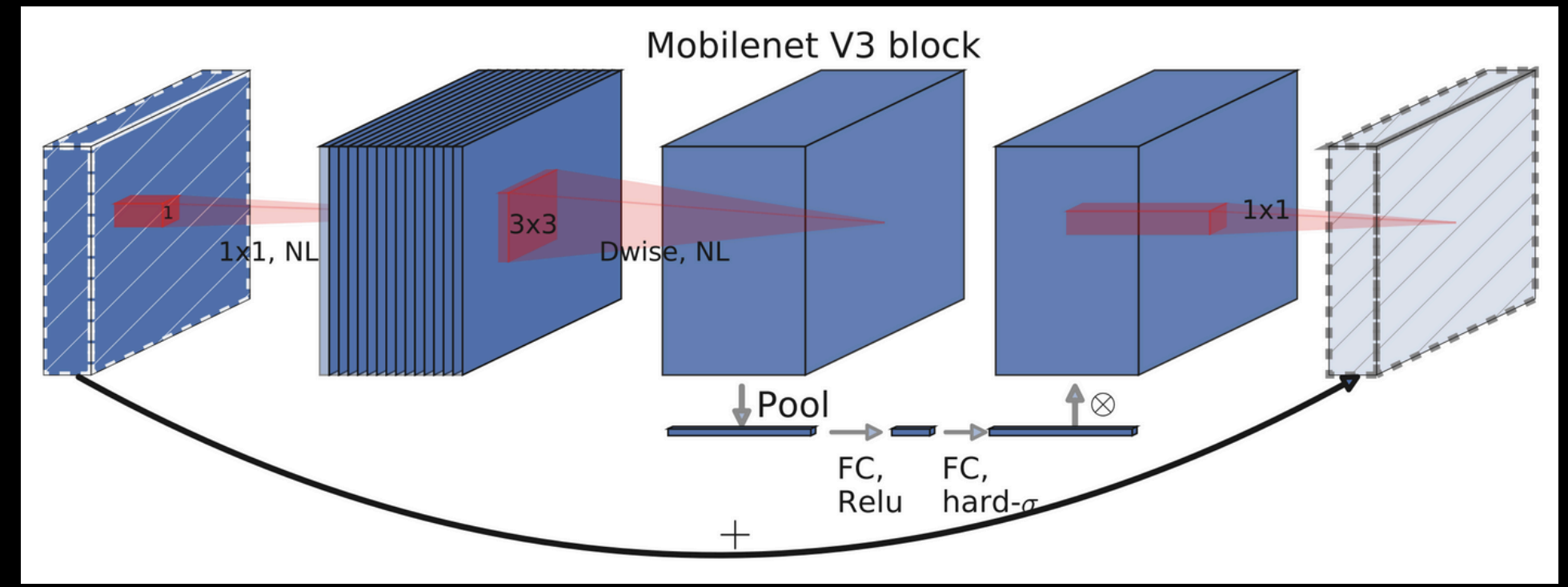
Epoch [33/200],
Train Loss: 10061.9254, Train SSIM: 0.4731,
Val Loss: 10088.3214, Val SSIM: 0.4161



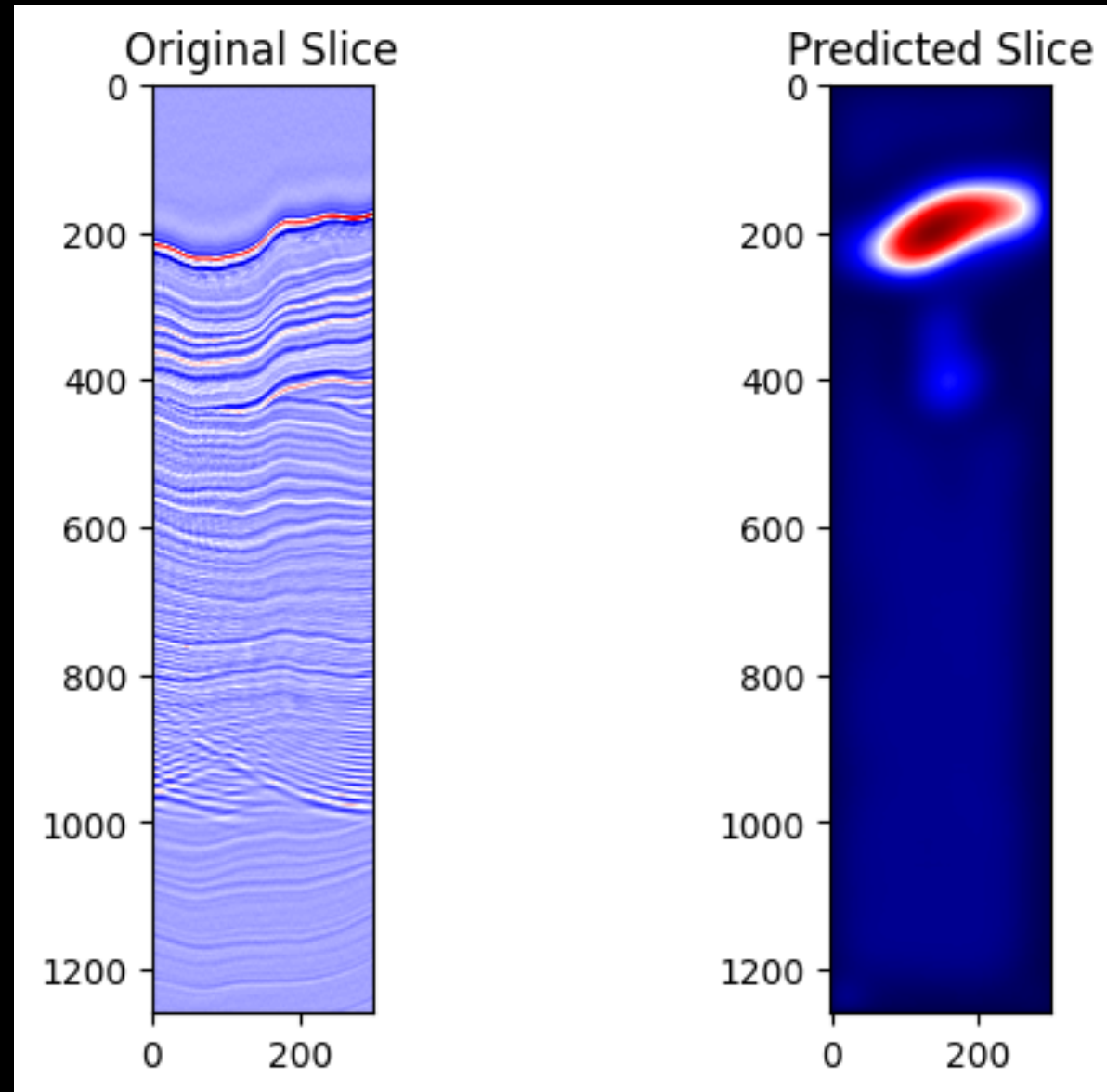
Epoch [34/200],
Train Loss: 10073.7707, Train SSIM: 0.4724,
Val Loss: 10088.3221, Val SSIM: 0.4161

MobileNet V3

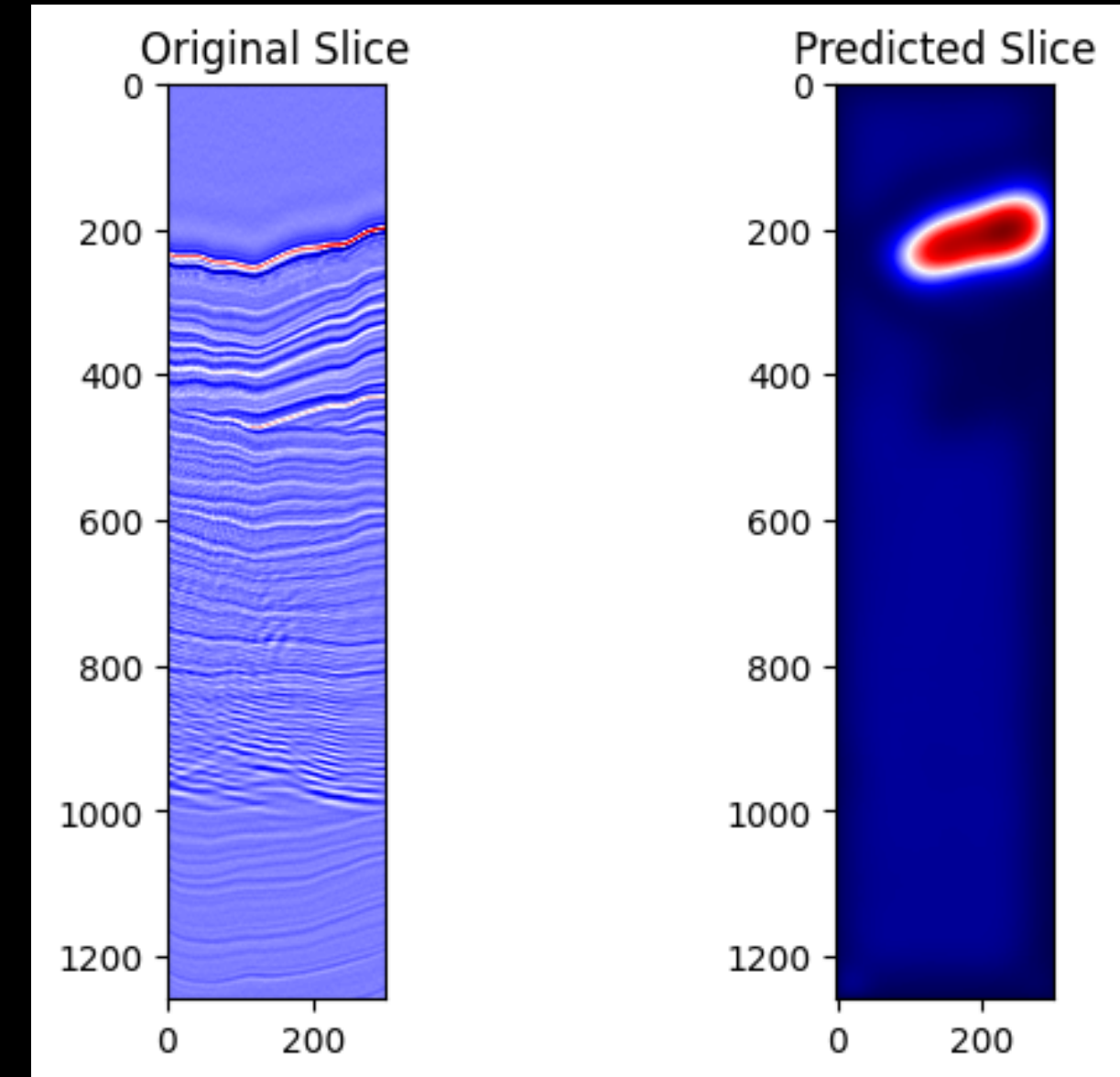
Model Design



Model Performance



Epoch [9/200],
Train Loss: inf, Train SSIM: nan,
Val Loss: inf, Val SSIM: nan



Epoch [8/200],
Train Loss: inf, Train SSIM: nan,
Val Loss: inf, Val SSIM: nan

Top Performing Model

DenseNet

Efficient V2

ResNext

Vision Transformer (ViT)

Max ViT

... we will work on later

Thanks You !

Slides Presented by

Priyanshu Sharma